

基于因果场景与滚动承诺的微网购电调度

摘要

针对微网负载、光伏与电价不确定性，本文以十分钟物理区间为统一尺度，将储能调度、非对称缺口损失与购电承诺建立为可实施的联合优化模型。数据仅来自题目附件，所有预测输入满足观测完成和预报发布时间约束。

问题一采用含效率、功率与储电量边界的线性规划，得到日购电费 35,126.95 元。问题二和问题四的日前模型把五倍价紧急采购直接纳入经验场景目标；历史负载、光伏和电价按同日联合取样，候选场景与终端惩罚只在一月选定。二月至十二月费用分别为 14,794,949.72 元和 15,623,134.83 元，相对均值预测控制分别减少 1,662,260.45 元和 1,668,740.06 元。

问题三及其波动电价扩展采用 24 小时滚动窗口。固定共同日前承诺后，0、6、12、18 点更新的 S3 方案相对仅 0 点更新的 S0 方案分别减少 123,506.61 元和 155,342.73 元；同时报告 S2、不同求解器和计费解释的结果，不把单一路径排序推广为信息价值定理。对真实承诺链逐笔结算，逐次调整相对最终净额分别多 96,160.53 元和 98,826.62 元。

为避免无历史首日的信息泄漏，一月采用明示的保守启动合同，并从 4464 个历史回放时段衔接二月储电量。以独立物理复算、共同基线、全年替代求解器回放、月度机制分析及块长度敏感性检验形成证据链。结果支持在当前数据与计费解释下显式处理尾部缺口损失和日内信息更新；单个有限窗口的最优解不等于全年策略最优，经验尾部统计也不能保证未知极端场景下的供电表现。

关键词：微网调度；非对称缺口成本；经验场景；非前视约束；滚动优化

1 问题与信息边界

题目[1]中的微网由小区负载、光伏电源、储能和外部电网构成。问题一给定固定日型，要求0点制定计划且首尾储电量相同；问题二在负载和光伏变化下引入五倍价紧急购电；问题三允许使用每天四次发布的未来24小时光伏预报调整采购；问题四再考虑未知未来实时电价。除完整工作簿外，论文需报告四个指定日期的十分钟采购、四小时储能、端点储电量及紧急购电区间。

附件2—4覆盖2025年365天，正式结果表覆盖2月1日至12月31日334天，每天144段，共48,096段。1月1日6000 kWh是题给初值，不能不经运行过程移到2月。本文将一月用于有明示规则的设备启动和离线参数验证；正式比较自二月开始，所有方案读取同一启动末状态。

令 t 表示区间起点， $\Delta t=1/6$ h。源数据 00:10 解释为[00:00,00:10)的代表功率，依次至 24:00。该右端点代表值解释是统一采用的离散化约定。原模板的 0:10–0:20 等文字与列序存在偏移，因此保存位置键、源时间戳、物理区间、模板标签四个字段；计算与论文按物理键，官方表按原位置保存，并另附映射页。六个指定区间的起始位置分别为 61、73、85、97、109、121。

预报“提前1小时”对应发布后第一个整点，本文将该值作为前一个小时六个十分钟区间的代表值；例如6点发布的第1小时值用于6:00–7:00，18点的第7小时值用于次日0:00–1:00。此零阶保持离散化不使用未来实际插值。凌晨、跨日和全天终点采用同一映射。

信息集包括决策时刻以前已完成的观测、截至该时刻发布的预报及已形成的合同。允许观测在物理区间结束后立即可得；若存在实际通信延迟，应在可用时刻中加入延迟量。

$F_\tau = \sigma(\text{观测结束时刻} \leq \tau; \text{预报发布时间} \leq \tau; \text{既有采购承诺})$ (1)

计划采购和储能控制必须可由 F_τ 决定。当前或未来的实际负载、出力与电价只进入事后执行评价，不能作为当前优化输入。完全信息下界单独命名为事后对照。

1.1 方法依据与本题贡献

线性规划提供物理可行域和费用目标的求解基础[2]；样本平均近似以有限场景表示不确定量[3]；滚动控制在新观测到达后更新有限窗口决策[4]。在上述方法基础上，本文将五倍缺口代价、联合历史误差和储能跨时段耦合纳入同一规划，以共同日前合同控制更新频次比较的初始条件，并记录每次承诺变化以核验净额结算与逐次结算。

本文从决策可实施性、物理与计费结果可复算性、相对可行基线的经济改善三个方面评价模型。数值核验与经济性比较分别提供证据；有限窗口的求解最优性不代表全年策略最优性。

1.2 技术路线

原始附件先按日期和物理区间对齐并统一量纲；每次决策截取已完成历史和已发布预报，生成联合场景；在储电量和承诺约束下求解，仅执行至下一允许更新时点；区间结束后记录实际缺口、费用和下一状态。训练期选择、正式回放、对照实验及文稿生成分别留存输出，预测场景与实测数据分别保存。本文评价基于历史数据回放，因果信息检查用于验证逐次决策的信息可用边界。

2 假设与符号

微网视为单母线系统，忽略未提供参数的线路损耗、储能自放电和寿命折旧；不引入售电收入。外网计划和紧急采购均非负，题面未给外网采购容量上限。允许光伏和已付费富余电量不利用，计入未用能量而非负采购。充电与放电效率各为 0.9，往返效率为 0.81。充放电量均定义在交流母线侧。

主结算采用取消部分退回原购电费、保留 50%违约费的解释；新增量总价为 1.5 倍。价格按交付物理区间的固定价或实际价计，采购时用其因果预测。题面对取消退费 and 多次结算并不完全明确，因此本文同时报告不退原价及逐次收费的同路径对照；不同解释下的金额不能称为同一优化问题的最优值。

表 1 假设与符号

符号	含义	单位
L_t, V_t	负载与光伏区间电量	kWh
B_t, A_t	0 点计划量与当前最终承诺	kWh
C_t, D_t	充电与放电母线电量	kWh
E_t	区间开始时储电量	kWh
Q_t, U_t	紧急采购与未用富余	kWh
p_t	交付区间电价	元/kWh
$H\tau, s, N\tau$	预测窗口、场景索引、可用场景数	—
G_t	窗口内共享计划采购量	kWh
Δt	离散区间长度 (1/6)	h
$\eta c, \eta d$	充电效率与放电效率 (均为 0.9)	—
λ, ε	终端储电量惩罚与数值正则系数	元/kWh

3 物理约束与问题一

执行层由实际负载和光伏重新计算紧急缺口及未用能量。储能状态使用实际执行的充放电动作递推，前一日末值等于后一日初值。

$$A_t + V_t + D_t + Q_t = L_t + C_t + U_t; \quad Q_t, U_t \geq 0 \quad (2)$$

$$E_{t+1} = E_t + 0.9 C_t - \frac{D_t}{0.9} \quad (3)$$

$$1200 \leq E_t \leq 10800; \quad 0 \leq C_t, D_t \leq \frac{5000}{6} \quad (4)$$

物理设备不允许同时充放电。线性松弛求解后令 $z=0.9C-D/0.9$ ，将动作替换为 $C'=\max(z,0)/0.9$ 、 $D'=0.9\max(-z,0)$ 。该变换保持储电量变化、降低循环损耗和所需供电，在允许免费处置富余、没有最小功率约束的本模型下不会破坏供需可行性。程序逐段验证转换后的互斥、状态、功率及情景供电约束，而不仅检查求解器返回成功。

上述互斥处理具有支配关系。若 $z \geq 0$ ，则 $C'=C-D/(\eta c \eta d)$ 、 $D'=0$ ，减少的母线需求为 $D(1/(\eta c \eta d)-1) \geq 0$ ；若 $z < 0$ ，则 $C'=0$ 、 $D'=D-\eta c \eta d C$ ，减少的母线需求为 $C(1-\eta c \eta d) \geq 0$ 。两种情形均有 $C' \leq C$ 、 $D' \leq D$ ，故额定功率上界保留，储电量轨迹不变，采购需求不增。由于本题允许不用富余且不设最小充放功率，此变换给出不劣的可行互斥解。若另加禁止弃电、负电价收益或最小运行时间等条件，需要重新论证，不能直接沿用该结论。

问题一以 $Q=0$ 、 $A=B$ 建立确定性线性规划（LP），最小化 ΣpB ，并施加 $E_0=E_{144}=6000$ 。另按原附件独立构造约束矩阵重新求解，核对最优目标。存在并列最优时，不要求逐段动作唯一。无储能对照固定 $C=D=0$ ，只购入 $\max(L-V,0)$ 。

表 2 确定性调度与无储能对照

方案	全天购电量/kWh	全天费用/元
储能 LP	59,482.6990	35,126.95
无储能	61,789.9354	48,052.05

4 不对称损失与日前随机规划

4.1 联合经验场景

场景仅来自决策时已完整观测的最近 20 条 24 小时轨迹；不足 20 条时使用全部可用轨迹，并记录实际样本数。负载、光伏误差和价格保持同一历史日期配对，以避免独立拼接破坏相关结构。以下两种场景模型在一月内比较，不在正式期根据结果挑选。

原始历史候选直接使用历史联合负载、光伏和价格轨迹。因果残差候选先以决策日前最近 4 个同星期日均值为中心；不存在同星期历史时取最近最多 4 个已完成日。每一历史误差也必须相对当时仅用更早日期生成的预测计算，不能用当前的拟合曲线回填过去。这遵循逐次向前验证的时间切分原则[5]。

$$\hat{X}_t^s = \max(0, \hat{X}_{t|\tau} + (X_{h^s} - \hat{X}_{h^s|\tau_h})) \quad (5)$$

式中 X 可为负载、光伏或波动电价；固定价格使用给定曲线。负值预测截断为零，这是预测分布的非负约束，不改变原始观测。没有先验基线的首日不进入残差样本。跨日预测对次日目标使用对应星期属性，但统计截止仍是当前决策日之前。

问题三/四-3 的光伏中心使用当次正式发布预报，场景误差为同一历史发布时间对应的整段实际小时均值减原预报；历史轨迹结束时刻不得晚于当前决策时刻。两个候选都保留这种发布时间对齐。场景是由真实历史误差构成的预测样本，并非声称未来实际会等于某一历史日。

4.2 目标与非前视约束

在给定 $F\tau$ 的窗口内，对所有场景共享采购 G 、充电 C 、放电 D 和储电量；只有场景紧急量 Q^s 允许不同。未来储能动作也不选择“最有利的场景分支”作为承诺，这比只共享第一个执行块更保守，但消除了场景信息先于观测到达的隐性假设。窗口解只执行至下一次允许更新；新信息到达后重新优化。

$$G_t + D_t - C_t + Q_t^s \geq L_t^s - V_t^s; \quad Q_t^s \geq 0, \quad \forall s, t \in H_\tau \quad (6)$$

$$G_t^s = G_t, \quad C_t^s = C_t, \quad D_t^s = D_t, \quad E_t^s = E_t, \quad \forall s \quad (7)$$

$$\min \frac{1}{N_\tau} \sum_s \sum_t [p_t^s G_t + 5 p_t^s Q_t^s] + \lambda |E_{\text{末}} - 6000| + \varepsilon \sum_t (C_t + D_t) \quad (8)$$

式（8）用于未承诺的日前采购；已承诺的当前日区间改用第 5 节调整费用。 λ 为场景平均价格中位数乘惩罚倍数； $\varepsilon=10^{-7}$ 元/kWh 用于削弱无意义循环和并列解差异。它是明确的数值正则，实际费用表不计该项；不能据微小目标差宣称经济优势。真实评价终点强制 $E_{\text{末}}=6000$ ，中途分段保存状态不触发终端重置。

紧急成本在优化中与计划费同时权衡。无储能单期下，若净需求分布连续且价格固定正值，最小化 $pq+5pE[(N-q)^+]$ 给出 $F_N(q)=0.8$ 。该推导解释损失不对称，不能把 80% 分位数机械替代含储能和跨时段耦合的式（8）。本文使用完整 LP 进行正式调度。更具体地，单期目标对采购 q 的导数为 $p-5p \cdot \Pr(N>q)$ ，令其为零得到 0.8 分位点；离散经验分布取对应分位区间即可。若价格和缺口相关，期望应对联合分布计算，不能先把价格与需求独立平均。

4.3 训练与基线

一月 8—31 日用于比较两个场景候选和惩罚倍数 0.5、1、1.5，四类模型合计实际费用最小者被锁定；验证子实验使用相同 6000 首尾储电量。相同目标并列时按预先列出的顺序取首项。正式期不重新选候选或惩罚倍数。选择结果为逐日前推因果残差场景、倍数 1.0。

表 3 一月验证期候选策略费用

场景候选	惩罚倍数	一月验证合计费用/元
因果残差	0.5	6,296,205.83
因果残差	1.0	6,291,638.88
因果残差	1.5	6,291,638.88
原始历史	0.5	6,578,596.29
原始历史	1.0	6,575,237.21
原始历史	1.5	6,575,237.21

正式比较包括均值预测确定性采购和同场景无储能控制。前者将场景折叠为一个均值场景，后者固定充放电为零；采用共同日期、起始状态和最终状态。完全信息下界一次性使用正式期实际曲线求解，仅为事后最优信息对照。在线策略与其差额包含预测不足、有限场景、窗口和算法选择，不能全部解释为纯信息价值。

表 4 日前采购与可行基线

模式	方案	正式期总费用/元	紧急购电量/kWh
Q2	联合场景风险采购	14,794,949.72	304,382.5519
Q2	均值预测	16,457,210.18	1,084,623.2193
Q2	无储能	18,581,927.00	243,596.7610
Q2	完全信息事后下界	12,230,383.65	0.0000
Q4-2	联合场景风险采购	15,623,134.83	301,847.5698
Q4-2	均值预测	17,291,874.90	1,084,623.2193
Q4-2	无储能	19,297,857.40	244,173.5421
Q4-2	完全信息事后下界	12,783,021.74	0.0000

5 滚动调整与承诺链结算

5.1 共同日前合同的受控比较

先用只在 0 点求解的参考策略在线形成每一天的 B 及其参考状态。该参考过程本身只使用当日允许信息，可与实际调度并行运行。S0、S1、S2、S3 分别允许在 {0}、{0,6}、{0,6,12}、{0,6,12,18} 求解，但四组读取完全相同的 B。其他策略 0 点保持 B 锁定，只安排自身当前储电量下的储能；随后在允许时刻调整尚未交付的承诺。实际状态因过去更新而产生的差异属于处理结果，不另注入免费库存。

每次求解使用未来 24 小时；当前日剩余区间形成正式承诺，次日部分仅用于储能前瞻，不进入当前日调整账本。年末窗口按评价终点截断。已经执行的时段不得重写。S3 作为题目完整四时点机制的正式结果表口径；费用最低的受控方案另如实报告，不将“机制最完整”当作最优证明。

5.2 主结算与逐次结算

令 $x=A-B$ ，下调取消部分退原价后保留 50% 违约费，上调新增量按 1.5 倍价计。主费用与逐次费用分别为：

$$f_t(x) = p_t x + 0.5 p_t |x| \quad (9)$$

$$J_{\text{净}} = \sum_t [p_t B_t + f_t(A_t^{\text{净}} - B_t) + 5 p_t Q_t] \quad (10)$$

$$J_{\text{逐次}} = \sum_t [p_t B_t + \sum_k f_t(A_t^k - A_t^{k-1}) + 5 p_t Q_t] \quad (11)$$

两式严格使用同一正式承诺链和同一执行轨迹。每条承诺记录发布时间、决策时储电量、交付物理键、旧值、新值和可用历史截止。以价格 1 元、计划 100 kWh 说明：100→80 的主费用为 90 元，100→120 为 130 元；100→80→100 的净变化为 0，逐次费用比净额多 20 元。

在某交付时段各次调整均采用相同非负价格的本文口径下，由三角不等式：

$$J_{\text{逐次}} - J_{\text{净}} = 0.5 \sum_t p_t [\sum_k |A_t^k| - |\sum_k A_t^k|] \geq 0 \quad (12)$$

仅一次有效调整时可相等，因此审计不要求两种费用必须不同。若采用每次下单瞬时价格而非交付区间价，应重新定义合同价格和决策目标，上式不能直接套用。本文将上述差异作为计费假设的适用边界。

表 5 同一承诺链的不同结算解释

模式	最终净额/元	同链逐次费用/元	不退原价费用/元
Q2	14,794,949.72	14,794,949.72	14,794,949.72
Q3	15,632,485.48	15,728,646.00	16,097,957.53
Q4-2	15,623,134.83	15,623,134.83	15,623,134.83
Q4-3	16,455,780.60	16,554,607.22	16,936,032.61

不退还取消部分原价的对照在相同路径上额外加 $\Sigma p(B-A \text{ 末})^*$ ；它反映计费解释变化，未重新优化策略，故不能称为另一制度下的最优策略收益。

5.3 信息价值与实际收益的区别

如果扩大的信息集允许完全忽略新增预报，且优化覆盖相同状态合同下的全部可行策略，则较丰富信息的最优期望费用不会更高：旧策略仍属于新策略集合。这是可行策略集合包含关系的结果，而不是某次实现路径上的费用保证。本文采用有限窗口、有限经验场景和数值求解，因此既不保证每次更新降低实际费用，也不以 S2 略低于 S3 否定信息的一般价值。是否引入更新应依据受控费用、缺口和实现负担共同判断。

6 连续启动与因果检验

缺少首日前史时，本文不从首日完整实际曲线构造 0 点预测。一月采用共同设备启动策略：储能保持待机，储电量按零充放电逐段递推；首日计划采购为零，实际负载减光伏的正缺口由紧急合同供给，负缺口弃置。随后各日按截至前一日已完成的经验净负载与价格进行单期采购。价格变化时以历史价格为权的经验 80%分位数求解采购量，一月仍不操作储能。所有计划、实际紧急量和费用逐段记录。

该启动合同提供缺少首日前史时的可实施基线，其费用单独列示，未进行一月策略最优性论证。由于每段 $C=D=0$ ，一月末 6000 kWh 由题给一月初值连续递推得到。训练候选回放属于独立的离线参数选择实验，与用于衔接正式期初始状态的启动策略分别保存。

表 6 一月启动与二月状态衔接

模式	一月启动费用/元	一月末储电量/kWh	二月初储电量/kWh
Q2	2,637,166.81	6,000.0000	6,000.0000
Q4-2	2,939,367.99	6,000.0000	6,000.0000
Q3	2,637,166.81	6,000.0000	6,000.0000
Q4-3	2,939,367.99	6,000.0000	6,000.0000

未来信息扰动检验保持决策时已知数据不变，只改其后的实际负载、光伏和实时价，要求场景及当前购电/充放电决策不变；同时验证更晚发布的预报不会影响更早决策。对于前一日同一发布时间的 24 小时残差，其完整观测恰截至本日同一时刻，可在立即观测假设下使用；不是仅比较日期大小就认定因果性。

分段复验比较连续运行与在中途保存储电量后继续运行，不能因传入了临时end_date就额外强制储电量恢复。年末终端和训练子实验终端采用显式评价端点，普通中途截断不改变控制规则。

7 结果与稳健性边界

表 7 正式期费用构成

模式	计划费/元	调整净费/元	紧急费/元	总费用/元
Q2	13,628,630.29	0.00	1,166,319.43	14,794,949.72
Q4-2	14,398,125.99	0.00	1,225,008.84	15,623,134.83
Q3	13,787,650.62	-94,795.02	1,939,629.88	15,632,485.48
Q4-3	14,529,578.19	-76,763.21	2,002,965.61	16,455,780.60

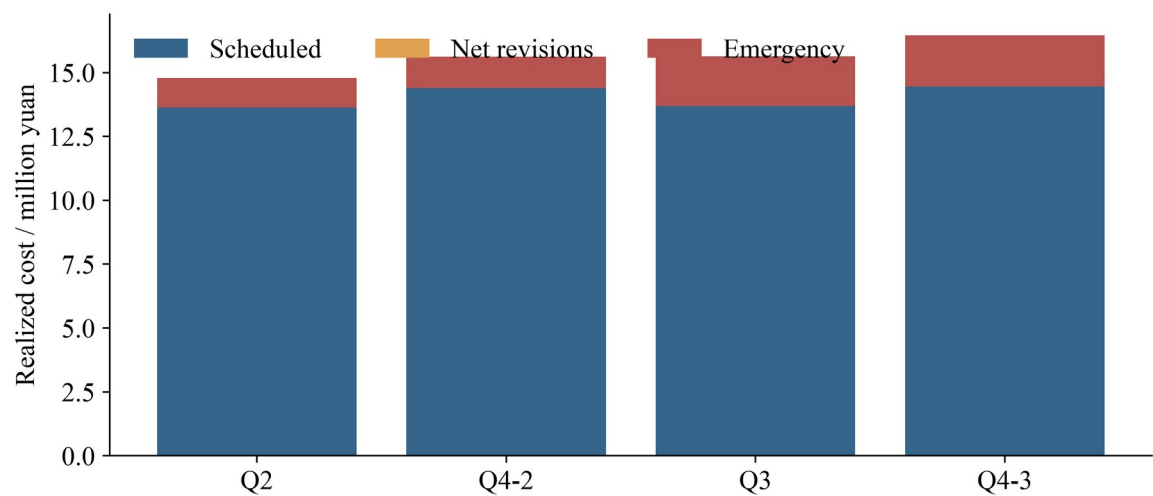


图 1 四类正式结果的计划、调整与紧急费用

表 8 更新频次的费用与紧急购电量

模式	时点策略	主费用/元	逐次费用/元	紧急购电量/kWh
Q3	S0	15,755,992.09	15,755,992.09	480,127.6923
Q3	S1	15,719,285.75	15,719,285.75	458,877.2938
Q3	S2	15,632,278.97	15,728,445.92	461,452.2838
Q3	S3	15,632,485.48	15,728,646.00	461,512.7475
Q4-3	S0	16,611,123.32	16,611,123.32	491,220.5871
Q4-3	S1	16,551,767.48	16,551,767.48	458,919.9717
Q4-3	S2	16,455,768.85	16,554,595.47	461,095.1851
Q4-3	S3	16,455,780.60	16,554,607.22	461,100.4099

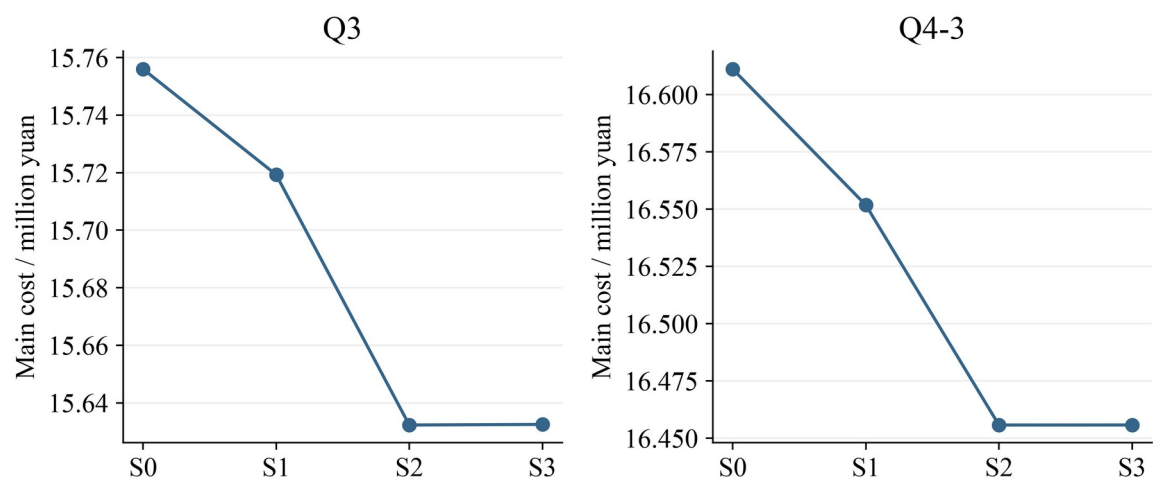


图 2 共同日前合同下的更新频次比较

Q3 在本样本共同日前合同下，主费用最低为 S2，费用 15,632,278.97 元；S3 相对 S0 减少 123,506.61 元。该差额来自本算法在更新信息下选择的后续采购和储能，不能推广成信息本身必然有害或必然有益。

Q4-3 在本样本共同日前合同下，主费用最低为 S2，费用 16,455,768.85 元；S3 相对 S0 减少 155,342.73 元。该差额来自本算法在更新信息下选择的后续采购和储能，不能推广成信息本身必然有害或必然有益。

逐日配对差参考块自助法处理局部时间相关的思路[6]，使用 7 天移动块、2000 次重采样估计区间，重采样对象是历史回放得到的逐日配对费用差。该区间描述这一个个年份样本中的时序变动，并不排除有限样本偏差或分布漂移。

表 9 配对费用差及区间估计

模式	比较	配对差额/元	95%区间下端/元	95%区间上端/元
Q3	S1-S0	-36,706.34	-94,800.29	24,743.50
Q3	S2-S0	-123,713.12	-184,599.51	-64,757.47
Q3	S3-S0	-123,506.61	-183,617.04	-62,973.55
Q4-3	S1-S0	-59,355.84	-120,183.32	9,131.22
Q4-3	S2-S0	-155,354.47	-212,721.58	-91,199.75
Q4-3	S3-S0	-155,342.73	-216,836.95	-92,212.38

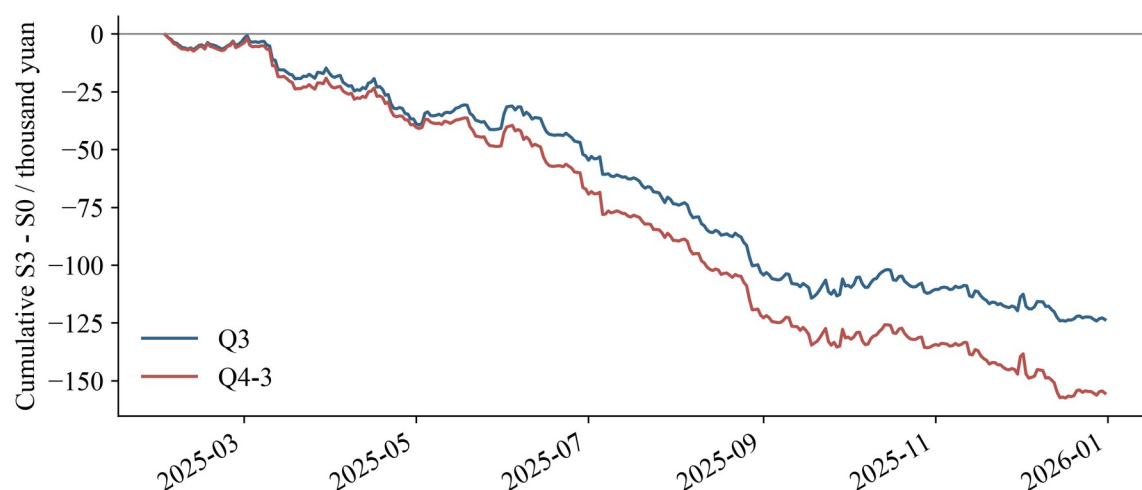


图 3 两种电价下 S3 相对 S0 累计费用差

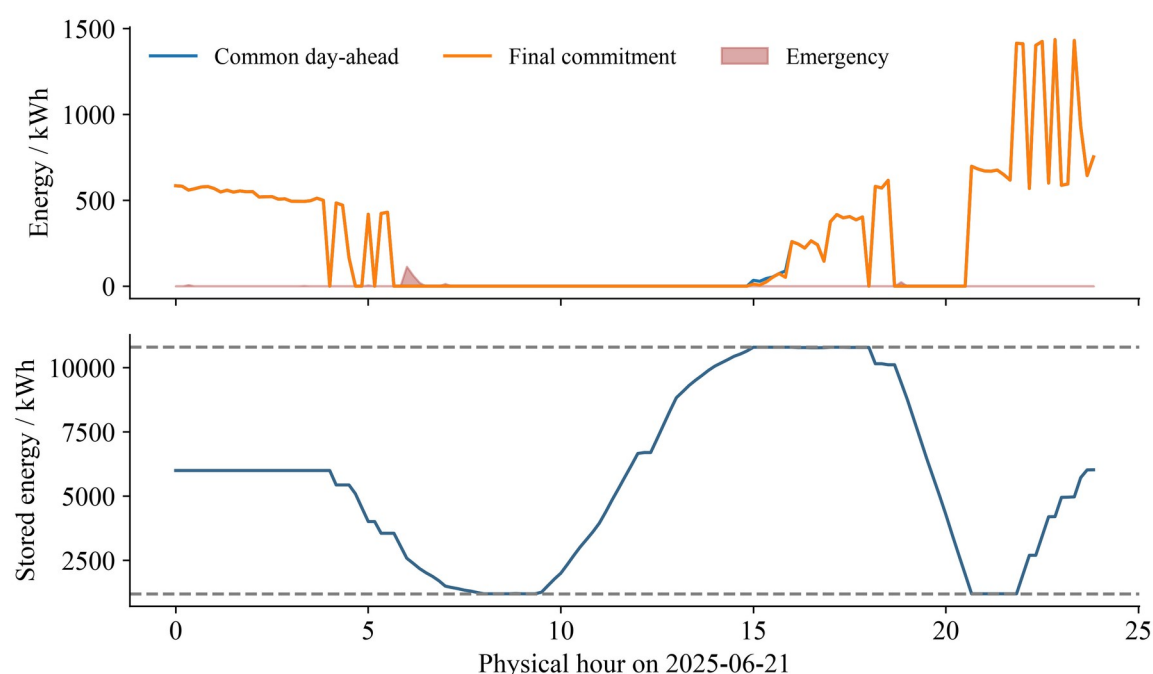


图 4 六月二十一日问题三的调度轨迹与储电量

以四个指定日期、两种电价、每天四次决策构成 32 个固定 LP 实例，分别用 HiGHS 自动、对偶单纯形和内点法复求。HiGHS 求解器及其对偶单纯形实现的来源见文献[7]。相同初始状态、场景、B 和窗口下比较目标及动作，检验可行性和并列解敏感性。该检验考察固定窗口的数值一致性，全年策略费用的比较另见第 7.3 节。

共完成 96 次固定实例复求。相对 HiGHS 自动法的目标最大绝对差为 $7.27596\text{e-}12$ 元，窗口采购量最大 L1 差为 292.46 kWh；最大储电量递推残差为 $1.023\text{e-}12$ kWh。目标接近而动作不一致时应视为并列解敏感性，其全年影响进一步通过第 7.3 节的连续回放检验。

风险评价仍有明确边界：有限历史场景不保证覆盖未来极端情况；负载/光伏/价格联合关系可能随季节改变；全部未来动作共享限制了场景补救自由度，可能增加保守性；有限窗口终端惩罚只是长期价值的近似。本文不引入附件未给出的外网限额或售电收益，也不把紧急购电量当作最终停电量。

7.1 缺口与尾部成本

紧急购电补足实际缺口，以下指标衡量额外采购暴露，不等同于停电概率。每日费用最高的 17 天定义为本样本上尾约 5%，其均值仅作描述性尾部指标；本文没有把目标函数改为 CVaR。

表 10 紧急购电与日费用尾部风险

模式	紧急时段占比	最高日费/元	上尾 17 日均费/元	紧急费占比
Q2	19.28%	85,284.12	71,024.39	7.88%
Q4-2	19.18%	96,029.77	83,271.29	7.84%
Q3	24.01%	78,340.53	71,191.53	12.41%
Q4-3	24.03%	93,916.64	84,263.76	12.17%

7.2 预测误差与月度机制

预测误差根据历史回放各执行时段采用的场景均值计算。问题三包含多个预报发布时间，其误差与问题二的差异同时受信息可用时点和模型影响。平均绝对误差 $MAE = \text{mean}(|\text{预测} - \text{实测}|)$ ，均方根误差 $RMSE = \sqrt{\text{mean}((\text{预测} - \text{实测})^2)}$ ，偏差为 $\text{mean}(\text{预测} - \text{实测})$ ；各指标与被预测变量使用相同单位。夜间光伏及接近零的观测不适合采用平均绝对百分比误差（MAPE）。

表 11 执行时刻预测误差

模式	变量	MAE	RMSE	偏差
Q2	负载/kWh	27.0520	36.3298	-0.4195
Q2	光伏/kWh	30.7103	61.3999	1.3741
Q4-2	负载/kWh	27.0520	36.3298	-0.4195
Q4-2	光伏/kWh	30.7103	61.3999	1.3741
Q4-2	电价/(元/kWh)	0.0423	0.0582	-0.0011
Q3	负载/kWh	27.0520	36.3298	-0.4195
Q3	光伏/kWh	35.0775	59.2895	0.3130
Q4-3	负载/kWh	27.0520	36.3298	-0.4195
Q4-3	光伏/kWh	35.0775	59.2895	0.3130
Q4-3	电价/(元/kWh)	0.0423	0.0582	-0.0011

月度统计见图 5：成本波动同时取决于负荷、电价和充放电轨迹，不能仅由一个误差指标解释。

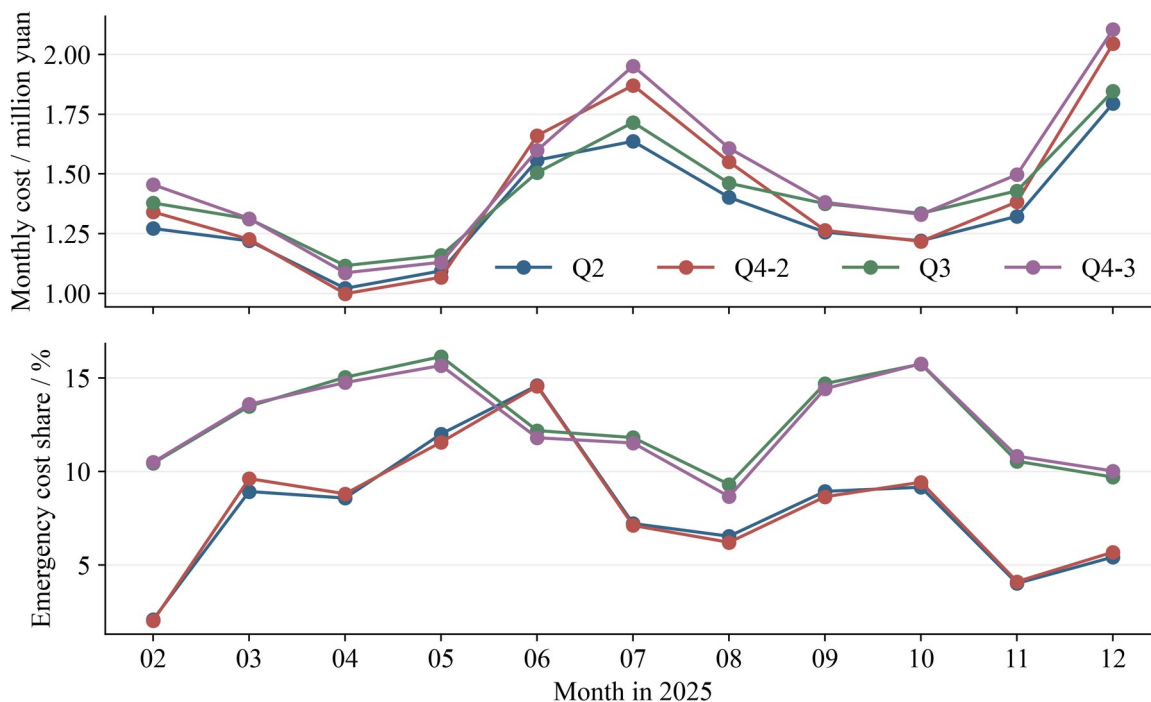


图 5 正式期月度费用与紧急采购占比

Q2 月总费最高为 2025-12，1,795,177.79 元；紧急费用占比最高为 2025-06，14.59%。

Q4-2 月总费最高为 2025-12，2,045,204.52 元；紧急费用占比最高为 2025-06，14.55%。

Q3 月总费最高为 2025-12，1,846,391.96 元；紧急费用占比最高为 2025-05，16.14%。

Q4-3 月总费最高为 2025-12，2,104,869.85 元；紧急费用占比最高为 2025-10，15.76%。

7.3 全年替代求解器回放

在补充回放前固定协议，以相同一月末储电量、同一组 S0 参考日前承诺和一月已选定参数，分别对两种电价的 S2、S3 运行 HiGHS 内点法。每组连续回放 334 天，不在每日重置状态；保留每段动作并从原始附件重算费用。四组结果全部报告，不据本实验更换正式方案。

表 12 全年替代求解器回放费用

模式	求解方法	S2 费用/元	S3 费用/元	S3 减 S2/元
Q3	highs	15,632,278.97	15,632,485.48	206.51
Q3	highs-ipm	15,615,324.64	15,618,759.04	3,434.40
Q4-3	highs	16,455,768.85	16,455,780.60	11.75
Q4-3	highs-ipm	16,456,829.52	16,457,060.93	231.40

Q3 中，默认方法的 S3 减 S2 为 206.51 元，内点法为 3,434.40 元。两种方法的排序方向一致，但差距大小随数值方法变化。

Q4-3 中，默认方法的 S3 减 S2 为 11.75 元，内点法为 231.40 元。两种方法的排序方向一致，但差距大小随数值方法变化。

7.4 时间相关与块长度敏感性

在同一 334 天费用差上，预先列出 3、7、14、28 天四种移动块长度，每种重采样 2000 次，使用固定随机种子 20260913。表中列 S3 减 S2，正数表示 S3 较贵；完整 S3 减 S0 区间也随支撑材料交付。年份存在季节变化，块重采样不保证严格平稳条件，区间用于判断样本内结论对重采样尺度的敏感性，不解释为下一年收益保证。

表 13 分块重采样对块长度的敏感性

模式	块长/天	实际差额/元	95%下端/元	95%上端/元
Q3	3	206.51	-306.79	942.80
Q3	7	206.51	-303.02	921.92
Q3	14	206.51	-241.45	956.14
Q3	28	206.51	-221.26	944.08
Q4-3	3	11.75	-96.55	108.69
Q4-3	7	11.75	-71.06	85.97
Q4-3	14	11.75	-70.47	81.74
Q4-3	28	11.75	-60.92	83.44

四种块长度下，S3 相对 S0 的费用差区间均低于零，S3 相对 S2 的区间均跨零。日内更新具有本样本经济价值，但尚不足以认定增加 18 点更新严格改善经济性。将更新计算或交易操作成本加入后，还需重新计算其盈亏阈值。

8 证据链与结论

证据按“原始附件及哈希—物理时间映射—历史完成时刻和发布预报—当前源码与已解析配置—逐时段执行与承诺链—工作簿及论文数值”保存。独立审计直接从原始 Excel 取实际负载、光伏及价格，重新计算状态和费用；另建问题一 LP，不以模型自带 PASS 或求解器状态代替数学核验。文档构建从持久化正文与补充诊断生成器读取内容，全部表格由对应冻结值生成；生成后再核对指定区间和四小时聚合。

本文独立检查共 237 项，失败 0 项。检查覆盖逐时段原数据匹配、物理区间、设备边界、真实缺口及费用、正式承诺连续性、共同日前合同和一月衔接。测试还覆盖未来观测隔离、分段状态不重置、不同充放电效率和年末边界。该结果是所列检查范围内的证据，不是模型最优性或外推有效性的证明。

问题一可用线性规划实现题设物理边界内的确定性调度。问题二/四-2 的采购必须显式面对五倍缺口损失，点预测均值并非充分的经济决策准则。对问题三/四-3，固定共同日前承诺后才可讨论更新频次；结果的正负均应如实保留。逐次收费必须从真实承诺链计算，其同路径差额受式（12）约束。最后，一月启动策略、正式期初值及终端必须与同一状态合同一致，才能使费用和策略比较具有可解释性。

AI 工具使用声明

本轮使用 OpenAI Codex 辅助审查既有程序、实现修订、组织实验、复算结果和生成文档。保留原始输入，不编造观测数据或计算结果；预测场景与统计重采样均明确标记为数学估计。此前 AI 会话的具体型号和完整操作记录无法由现存项目全部核验，不猜填。参赛队仍需对题意假设、模型取舍、代码、数值和最终提交内容进行人工审阅。详细范围和记录索引见 AI 工具使用详情.pdf。

参考文献

- [1] 全国大学生数学建模竞赛组委会. 2026 年高教社杯全国大学生数学建模竞赛 C 题：微网与外部电网电力调控策略及附件[Z]. 2026.
- [2] BOYD S, VANDENBERGHE L. Convex Optimization[M]. Cambridge: Cambridge University Press, 2004. <https://web.stanford.edu/~boyd/cvxbook/>.
- [3] SHAPIRO A, DENTCHEVA D, RUSZCZYNSKI A. Lectures on Stochastic Programming: Modeling and Theory[M]. 3rd ed. Philadelphia: SIAM, 2021. DOI:10.1137/1.9781611976595.
- [4] RAWLINGS J B, MAYNE D Q, DIEHL M M. Model Predictive Control: Theory, Computation, and Design[M]. 2nd ed. Santa Barbara: Nob Hill Publishing, 2017. <https://sites.engineering.ucsb.edu/~jbrow/mpc/>.
- [5] HYNDMAN R J, ATHANASOPOULOS G. Forecasting: Principles and Practice[M/OL]. 3rd ed. Melbourne: OTexts, 2021[2026-09-12]. <https://otexts.com/fpp3/tscv.html>.
- [6] KUNSCH H R. The jackknife and the bootstrap for general stationary observations[J]. The Annals of Statistics, 1989, 17(3): 1217-1241. DOI:10.1214/aos/1176347265.
- [7] HUANGFU Q, HALL J A J. Parallelizing the dual revised simplex method[J]. Mathematical Programming Computation, 2018, 10(1): 119-142. DOI:10.1007/s12532-017-0130-5.

附录

附录 1 数据与题面指定计算结果

表格按物理区间抽取。问题三及问题四-3 均为 S3 完整机制口径；表中同时列 0 点计划和最终承诺，不将两种量混用。所有电量单位为 kWh，费用单位为元。全精度参与计算，四位小数只在展示时舍入；紧急表按每天连续正缺口区间聚合。原始 Excel、完整冻结明细和中间计算表随配套支撑材料交付，文件索引见附录 2；完整数据不以抽样展示表替代。

Q1 2025-01-01 指定结果

表 14 指定物理区间购电量

物理区间	0 点计划量	最终承诺量
10:00-10:10	0.0000	0.0000
12:00-12:10	480.4125	480.4125
14:00-14:10	0.0000	0.0000
16:00-16:10	445.4317	445.4317
18:00-18:10	531.8940	531.8940
20:00-20:10	0.0000	0.0000

表 15 全天购电及费用汇总

全天计划量	全天最终量	全天总购电费
59,482.6990	59,482.6990	35,126.95

表 16 指定物理区间储能动作

物理区间	充电量	放电量
0:00-4:00	4,500.0000	0.0000
4:00-8:00	833.3333	6,365.8412
8:00-12:00	4,787.9643	1,702.9970
12:00-16:00	5,286.0352	91.1014
16:00-20:00	0.0000	5,780.1319
20:00-24:00	5,333.3333	2,859.8681

表 17 日初与日末储能状态

0 点储电量	24 点储电量
6,000.0000	6,000.0000

Q2 2025-03-20 指定结果

表 18 指定物理区间购电量

物理区间	0 点计划量	最终承诺量
10:00-10:10	0.0000	0.0000
12:00-12:10	662.0628	662.0628
14:00-14:10	0.0000	0.0000
16:00-16:10	521.3045	521.3045
18:00-18:10	701.0510	701.0510
20:00-20:10	0.0000	0.0000

表 19 全天购电及费用汇总

全天计划量	全天最终量	全天总购电费
71,611.5551	71,611.5551	45,330.57

表 20 指定物理区间储能动作

物理区间	充电量	放电量
0:00-4:00	4,500.0000	0.0000
4:00-8:00	833.3333	4,986.8567
8:00-12:00	5,898.7854	3,653.1433
12:00-16:00	5,481.1266	577.7287
16:00-20:00	0.0000	5,666.2340
20:00-24:00	5,333.3333	2,973.7660

表 21 日初与日末储能状态

0 点储电量	24 点储电量
6,000.0000	6,000.0000

表 22 连续紧急购电区间

紧急购电物理区间	紧急购电量
0:20-0:30	19.0054
1:10-1:30	42.3965
2:30-2:40	0.8868
3:00-3:40	106.3099
3:50-4:00	0.7579
4:50-5:20	40.5585
5:50-6:00	9.6804
8:20-8:30	0.0281
15:30-16:50	136.6221
18:40-19:10	79.7381
20:00-20:20	20.2454
21:20-21:30	9.9067
23:30-23:50	43.2042

Q2 2025-06-21 指定结果

表 23 指定物理区间购电量

物理区间	0 点计划量	最终承诺量
10:00-10:10	0.0000	0.0000
12:00-12:10	0.0000	0.0000
14:00-14:10	0.0000	0.0000
16:00-16:10	270.9544	270.9544
18:00-18:10	0.0000	0.0000
20:00-20:10	0.0000	0.0000

表 24 全天购电及费用汇总

全天计划量	全天最终量	全天总购电费
44,317.3693	44,317.3693	25,430.97

表 25 指定物理区间储能动作

物理区间	充电量	放电量
0:00-4:00	925.2727	0.0000
4:00-8:00	833.3333	2,546.3943
8:00-12:00	4,007.4024	32.7666
12:00-16:00	2,796.9521	36.8372
16:00-20:00	0.0000	5,869.8662
20:00-24:00	5,333.3333	2,770.1338

表 26 日初与日末储能状态

0 点储电量	24 点储电量
6,000.0000	6,000.0000

表 27 连续紧急购电区间

紧急购电物理区间	紧急购电量
0:20-0:30	7.1975
3:20-3:30	2.6717
6:00-6:10	27.2964

Q2 2025-09-23 指定结果

表 28 指定物理区间购电量

物理区间	0 点计划量	最终承诺量
10:00-10:10	0.0000	0.0000
12:00-12:10	389.9034	389.9034
14:00-14:10	0.0000	0.0000
16:00-16:10	495.7207	495.7207
18:00-18:10	756.8145	756.8145
20:00-20:10	0.0000	0.0000

表 29 全天购电及费用汇总

全天计划量	全天最终量	全天总购电费
64,683.9027	64,683.9027	54,413.55

表 30 指定物理区间储能动作

物理区间	充电量	放电量
0:00-4:00	4,500.0000	0.0000
4:00-8:00	833.3333	6,370.3856
8:00-12:00	5,660.4829	2,269.6144
12:00-16:00	5,234.9501	185.3007
16:00-20:00	0.0000	5,560.6287
20:00-24:00	5,333.3333	3,079.3713

表 31 日初与日末储能状态

0 点储电量	24 点储电量
6,000.0000	6,000.0000

表 32 连续紧急购电区间

紧急购电物理区间	紧急购电量
0:30-1:20	171.6364
1:30-2:40	179.8680
3:10-3:20	7.2253
3:40-4:00	38.1843
4:20-4:30	6.3537
4:50-5:20	64.3631
5:40-5:50	17.4609
6:00-6:30	39.9485
6:50-7:10	46.4634
9:20-11:10	687.3832
12:10-12:20	34.5086
14:20-15:30	393.8956
15:50-17:50	639.8084
18:00-18:50	216.4812
19:00-19:30	100.8129
19:50-22:00	586.8745
22:10-22:40	92.9401
23:00-23:40	95.4675

Q2 2025-12-21 指定结果

表 33 指定物理区间购电量

物理区间	0 点计划量	最终承诺量
10:00-10:10	0.0000	0.0000
12:00-12:10	1,060.2059	1,060.2059
14:00-14:10	0.0000	0.0000
16:00-16:10	904.1864	904.1864
18:00-18:10	731.0644	731.0644
20:00-20:10	0.0000	0.0000

表 34 全天购电及费用汇总

全天计划量	全天最终量	全天总购电费
103,683.5072	103,683.5072	67,672.11

表 35 指定物理区间储能动作

物理区间	充电量	放电量
0:00-4:00	4,500.0000	0.0000
4:00-8:00	1,666.6667	1,508.3333
8:00-12:00	5,833.3333	7,806.6667
12:00-16:00	8,333.3333	2,835.0000
16:00-20:00	0.0000	5,614.8196
20:00-24:00	5,333.3333	3,025.1804

表 36 日初与日末储能状态

0 点储电量	24 点储电量
6,000.0000	6,000.0000

表 37 连续紧急购电区间

紧急购电物理区间	紧急购电量
3:00-3:10	8.2484
3:30-3:50	25.2708
11:00-11:10	1.2873
17:20-17:30	6.0454
18:40-19:00	6.6983
19:40-20:00	59.0221

Q3 2025-03-20 指定结果

表 38 指定物理区间购电量

物理区间	0 点计划量	最终承诺量
10:00-10:10	0.0000	0.0000
12:00-12:10	850.5195	572.9210
14:00-14:10	0.0000	0.0000
16:00-16:10	691.5551	670.3114
18:00-18:10	701.0163	701.0163
20:00-20:10	0.0000	0.0000

表 39 全天购电及费用汇总

全天计划量	全天最终量	全天总购电费
73,153.5596	68,035.7056	53,295.77

表 40 指定物理区间储能动作

物理区间	充电量	放电量
0:00-4:00	4,486.8644	0.0000
4:00-8:00	833.3333	6,288.2413
8:00-12:00	5,443.1711	2,351.7587
12:00-16:00	6,101.9723	711.5661
16:00-20:00	0.0000	5,666.2340
20:00-24:00	5,333.3333	2,973.7660

表 41 日初与日末储能状态

0 点储电量	24 点储电量
6,011.8220	6,000.0000

表 42 连续紧急购电区间

紧急购电物理区间	紧急购电量
0:20-0:30	12.6675
1:10-1:30	41.4065
2:30-2:40	0.8868
3:00-3:40	106.3099
3:50-4:00	0.7579
4:50-5:20	40.5518
5:50-6:40	283.9429
7:00-7:30	148.6680
8:00-8:40	389.2008
9:00-9:20	231.9733
10:00-10:30	75.2595
11:00-11:50	276.4465
12:30-12:50	185.5360
13:10-14:00	102.8504
14:50-15:00	91.4396
15:40-16:00	185.6210
16:30-17:00	215.6824
17:50-18:00	1.6308
18:40-19:10	79.8939
20:00-20:20	20.2454
21:20-21:30	9.9067
23:30-23:50	43.2042

Q3 2025-06-21 指定结果

表 43 指定物理区间购电量

物理区间	0 点计划量	最终承诺量
10:00-10:10	0.0000	0.0000
12:00-12:10	0.0000	0.0000
14:00-14:10	0.0000	0.0000
16:00-16:10	259.5906	259.5906
18:00-18:10	0.0000	0.0000
20:00-20:10	0.0000	0.0000

表 44 全天购电及费用汇总

全天计划量	全天最终量	全天总购电费
39,027.9639	38,924.9185	23,844.92

表 45 指定物理区间储能动作

物理区间	充电量	放电量
0:00-4:00	0.0000	0.0000
4:00-8:00	0.0000	4,320.0000
8:00-12:00	6,083.2092	9.2682
12:00-16:00	4,594.8997	0.0000
16:00-20:00	27.5462	5,893.4659
20:00-24:00	5,366.1146	2,768.8465

表 46 日初与日末储能状态

0 点储电量	24 点储电量
6,000.0000	6,029.5032

表 47 连续紧急购电区间

紧急购电物理区间	紧急购电量
0:20-0:30	7.1975
3:20-3:30	2.6720
5:00-5:10	5.6007
6:00-6:30	195.8855
7:00-7:10	13.9667
18:50-19:00	24.3090

Q3 2025-09-23 指定结果

表 48 指定物理区间购电量

物理区间	0 点计划量	最终承诺量
10:00-10:10	0.0000	0.0000
12:00-12:10	556.9921	435.2244
14:00-14:10	0.0000	0.0000
16:00-16:10	697.7418	697.7418
18:00-18:10	755.3255	755.3255
20:00-20:10	0.0000	0.0000

表 49 全天购电及费用汇总

全天计划量	全天最终量	全天总购电费
71,018.2496	67,255.4981	56,892.92

表 50 指定物理区间储能动作

物理区间	充电量	放电量
0:00-4:00	4,469.4905	0.0000
4:00-8:00	832.9685	5,845.3760
8:00-12:00	6,055.9581	2,821.5471
12:00-16:00	5,546.0894	757.6585
16:00-20:00	16.6471	5,574.1128
20:00-24:00	5,357.0539	3,079.3713

表 51 日初与日末储能状态

0 点储电量	24 点储电量
6,057.7015	6,021.3485

表 52 连续紧急购电区间

紧急购电物理区间	紧急购电量
0:30-1:20	164.5246
1:30-2:40	179.8680
3:10-3:20	7.2253
3:40-4:00	38.1843
4:20-4:30	6.3539
4:50-5:10	54.8209
5:40-5:50	17.4609
6:00-6:30	133.6420
7:00-7:30	248.2242
8:00-8:30	163.7542
10:00-10:10	34.2439
11:00-11:10	21.9809
11:40-11:50	1.5358
12:50-13:00	12.1870
13:30-14:00	174.9619
14:40-15:00	156.3889
15:40-16:00	219.8262
16:30-17:00	178.4692
17:20-18:50	511.2547
19:00-19:30	100.8152
19:50-22:00	586.8756
22:10-22:40	92.9401
23:00-23:50	99.7459

Q3 2025-12-21 指定结果

表 53 指定物理区间购电量

物理区间	0 点计划量	最终承诺量
10:00-10:10	0.0000	0.0000
12:00-12:10	1,006.9192	1,006.9192
14:00-14:10	0.0000	0.0000
16:00-16:10	965.4095	965.4095
18:00-18:10	731.0633	731.0633
20:00-20:10	0.0000	0.0000

表 54 全天购电及费用汇总

全天计划量	全天最终量	全天总购电费
101,836.4072	99,908.8762	68,839.39

表 55 指定物理区间储能动作

物理区间	充电量	放电量
0:00-4:00	4,475.2658	0.0000
4:00-8:00	1,647.9174	1,600.4611
8:00-12:00	5,833.3333	7,465.6904
12:00-16:00	8,213.1269	2,975.9722
16:00-20:00	0.4947	5,614.8196
20:00-24:00	5,362.9476	3,025.1804

表 56 日初与日末储能状态

0 点储电量	24 点储电量
6,027.0131	6,026.6528

表 57 连续紧急购电区间

紧急购电物理区间	紧急购电量
3:00-3:10	8.2484
3:30-3:50	25.2708
7:00-7:20	85.2606
8:00-8:10	36.6391
9:00-9:20	57.0566
11:00-12:00	687.2493
12:50-13:00	3.0959
13:50-14:00	20.9081
14:50-15:00	11.7177
15:50-16:00	55.8293
16:40-17:00	91.7416
17:20-17:30	6.2231
18:40-19:00	6.6997
19:40-20:00	59.0221

Q4-2 2025-03-20 指定结果

表 58 指定物理区间购电量

物理区间	0 点计划量	最终承诺量
10:00-10:10	0.0000	0.0000
12:00-12:10	662.0628	662.0628
14:00-14:10	0.0000	0.0000
16:00-16:10	521.3045	521.3045
18:00-18:10	701.0510	701.0510
20:00-20:10	0.0000	0.0000

表 59 全天购电及费用汇总

全天计划量	全天最终量	全天总购电费
71,751.2395	71,751.2395	46,667.50

表 60 指定物理区间储能动作

物理区间	充电量	放电量
0:00-4:00	3,333.3333	0.0000
4:00-8:00	2,833.3333	5,283.2858
8:00-12:00	5,898.7854	3,704.8972
12:00-16:00	5,101.5338	597.0756
16:00-20:00	0.0000	5,666.2340
20:00-24:00	5,333.3333	2,973.7660

表 61 日初与日末储能状态

0 点储电量	24 点储电量
6,000.0000	6,000.0000

表 62 连续紧急购电区间

紧急购电物理区间	紧急购电量
0:20-0:30	19.0054
1:10-1:30	42.3965
2:30-2:40	0.8868
3:00-3:40	109.9422
3:50-4:00	0.7579
4:50-5:20	42.8569
5:50-6:00	9.6804
15:30-16:50	136.6221
18:40-19:10	79.7381
20:00-20:20	20.2454
21:20-21:30	9.9067
23:30-23:50	43.2042

Q4-2 2025-06-21 指定结果

表 63 指定物理区间购电量

物理区间	0 点计划量	最终承诺量
10:00-10:10	0.0000	0.0000
12:00-12:10	0.0000	0.0000
14:00-14:10	0.0000	0.0000
16:00-16:10	270.9544	270.9544
18:00-18:10	496.1851	496.1851
20:00-20:10	0.0000	0.0000

表 64 全天购电及费用汇总

全天计划量	全天最终量	全天总购电费
44,903.6591	44,903.6591	22,307.40

表 65 指定物理区间储能动作

物理区间	充电量	放电量
0:00-4:00	0.0000	0.0000
4:00-8:00	2,247.6359	2,559.5579
8:00-12:00	4,303.0422	40.1968
12:00-16:00	2,062.8868	57.2328
16:00-20:00	0.0000	5,869.8662
20:00-24:00	6,263.5118	3,523.5784

表 66 日初与日末储能状态

0 点储电量	24 点储电量
6,000.0000	6,000.0000

表 67 连续紧急购电区间

紧急购电物理区间	紧急购电量
0:20-0:30	7.1975
3:20-3:30	2.6717
6:00-6:10	27.2964

Q4-2 2025-09-23 指定结果

表 68 指定物理区间购电量

物理区间	0 点计划量	最终承诺量
10:00-10:10	0.0000	0.0000
12:00-12:10	389.9034	389.9034
14:00-14:10	0.0000	0.0000
16:00-16:10	495.7207	495.7207
18:00-18:10	0.0000	0.0000
20:00-20:10	0.0000	0.0000

表 69 全天购电及费用汇总

全天计划量	全天最终量	全天总购电费
64,704.0238	64,704.0238	56,947.39

表 70 指定物理区间储能动作

物理区间	充电量	放电量
0:00-4:00	5,333.3333	0.0000
4:00-8:00	0.0000	6,352.9335
8:00-12:00	5,691.2229	2,287.0665
12:00-16:00	5,038.1602	50.8003
16:00-20:00	0.0000	6,312.6995
20:00-24:00	5,333.3333	2,327.3005

表 71 日初与日末储能状态

0 点储电量	24 点储电量
6,000.0000	6,000.0000

表 72 连续紧急购电区间

紧急购电物理区间	紧急购电量
0:30-1:20	171.8013
1:30-2:40	179.8680
3:10-3:20	7.2253
3:40-4:00	38.1843
4:20-4:30	6.3537
4:50-5:20	70.1346
5:40-5:50	17.4609
6:00-6:30	39.9485
6:50-7:10	46.4634
9:20-11:10	658.6726
12:10-12:20	34.5086
14:20-15:30	411.5260
15:50-17:50	639.8084
18:00-18:50	216.4812
19:00-19:30	100.8129
19:50-22:00	586.8746
22:10-22:40	92.9401
23:00-23:40	97.1327

Q4-2 2025-12-21 指定结果

表 73 指定物理区间购电量

物理区间	0 点计划量	最终承诺量
10:00-10:10	0.0000	0.0000
12:00-12:10	1,060.2059	1,060.2059
14:00-14:10	0.0000	0.0000
16:00-16:10	904.1864	904.1864
18:00-18:10	731.0644	731.0644
20:00-20:10	0.0000	0.0000

表 74 全天购电及费用汇总

全天计划量	全天最终量	全天总购电费
103,577.5086	103,577.5086	79,771.49

表 75 指定物理区间储能动作

物理区间	充电量	放电量
0:00-4:00	5,333.3333	0.0000
4:00-8:00	0.0000	833.3333
8:00-12:00	5,833.3333	7,806.6667
12:00-16:00	8,696.6951	3,129.3230
16:00-20:00	0.0000	5,614.8196
20:00-24:00	5,333.3333	3,025.1804

表 76 日初与日末储能状态

0 点储电量	24 点储电量
6,000.0000	6,000.0000

表 77 连续紧急购电区间

紧急购电物理区间	紧急购电量
3:00-3:10	8.2484
3:30-3:50	25.2708
11:00-11:10	1.2873
17:20-17:30	6.0454
18:40-19:00	6.6983
19:40-20:00	59.0221

Q4-3 2025-03-20 指定结果

表 78 指定物理区间购电量

物理区间	0 点计划量	最终承诺量
10:00-10:10	0.0000	0.0000
12:00-12:10	850.5195	570.8778
14:00-14:10	309.1426	0.0000
16:00-16:10	691.5551	687.3114
18:00-18:10	701.0163	701.0163
20:00-20:10	0.0000	0.0000

表 79 全天购电及费用汇总

全天计划量	全天最终量	全天总购电费
72,867.5694	68,055.5090	54,809.53

表 80 指定物理区间储能动作

物理区间	充电量	放电量
0:00-4:00	3,333.3333	0.0000
4:00-8:00	2,000.0000	6,278.9662
8:00-12:00	5,443.1711	2,361.0338
12:00-16:00	6,101.9723	711.5661
16:00-20:00	0.0000	5,666.2340
20:00-24:00	5,328.5294	2,976.6885

表 81 日初与日末储能状态

0 点储电量	24 点储电量
6,000.0000	5,992.4292

表 82 连续紧急购电区间

紧急购电物理区间	紧急购电量
0:20-0:30	19.0054
1:10-1:30	42.3965
2:30-2:40	0.8868
3:00-3:40	109.9422
3:50-4:00	0.7579
4:50-5:20	42.9154
5:50-6:40	278.5855
7:00-7:30	148.6680
8:00-8:40	382.4816
9:00-9:20	231.9733
10:00-10:30	75.2595
11:00-11:50	274.2590
12:30-12:50	181.9861
13:10-14:00	102.8504
14:50-15:00	91.4396
15:30-16:00	167.7821
16:30-17:00	192.2386
17:50-18:00	12.0391
18:40-19:10	79.8939
20:00-20:20	20.2454
21:20-21:30	9.9067
23:30-23:50	43.2042

Q4-3 2025-06-21 指定结果

表 83 指定物理区间购电量

物理区间	0 点计划量	最终承诺量
10:00-10:10	0.0000	0.0000
12:00-12:10	0.0000	0.0000
14:00-14:10	0.0000	0.0000
16:00-16:10	259.5906	259.5906
18:00-18:10	572.3498	572.3498
20:00-20:10	0.0000	0.0000

表 84 全天购电及费用汇总

全天计划量	全天最终量	全天总购电费
39,488.2755	39,426.6223	20,827.64

表 85 指定物理区间储能动作

物理区间	充电量	放电量
0:00-4:00	0.0000	3,540.3852
4:00-8:00	1,669.7950	2,099.7834
8:00-12:00	6,083.2092	9.2682
12:00-16:00	4,594.8997	0.0000
16:00-20:00	27.5462	5,892.1757
20:00-24:00	6,263.5118	3,521.6217

表 86 日初与日末储能状态

0 点储电量	24 点储电量
5,964.0385	6,002.1773

表 87 连续紧急购电区间

紧急购电物理区间	紧急购电量
0:20-0:30	7.1975
3:20-3:30	2.6720
5:00-5:10	5.6007
6:00-6:30	195.8855
7:00-7:10	13.9667
18:50-19:00	24.3090

Q4-3 2025-09-23 指定结果

表 88 指定物理区间购电量

物理区间	0 点计划量	最终承诺量
10:00-10:10	66.3309	0.0000
12:00-12:10	556.9921	435.2244
14:00-14:10	211.4517	157.5829
16:00-16:10	697.7418	697.7418
18:00-18:10	0.0000	0.0000
20:00-20:10	0.0000	0.0000

表 89 全天购电及费用汇总

全天计划量	全天最终量	全天总购电费
70,773.8684	67,327.8786	59,667.96

表 90 指定物理区间储能动作

物理区间	充电量	放电量
0:00-4:00	5,333.3333	0.0000
4:00-8:00	0.0000	5,987.9969
8:00-12:00	6,042.8171	2,652.0031
12:00-16:00	5,200.6404	467.2005
16:00-20:00	14.2894	6,324.2739
20:00-24:00	5,333.3333	2,327.3005

表 91 日初与日末储能状态

0 点储电量	24 点储电量
6,000.0000	6,000.0000

表 92 连续紧急购电区间

紧急购电物理区间	紧急购电量
0:30-1:20	171.8013
1:30-2:40	179.8680
3:10-3:20	7.2253
3:40-4:00	38.1843
4:20-4:30	6.3539
4:50-5:20	70.1352
5:40-5:50	17.4609
6:00-6:30	133.6420
7:00-7:30	236.0701
8:00-8:30	163.7542
10:00-10:10	34.2439
11:00-11:20	26.2146
11:40-11:50	2.4173
13:30-14:00	174.1958
14:40-15:00	192.0331
15:40-16:00	193.0308
16:30-17:00	187.2556
17:20-18:50	511.2547
19:00-19:30	100.8152
19:50-22:00	586.8756
22:10-22:40	92.9401
23:00-23:40	97.1327

Q4-3 2025-12-21 指定结果

表 93 指定物理区间购电量

物理区间	0 点计划量	最终承诺量
10:00-10:10	0.0000	0.0000
12:00-12:10	1,006.9192	1,006.9192
14:00-14:10	0.0000	0.0000
16:00-16:10	965.4095	965.4095
18:00-18:10	731.0633	731.0633
20:00-20:10	0.0000	0.0000

表 94 全天购电及费用汇总

全天计划量	全天最终量	全天总购电费
101,586.3615	99,749.3345	81,948.38

表 95 指定物理区间储能动作

物理区间	充电量	放电量
0:00-4:00	5,314.1330	0.7858
4:00-8:00	0.0000	938.9771
8:00-12:00	5,833.3333	7,462.1538
12:00-16:00	8,211.2025	2,975.3438
16:00-20:00	0.4947	5,614.8196
20:00-24:00	5,343.6832	3,025.1804

表 96 日初与日末储能状态

0 点储电量	24 点储电量
6,018.1534	6,009.3148

表 97 连续紧急购电区间

紧急购电物理区间	紧急购电量
3:00-3:10	8.2484
3:30-3:50	25.2708
7:00-7:20	85.2606
8:00-8:10	36.6391
9:00-9:20	57.0566
11:00-12:00	767.0957
12:50-13:00	3.0959
13:50-14:00	20.9081
14:50-15:00	11.7177
15:50-16:00	55.8293
16:40-17:00	91.7416
17:20-17:30	6.2231
18:40-19:00	6.6997
19:40-20:00	59.0221

附录 2 运行环境与完整源代码

运行入口为支撑材料/run_pipeline.sh。原始附件路径在 config/final.yaml 中配置，也可用 CUMCM_SOURCE_ROOT 覆盖。源数据校验、模型训练、正式期运行、独立数值审核、工作簿、图、论文及最终清单按依赖顺序生成，任一必要检查失败即停止。已交付包同时包含压缩冻结数据、当前源码及参数来源。年度费用与训练费用分开，三种结算口径分栏。已记录环境为 Python 3.12.14、NumPy 2.3.5、SciPy 1.16.1、pandas 2.2.3 及 openpyxl 3.1.5，完整版本见 evidence/environment_versions.json。配套包中“原始附件”保存输入 Excel；“支撑材料/frozen/columns.tar.xz”保存压缩列式明细，使用 scripts/unpack_frozen.py 恢复 CSV。处理后的逐时段输入、决策、状态及费用与中间统计分别保存在 frozen、evidence 及 research 目录。原文件与完整精度值均保留。

表 98 数据与复算证据索引

证据	位置
正式明细	frozen/q*_dispatch.csv（解包后）
真实承诺链	frozen/decision_versions.csv（解包后）
训练与解析配置	evidence/january_parameter_validation.csv 及 frozen/resolved_config.json
独立复核	audit/independent_validation.json
源码与输入校验	evidence/source_provenance.json
可执行入口	run_pipeline.sh
数学约束测试	tests/test_closed_loop.py

支撑材料的必要文件按功能列示如下；每个源码文件在本附录逐一给出，完整文件哈希清单见包内 SHA256SUMS.json。

表 99 支撑材料目录与用途

目录或文件	内容
src/model	全部模型与数据加载程序
src/audit	独立逐时段及工作簿审核程序
scripts	运行、补充实验、图表、文稿、打包程序
tests	数学约束和未来观测扰动测试
config/final.yaml 与 requirements.txt	配置及依赖
frozen 与 workbooks	原正式结果及五份指定工作簿
research	补充全年回放、尾部、月度、误差与统计诊断
evidence 与 audit	来源、源码附录清单及复验记录
charts 与 logs	图像及运行日志
AI 工具使用详情.pdf	真实使用范围及仍需人工核验事项

2.1 完整源程序与运行配置

以下逐文件收录当前工程自编的全部 Python、JavaScript、Shell 程序和配置。第三方依赖按 requirements.txt 安装；工作簿导出还需 Node 与 @oai/artifact-tool，文档转换需配套 LibreOffice 运行环境。程序以支撑材料内的原始文件运行，版面换行不改变源文件。独立数值复验仅依赖 Python，不要求文档转换环境。

config/final.yaml

SHA256 49e195de542a215f9483bbae47b362703bd7eeb51177439936b3cecd42c06570

```
{
  "schema_version": "3.1-closed-loop",
  "source_root": "input",
  "output_root": "outputs/c_final_v1",
  "study": {
    "year": 2025,
    "training_start": "2025-01-01",
    "training_end": "2025-01-31",
    "official_start": "2025-02-01",
    "official_end": "2025-12-31",
    "initial_soc_kwh": 6000.0,
    "carry_soc_from_january": true,
    "terminal_soc_kwh": 6000.0,
    "startup_policy": "January_idle_battery_empirical_procurement"
  },
  "storage": {
    "capacity_kwh": 12000.0,
    "soc_min_kwh": 1200.0,
    "soc_max_kwh": 10800.0,
    "power_kw": 5000.0,
    "charge_efficiency": 0.9,
    "discharge_efficiency": 0.9,
    "interval_minutes": 10
  },
  "forecast": {
    "scenario_count": 20,
    "scenario_history": "latest_completed_24h_trajectories",
    "pv_hour_mapping": "right_endpoint_zero_order_hold",
    "scenario_model_candidates": [
      "raw_history",
      "causal_residual"
    ],
    "scenario_model": "raw_history"
  },
  "rolling": {
    "horizon_hours": 24,
    "settlement": "final_relative_to_common_00_plan",
    "sequential_sensitivity": "same_formal_commitment_chain",
    "soft_penalty_multipliers": [
      0.5,
```

```

    1.0,
    1.5
  ],
  "selected_soft_penalty_multiplier": 0.5,
  "selection_method": "minimum combined January 8-31 holdout cost across four modes;
equal 6000 endpoints",
  "terminal_rule": "soft_absolute_deviation; forced only at explicit evaluation endpoint"
},
"cost": {
  "down_adjustment_multiplier": 0.5,
  "up_adjustment_multiplier": 1.5,
  "emergency_multiplier": 5.0,
  "throughput_tiebreak_yuan_per_kwh": 1e-07
},
"audit": {
  "numeric_tolerance": 1e-07,
  "feasibility_tolerance": 1e-08,
  "display_decimals": 4,
  "emergency_display_omit_threshold_kwh": 5e-05
},
"experiments": {
  "official_arm": "joint_empirical",
  "official_scenario": "S3",
  "schedules": {
    "S0": [
      0
    ],
    "S1": [
      0,
      6
    ],
    "S2": [
      0,
      6,
      12
    ],
    "S3": [
      0,
      6,
      12,
      18
    ]
  }
},
"controls": [
  "mean_forecast",
  "no_storage"
],
"solver_methods": [
  "highs",
  "highs-ds",

```

```

    "highs-ipm"
  ],
  "representative_dates": [
    "2025-03-20",
    "2025-06-21",
    "2025-09-23",
    "2025-12-21"
  ]
}
}

```

requirements.txt

```

SHA256 6566f256706346dc54cc9e02a45794ab938bc296a5878ca670f886052e1702a3
numpy==2.3.5
scipy==1.16.1
pandas==2.2.3
openpyxl==3.1.5
python-docx==1.2.0
pypdf==6.10.0
Pillow==12.3.0
matplotlib==3.11.2

```

run_pipeline.sh

```

SHA256 af6e9596d23433b106aeb6a2856e0bed0e0a027152d85c84d0f5acb544eadd90
#!/usr/bin/env bash
set -euo pipefail
PROJECT_DIR="$(cd "$(dirname "$0")" && pwd)"
cd "$PROJECT_DIR"
RUNTIME_BASE="${HOME}/.cache/codex-runtimes/codex-primary-runtime/dependencies"
PY_BIN="${CODEX_PYTHON_BIN:-${RUNTIME_BASE}/python/bin/python3}"
NODE_BIN="${CODEX_NODE_BIN:-${RUNTIME_BASE}/node/bin/node}"
export PYTHONPATH="$PROJECT_DIR/work/python_deps:$PROJECT_DIR${PYTHONPATH:+:$PYTHONPATH}"
FINAL="outputs/c_final_v1"
mkdir -p "$FINAL/logs" "$FINAL/qa" work/fontconfig
if [[ ! -e node_modules ]]; then ln -s "$RUNTIME_BASE/node/node_modules"
node_modules; fi
if [[ "${1:-}" != "--skip-model" ]]; then
  "$PY_BIN" -m unittest discover -s tests -v 2>&1 | tee
"$FINAL/logs/constraint_and_causality_tests.log"
  "$PY_BIN" -u scripts/run_model.py --scope all 2>&1 | tee
"$FINAL/logs/repaired_model.log"
fi
"$PY_BIN" scripts/analyze_results.py 2>&1 | tee "$FINAL/logs/analysis.log"
"$PY_BIN" -u scripts/strengthen_evidence.py 2>&1 | tee
"$FINAL/logs/strengthen_evidence.log"
"$PY_BIN" src/audit/audit_final.py 2>&1 | tee "$FINAL/logs/independent_model_audit.log"
"$NODE_BIN" scripts/build_workbooks.mjs 2>&1 | tee "$FINAL/logs/workbook_build.log"

```

```

"$PY_BIN" src/audit/audit_final.py --workbooks 2>&1 | tee
"$FINAL/logs/independent_workbook_audit.log"
"$PY_BIN" scripts/make_charts.py 2>&1 | tee "$FINAL/logs/charts.log"
"$PY_BIN" scripts/build_documents.py 2>&1 | tee "$FINAL/logs/documents.log"
if [[ -d /Library/Fonts ]]; then
    cat > work/fontconfig/fonts.conf <<'XML'
    <?xml version="1.0"?><!DOCTYPE fontconfig SYSTEM
    "fonts.dtd"><fontconfig><dir>/Library/Fonts</dir><dir>/System/Library/Fonts</
    dir><cachedir>/tmp/v3-font-cache</cachedir></fontconfig>
XML
    export FONTCONFIG_FILE="$PROJECT_DIR/work/fontconfig/fonts.conf"
fi
DOC_SKILL="${HOME}/.codex/plugins/cache/openai-primary-runtime/documents/
26.905.11957/skills/documents"
"$PY_BIN" "$DOC_SKILL/render_docx.py" "$FINAL/paper/C 题论文.docx" --output_dir
"$FINAL/qa/paper_render" --emit_pdf 2>&1 | tee "$FINAL/logs/paper_render.log"
"$PY_BIN" "$DOC_SKILL/render_docx.py" "$FINAL/paper/AI 工具使用详情.docx" --
output_dir "$FINAL/qa/ai_render" --emit_pdf 2>&1 | tee "$FINAL/logs/ai_render.log"
cp "$FINAL/qa/paper_render/C 题论文.pdf" "$FINAL/paper/C 题论文.pdf"
cp "$FINAL/qa/ai_render/AI 工具使用详情.pdf" "$FINAL/paper/AI 工具使用详情.pdf"
"$PY_BIN" scripts/final_quality.py 2>&1 | tee "$FINAL/logs/final_quality.log"
"$PY_BIN" scripts/write_improvement_report.py
"$PY_BIN" scripts/package_final.py 2>&1 | tee work/package.log
echo "Current-run artifacts are in $FINAL; synchronize only after visual review."

```

scripts/analyze_results.py

```

SHA256 40e3144113b35b4c7ad68969b78af1beac6d619ec616af748b7e5b15feaeefc1
from pathlib import Path
import sys,json,copy
from datetime import date
import pandas as pd,numpy as np
ROOT=Path(__file__).resolve().parents[1];sys.path.insert(0,str(ROOT))
from src.model.config import load_config
from src.model.data import load_data
from src.model.risk import cases,solve
from src.model.experiments import write_csv

def main():

cfg=load_config('config/final.yaml');F=Path(cfg['_output_root'])/'frozen';E=F.parent/'evidenc
e';cfg.update(json.loads((F/'resolved_config.json').read_text()));data=load_data(cfg);out=[];rn
g=np.random.default_rng(20260912)
    for mode in ['q3','q4_3']:
        base=pd.read_csv(F/f'{mode}_S0_daily.csv')
        for sc in ['S1','S2','S3']:
            d=pd.read_csv(F/f'{mode}_{sc}
            _daily.csv');delta=d.final_relative_cost_yuan.to_numpy()-
            base.final_relative_cost_yuan.to_numpy();samples=[]

```

```

    for _ in range(2000):

starts=rng.integers(0,len(delta)-6,size=(len(delta)+6)//7);indices=np.concatenate([np.arange(x,
x+7) for x in starts][:len(delta)];samples.append(float(delta[indices].sum()))

out.append(dict(mode=mode,comparison=sc+'_minus_S0',difference_yuan=float(delta.sum()),
,paired_days=len(delta),block_days=7,bootstrap_replicates=2000,ci_low=float(np.quantile(sa
mples,.025)),ci_high=float(np.quantile(samples,.975))))
    write_csv(F/'comparison_stats.csv',out)
    sensitivity=[];solver=[]
    for mode in ['q2','q3','q4_2','q4_3']:
        d=pd.read_csv(F/f'{mode}
_dispatch.csv');p=d.settlement_price.to_numpy();down=np.maximum(d.plan_00_grid_kwh-
d.final_adjusted_grid_kwh,0)

sensitivity.append(dict(mode=mode,main_cost=float(d.final_relative_cost_yuan.sum()),same
_chain_sequential_cost=float(d.sequential_cost_yuan.sum()),no_refund_same_path_cost=floa
t(d.final_relative_cost_yuan.sum()+np.dot(p,down)),note='same actions; no reoptimization;
delivery-time prices'))
        if mode in ['q2','q4_2']:continue
        for daystr in cfg['experiments']['representative_dates']:
            day=date.fromisoformat(daystr);dd=d[d.date==daystr]
            for issue in [0,6,12,18]:
                pos=issue*6;length=min(144,len(data.dates)*144-
(data.date_index[day]*144+pos));baseline=np.full(length,np.nan);baseline[:144-
pos]=dd.plan_00_grid_kwh.to_numpy()[pos:];lock=np.full(length,np.nan)
                if issue==0:lock[:144]=baseline[:144]
                cs=cases(data,cfg,day,issue,length,mode);initial=float(dd.soc_start_kwh.iloc[pos]);
sols=[]
                for method in ['highs','highs-ds','highs-ipm']:

r=solve(cs,initial,baseline,cfg,lock=lock,method=method);sols.append(r);solver.append(dict(
mode=mode,date=daystr,issue=issue,method=method,objective=r.objective,objective_differe
nce_vs_highs=r.objective-sols[0].objective,grid_l1_difference=float(abs(r.grid-
sols[0].grid).sum()),soc_max_difference=float(abs(r.soc_end-
sols[0].soc_end).max()),balance_residual=r.balance_residual,soc_residual=r.soc_residual))

write_csv(F/'settlement_sensitivity.csv',sensitivity);write_csv(F/'solver_instance_sensitivity.c
sv',solver)
    # Explicit diagnosis remains a replay, never a trained test-period parameter.
    reserve=[]
    for mode in ['q2','q4_2']:
        d=pd.read_csv(F/f'{mode}_dispatch.csv')
        for delta in [0,10,20,50]:
            q=np.maximum(d.actual_load_kwh+d.charge_kwh-d.actual_pv_kwh-
d.discharge_kwh-d.plan_00_grid_kwh-
delta,0);cost=float(np.sum(d.settlement_price*(d.plan_00_grid_kwh+delta)
+5*d.settlement_price*q));reserve.append(dict(mode=mode,delta_kwh=delta,total_cost=cost,
difference_vs_current=cost-
d.final_relative_cost_yuan.sum(),role='ex_post_diagnostic_not_parameter_selection'))

```

```

write_csv(F/'reserve_diagnostic.csv',reserve)
# Evidence source hashes include actual input bytes and active source code.
sources=[]
for p in (Path(cfg['_source_root'])/'C 题').rglob('*'):
    if
p.is_file():sources.append(dict(kind='original_input',path=str(p.relative_to(Path(cfg['_source_
root']))))),sha256=__import__('hashlib').sha256(p.read_bytes()).hexdigest())
    for folder in ['src','scripts','tests','paper','config']:
        for p in (ROOT/folder).rglob('*'):
            if p.is_file() and '__pycache__' not in
p.parts:sources.append(dict(kind='active_source',path=str(p.relative_to(ROOT)),sha256=__im
port__('hashlib').sha256(p.read_bytes()).hexdigest()))

(E/'source_provenance.json').write_text(json.dumps(sources,ensure_ascii=False,indent=2));pr
int('Analysis and exact-instance solver diagnostics complete',flush=True)

```

```

if __name__=='__main__':main()

```

scripts/build_documents.py

```

SHA256 cedb94b106d5a9eacd55d7753cb41a32b2222d08e9ff2284dc2efff480e70be9
#!/usr/bin/env python3
"""Inject frozen results into the paper and create the final DOCX sources."""
from __future__ import annotations

import csv
import json
import re
import shutil
from pathlib import Path

import numpy as np
import pandas as pd
from docx import Document
from docx.enum.section import WD_SECTION
from docx.enum.table import WD_ALIGN_VERTICAL, WD_TABLE_ALIGNMENT
from docx.enum.text import WD_ALIGN_PARAGRAPH, WD_BREAK,
WD_LINE_SPACING
from docx.oxml import OxmlElement
from docx.oxml.ns import qn
from docx.shared import Cm, Pt, RGBColor

ROOT = Path(__file__).resolve().parents[1]
FINAL = ROOT / "outputs" / "c_final_v1"
FROZEN = FINAL / "frozen"
PAPER = FINAL / "paper"
CHARTS = FINAL / "charts"
WORK = ROOT / "work" / "paper_build"

```

```

# Arial Unicode MS is available as a TrueType font to the bundled headless
# converter and contains the complete Simplified Chinese glyph set. macOS TTC
# family names such as Songti SC/Heiti SC are visible to desktop applications
# but are not resolved reliably by this conversion runtime.
BODY_FONT = "Arial Unicode MS"
HEADING_FONT = "Arial Unicode MS"

def clean_inline(value: str) -> str:
    return value.replace('**', '').replace("'", "")

def set_cell_shading(cell, fill: str) -> None:
    tc_pr = cell._tc.get_or_add_tcPr(); shd = tc_pr.find(qn("w:shd"))
    if shd is None: shd = OxmlElement("w:shd"); tc_pr.append(shd)
    shd.set(qn("w:fill"), fill)

def set_cell_borders(cell, color: str = "D9D9D9") -> None:
    tc_pr = cell._tc.get_or_add_tcPr(); borders = tc_pr.first_child_found_in("w:tcBorders")
    if borders is None: borders = OxmlElement("w:tcBorders"); tc_pr.append(borders)
    for edge in ("top", "left", "bottom", "right", "insideH", "insideV"):
        node = borders.find(qn(f"w:{edge}"))
        if node is None: node = OxmlElement(f"w:{edge}"); borders.append(node)
        node.set(qn("w:val"), "single"); node.set(qn("w:sz"), "4"); node.set(qn("w:color"), color)

def repeat_header(row) -> None:
    tr_pr = row._tr.get_or_add_trPr(); tbl_header = OxmlElement("w:tblHeader");
    tbl_header.set(qn("w:val"), "true"); tr_pr.append(tbl_header)

def add_page_number(paragraph) -> None:
    paragraph.alignment = WD_ALIGN_PARAGRAPH.CENTER
    run = paragraph.add_run(); begin = OxmlElement("w:fldChar");
    begin.set(qn("w:fldCharType"), "begin")
    instr = OxmlElement("w:instrText"); instr.set(qn("xml:space"), "preserve"); instr.text = "
PAGE "
    separate = OxmlElement("w:fldChar"); separate.set(qn("w:fldCharType"), "separate")
    end = OxmlElement("w:fldChar"); end.set(qn("w:fldCharType"), "end")
    run._r.extend([begin, instr, separate, end])

def configure(doc: Document) -> None:
    doc.core_properties.author = "
    doc.core_properties.last_modified_by = "
    section = doc.sections[0]; section.page_width = Cm(21); section.page_height = Cm(29.7)

```

```

    section.top_margin = Cm(2.5); section.bottom_margin = Cm(2.5); section.left_margin =
Cm(2.5); section.right_margin = Cm(2.5)
    styles = doc.styles
    normal = styles["Normal"]; normal.font.name = BODY_FONT; normal.font.size = Pt(10.5)
    normal._element.rPr.rFonts.set(qn("w: eastAsia"), BODY_FONT)
    normal.paragraph_format.line_spacing = 1.25; normal.paragraph_format.space_after =
Pt(3)
    for style_name, size, font in (("Title", 18, HEADING_FONT), ("Heading 1", 14,
HEADING_FONT), ("Heading 2", 12, HEADING_FONT), ("Heading 3", 11,
HEADING_FONT)):
        st = styles[style_name]; st.font.name = font; st.font.size = Pt(size); st.font.color.rgb =
RGBColor(0, 0, 0); st.font.bold = True
        st._element.rPr.rFonts.set(qn("w: eastAsia"), font); st.paragraph_format.keep_with_next
= True
        styles["Title"].paragraph_format.space_after = Pt(12)
    for st in styles:
        if st.type == 1:
            pr=st._element.find(qn('w:pPr'))
            if pr is not None:
                for border in list(pr.findall(qn('w:pBdr'))):pr.remove(border)
            table_style=styles.add_style('ResultTableText',1)
            table_style.font.name=BODY_FONT;table_style.font.size=Pt(8.5)
            table_style._element.get_or_add_rPr().rFonts.set(qn('w: eastAsia'),BODY_FONT)
            table_style.paragraph_format.line_spacing=1.0

table_style.paragraph_format.space_before=Pt(0);table_style.paragraph_format.space_after=P
t(0)
    add_page_number(section.footer.paragraphs[0])

def add_table(doc: Document, lines: list[str]) -> None:
    data = [[clean_inline(x.strip()) for x in line.strip().split("|")] for line in lines if not
re.match(r"^\s*|\?\s*?:?-\+", line)]
    if not data: return
    cols = max(map(len, data)); table = doc.add_table(rows=len(data), cols=cols);
table.alignment = WD_TABLE_ALIGNMENT.CENTER
    table.autofit = False
    for i, row in enumerate(data):
        for j in range(cols):
            cell = table.cell(i, j); cell.text = row[j] if j < len(row) else ""; cell.vertical_alignment =
WD_ALIGN_VERTICAL.CENTER; set_cell_borders(cell)
            for p in cell.paragraphs:
                p.style=doc.styles['ResultTableText']
                p.paragraph_format.space_after = Pt(0); p.alignment =
WD_ALIGN_PARAGRAPH.CENTER if j == 0 or i == 0 else
WD_ALIGN_PARAGRAPH.LEFT
                for run in p.runs:
                    run.font.name = BODY_FONT; run.font.size = Pt(8.5);
run._element.rPr.rFonts.set(qn("w: eastAsia"), BODY_FONT)
                    if i == 0: run.font.bold = True; run.font.color.rgb = RGBColor(255,255,255)

```

```

        if i == 0: set_cell_shading(cell, "1F4E78")
        elif i % 2 == 0: set_cell_shading(cell, "F4F7FA")
    for row in table.rows:
        prevent_split = OxmlElement('w:cantSplit')
        row._tr.get_or_add_trPr().append(prevent_split)
        for j, cell in enumerate(row.cells):
            cell.width = Cm(16 / cols)
    if len(data) <= 12:
        for row in table.rows[:-1]:
            for cell in row.cells:
                for p in cell.paragraphs: p.paragraph_format.keep_with_next = True
    repeat_header(table.rows[0])
    spacer=doc.add_paragraph();spacer.paragraph_format.line_spacing=Pt(3)
    spacer.paragraph_format.space_before=Pt(0);spacer.paragraph_format.space_after=Pt(4)

def markdown_to_docx(text: str, output: Path) -> None:
    doc = Document(); configure(doc)
    lines = text.splitlines(); i = 0; in_code = False; figure_no=0; table_no=0; current_heading=""
    while i < len(lines):
        line = lines[i]
        if line.startswith("` ` ") or line.startswith("` `"):
            in_code = not in_code; i += 1; continue
        if in_code:
            p = doc.add_paragraph()
            p.paragraph_format.space_after = Pt(0)
            p.paragraph_format.line_spacing = 1.05
            run = p.add_run(line if line else " ")
            run.font.name = "Consolas"; run.font.size = Pt(7.5)
            run._element.rPr.rFonts.set(qn("w: eastAsia"), BODY_FONT)
            i += 1; continue
        if line.startswith("|") and i + 1 < len(lines) and lines[i + 1].startswith("|"):
            table_no+=1
            caption=re.sub(r'^\d+(?:\.\d+)*\s+', "", current_heading)
            columns=[x.strip() for x in line.strip('|').split('|')]
            names={'场景候选': '一月验证期候选策略费用','方案': '确定性调度与无储能对照','物理区间': '指定物理区间购电量' if '0 点计划量' in columns else '指定物理区间储能动作','0 点 SOC': '日初与日末储能状态','全天计划量': '全天购电及费用汇总','紧急购电物理区间': '连续紧急购电区间'}
            caption=names.get(columns[0],caption)
            if columns[0]=='模式':
                caption={'方案': '日前采购与可行基线','最终净额': '同一承诺链的不同结算解释','一月启动费用': '一月启动与二月状态衔接','计划费': '正式期费用构成','时点策略': '更新频次的费用与紧急购电量','比较': '配对费用差及区间估计','紧急时段占比': '紧急购电与日费用尾部风险','变量': '执行时刻预测误差','求解方法': '全年替代求解器回放费用','块长/天': '分块重采样对块长度的敏感性'}.get(columns[1],caption)
            cap=doc.add_paragraph(f'表 {table_no} {caption}')

```

```

cap.alignment=WD_ALIGN_PARAGRAPH.CENTER;cap.paragraph_format.keep_with_next=True
    for run in cap.runs:run.font.size=Pt(9)
    block = []
    while i < len(lines) and lines[i].startswith("|"):
        block.append(lines[i]); i += 1
    add_table(doc, block); continue
fig = re.fullmatch(r"\\[FIGURE:([^\|+)](\\|+)]\\", line.strip())
if fig:
    figure_no+=1
    p = doc.add_paragraph(); p.alignment = WD_ALIGN_PARAGRAPH.CENTER
    p.paragraph_format.keep_with_next = False
    p.add_run().add_picture(str(CHARTS / fig.group(1)), width=Cm(15.2))
    p.add_run().add_break()
    cap=p;cap.add_run(f"图 {figure_no} {fig.group(2)}"); cap.alignment =
WD_ALIGN_PARAGRAPH.CENTER
    p.paragraph_format.keep_together = True
    for run in cap.runs: run.font.size = Pt(9); run.font.name = BODY_FONT;
run._element.rPr.rFonts.set(qn("w: eastAsia"), BODY_FONT)
    i += 1; continue
if line.startswith("# "):
    p = doc.add_paragraph(clean_inline(line[2:]), style="Title"); p.alignment =
WD_ALIGN_PARAGRAPH.CENTER
elif line.startswith("## "):
    heading = clean_inline(line[3:])
    current_heading=heading
    p=doc.add_paragraph(heading, style="Heading 1")
    if heading.startswith('1 ') or heading=='附录':
        p.paragraph_format.page_break_before=True
elif line.startswith("### "):
    current_heading=clean_inline(line[4:])
    p=doc.add_paragraph(clean_inline(line[4:]), style="Heading 2")
    if re.match(r'#### 附录[B-D]',line):p.paragraph_format.page_break_before=True
elif line.startswith("#### "):
    heading=clean_inline(line[5:])
    current_heading=heading
    p=doc.add_paragraph(heading, style="Heading 3")
elif re.match(r"^\d+\.\s+", line):
    doc.add_paragraph(clean_inline(re.sub(r"^\d+\.\s+", "", line)), style="List Number")
elif line.startswith("- "):
    doc.add_paragraph(clean_inline(line[2:]), style="List Bullet")
elif line.startswith("$ $") and line.endswith("$ $"):
    from native_equations import equation
    p = doc.add_paragraph(); p.alignment = WD_ALIGN_PARAGRAPH.CENTER
    number=int(re.search(r'((\d+))',line).group(1));p._p.append(equation(number))
elif not line.strip():
    pass
else:

```

```

p = doc.add_paragraph(); p.paragraph_format.first_line_indent = Pt(21)
p.paragraph_format.keep_together = False
run = p.add_run(clean_inline(line))
if re.match(r'^\[d+\] ',line):
    p.paragraph_format.first_line_indent=Pt(0)
    p.paragraph_format.line_spacing=1.05
    p.paragraph_format.space_after=Pt(3)
    p.paragraph_format.keep_together=True
    run.font.size=Pt(9)
i += 1
output.parent.mkdir(parents=True, exist_ok=True); doc.save(output)

```

```

def main():
    from paper_content import generate
    text,ai=generate()
    from research_content import enrich
    text,ai=enrich(text,ai)
    PAPER.mkdir(parents=True,exist_ok=True);WORK.mkdir(parents=True,exist_ok=True)
    (FINAL/"evidence/paper_source_frozen.md").write_text(text,encoding='utf-8')
    (FINAL/"evidence/AI 工具使用详情_冻结稿.md").write_text(ai,encoding='utf-8')
    markdown_to_docx(text,PAPER/"C 题论文.docx")
    from research_content import append_sources
    append_sources(PAPER/"C 题论文.docx")
    markdown_to_docx(ai,PAPER/"AI 工具使用详情.docx")
    print('Created current-run DOCX sources',flush=True)

```

```

if __name__=='__main__':main()

```

scripts/build_workbooks.mjs

```

SHA256 c5927746622d280f2e7c89dd7669a9965879636a55f8f3340d23ae1151803d88
import fs from "node:fs/promises";
import path from "node:path";
import { fileURLToPath } from "node:url";
import { FileBlob, SpreadsheetFile, Workbook } from "@oai/artifact-tool";

```

```

const projectRoot = path.resolve(path.dirname(fileURLToPath(import.meta.url)), "..");
const finalRoot = path.join(projectRoot, "outputs", "c_final_v1");
const frozenRoot = path.join(finalRoot, "frozen");
const workbookRoot = path.join(finalRoot, "workbooks");
const qaRoot = path.join(finalRoot, "qa", "workbook_renders");

```

```

function parseCsv(text) {
    const rows = [];
    let row = [], value = "", quoted = false;
    for (let i = 0; i < text.length; i++) {
        const ch = text[i];

```

```

if (quoted) {
  if (ch === '"' && text[i + 1] === '"') { value += '"'; i++; }
  else if (ch === '"') quoted = false;
  else value += ch;
} else if (ch === '"') quoted = true;
else if (ch === ',') { row.push(value); value = ''; }
else if (ch === '\n') { row.push(value.replace(/\r$/, '')); rows.push(row); row = []; value =
''; }
else value += ch;
}
if (value.length || row.length) { row.push(value.replace(/\r$/, '')); rows.push(row); }
if (rows.length && rows[0][0]?.charCodeAt(0) === 0xfeff) rows[0][0] = rows[0][0].slice(1);
const headers = rows.shift();
return rows.filter(r => r.some(v => v !== '')).map(r => Object.fromEntries(headers.map((h,
i) => [h, r[i] ?? ''])));
}

```

```

async function readCsv(name) {
  return parseCsv(await fs.readFile(path.join(frozenRoot, name), "utf8"));
}

```

```

function n(v) {
  const x = Number(v);
  if (!Number.isFinite(x)) throw new Error(`Expected numeric value, got $
{JSON.stringify(v)}`);
  return x;
}

```

```

function roundHalfUp(v, places = 4) {
  // Excel-facing values must follow the audit contract's decimal ROUND_HALF_UP
  // rule. Number.toFixed() rounds the already-binary IEEE-754 value and gives
  // the wrong result at exact decimal half units such as 2238.56265.
  const source = typeof v === "string" ? v.trim() : String(n(v));
  const match = source.match(/^[+-]?(\d+)(?:\.\d*)?(?:[eE](?:[+-]?\d+))?$/);
  if (!match) throw new Error(`Expected decimal value, got ${JSON.stringify(v)}`);
  const negative = match[1] === "-";
  const fraction = match[3] ?? "";
  const exponent = Number(match[4] ?? 0);
  let magnitude = BigInt(`${match[2]}${fraction}` || "0");
  const scale = fraction.length - exponent;
  if (scale <= places) {
    magnitude *= 10n ** BigInt(places - scale);
  } else {
    const divisor = 10n ** BigInt(scale - places);
    const remainder = magnitude % divisor;
    magnitude /= divisor;
    if (remainder * 2n >= divisor) magnitude += 1n;
  }
  const signed = negative ? -magnitude : magnitude;
  return Number(signed) / 10 ** places;
}

```

```
}
```

```
function rounded4(v) { return roundHalfUp(v, 4); }
```

```
function decimalParts(v) {  
  const source = typeof v === "string" ? v.trim() : String(n(v));  
  const match = source.match(/^([+-]?)(\d+)(?:\.(\d*))?(?:[eE]([+-]?\d+))?$$/);  
  if (!match) throw new Error(`Expected decimal value, got ${JSON.stringify(v)}`);  
  const fraction = match[3] ?? "";  
  const exponent = Number(match[4] ?? 0);  
  let magnitude = BigInt(`${match[2]}${fraction}` || "0");  
  let scale = fraction.length - exponent;  
  if (scale < 0) {  
    magnitude *= 10n ** BigInt(-scale);  
    scale = 0;  
  }  
  if (match[1] === "-") magnitude = -magnitude;  
  return { magnitude, scale };  
}
```

```
function decimalSum(values) {  
  const parts = values.map(decimalParts);  
  const scale = Math.max(0, ...parts.map(p => p.scale));  
  const total = parts.reduce(  
    (acc, p) => acc + p.magnitude * 10n ** BigInt(scale - p.scale),  
    0n,  
  );  
  const negative = total < 0n;  
  const digits = (negative ? -total : total).toString().padStart(scale + 1, "0");  
  if (scale === 0) return `${negative ? "-" : ""}${digits}`;  
  return `${negative ? "-" : ""}${digits.slice(0, -scale)}${digits.slice(-scale)}`;  
}  
function excelDate(s) { return new Date(`${s}T00:00:00Z`); }
```

```
function officialLabel(position) {  
  const start = position * 10;  
  const end = (position + 1) * 10;  
  const fmt = (mins) => {  
    if (mins >= 1440) return `${Math.floor((mins - 1440) / 60)}:${String((mins - 1440) % 60).padStart(2, "0")}${1}`;  
    return `${Math.floor(mins / 60)}:${String(mins % 60).padStart(2, "0")}`;  
  };  
  return `${fmt(start)}-${fmt(end)}`;  
}
```

```
function physicalLabel(startPosition, endPositionExclusive) {  
  const fmt = (mins) => `${Math.floor(mins / 60)}:${String(mins % 60).padStart(2, "0")}`;  
  return `${fmt((startPosition - 1) * 10)}-${fmt(endPositionExclusive * 10)}`;  
}
```

```

function groupByDate(rows) {
  const out = new Map();
  for (const row of rows) {
    if (!out.has(row.date)) out.set(row.date, []);
    out.get(row.date).push(row);
  }
  for (const rs of out.values()) rs.sort((a, b) => n(a.position) - n(b.position));
  return out;
}

function validateDispatch(rows, name, expectedDays) {
  if (rows.length !== expectedDays * 144) throw new Error(` ${name}: expected $
{expectedDays * 144} rows, got ${rows.length}`);
  const byDate = groupByDate(rows);
  if (byDate.size !== expectedDays) throw new Error(` ${name}: expected $ {expectedDays}
dates, got $ {byDate.size}`);
  const referenceLabels = [...byDate.values()][0].map(r => r.official_label);
  if (referenceLabels.some(v => !v) || new Set(referenceLabels).size !== 144) {
    throw new Error(` ${name}: official labels must contain 144 nonblank unique values`);
  }
  for (const [date, rs] of byDate) {
    if (rs.length !== 144) throw new Error(` ${name} ${date}: expected 144 rows, got $
{rs.length}`);
    for (let i = 0; i < 144; i++) {
      if (n(rs[i].position) !== i + 1) throw new Error(` ${name} ${date}: bad position at $ {i + 1}
`);
      if (rs[i].official_label !== referenceLabels[i]) throw new Error(` ${name} ${date}: official
label mismatch at $ {i + 1}`);
    }
  }
  return byDate;
}

function styleSheet(sheet, usedRange, numericRanges = []) {
  // Keep the template's cell boundaries visible. Several adjacent fields
  // (date/time and charge/discharge/time) otherwise read as one value in the
  // rendered workbook even though the underlying cells are correct.
  sheet.showGridLines = true;
  usedRange.format.font = { name: "Arial", size: 10, color: "#1111827" };
  usedRange.format.verticalAlignment = "center";
  const header = usedRange.getRow(0);
  header.format.fill = "#1F4E78";
  header.format.font = { name: "Arial", size: 10, bold: true, color: "#FFFFFF" };
  header.format.horizontalAlignment = "center";
  header.format.wrapText = true;
  header.format.rowHeight = 30;
  for (const range of numericRanges) range.format.numberFormat = "0.0000";
}

```

```

function addPlanSheet(wb, name, byDate, valueField, costField) {
  const sheet = wb.worksheets.add(name);
  const labels = [...byDate.values()][0].map(r => r.official_label);
  const matrix = [["日期\\时间", ...labels, "全天购电量", "全天购电费"]];
  for (const [date, rows] of byDate) {
    matrix.push([
      excelDate(date),
      ...rows.map(r => rounded4(r[valueField])),
      rounded4(decimalSum(rows.map(r => r[valueField]))),
      rounded4(rows.reduce((s, r) => s + n(r[costField]), 0)),
    ]);
  }
  const used = sheet.getRangeByIndexes(0, 0, matrix.length, matrix[0].length);
  used.values = matrix;
  styleSheet(sheet, used, [sheet.getRangeByIndexes(1, 1, matrix.length - 1, matrix[0].length - 1)]);
  sheet.getRangeByIndexes(1, 0, matrix.length - 1, 1).format.numberFormat = "yyyy-mm-dd";
  sheet.freezePanes.freezeRows(1);
  sheet.freezePanes.freezeColumns(1);
  sheet.getRange(`A1:A${matrix.length}`).format.columnWidth = 12;
  sheet.getRange(`B1:EO${matrix.length}`).format.columnWidth = 10;
  sheet.getRange(`EP1:EQ${matrix.length}`).format.columnWidth = 15;
  return sheet;
}

```

```

function addQ1Sheets(wb, rows) {
  const plan = wb.worksheets.add("计划购电量");
  // result1's official template uses a different final label spelling from
  // the annual templates; preserve that literal template label.
  const planMatrix = [["时间段", "购电量"], ...rows.map((r, i) => [
    officialLabel(i + 1),
    rounded4(r.plan_00_grid_kwh),
  ])];
  const pUsed = plan.getRangeByIndexes(0, 0, planMatrix.length, 2);
  pUsed.values = planMatrix;
  styleSheet(plan, pUsed, [plan.getRange(`B2:B${planMatrix.length}`)]);
  plan.freezePanes.freezeRows(1);
  plan.getRange(`A1:A${planMatrix.length}`).format.columnWidth = 19;
  plan.getRange(`B1:B${planMatrix.length}`).format.columnWidth = 14;

```

```

const charge = wb.worksheets.add("充放电量");
const cm = [["时间段", "充电量", "放电量", "时刻", "储电量"]];
for (let block = 0; block < 6; block++) {
  const rs = rows.slice(block * 24, (block + 1) * 24);
  cm.push([
    `${block * 4}:00-${(block + 1) * 4}:00`,
    rounded4(decimalSum(rs.map(r => r.charge_kwh))),
  ]);
}

```

```

        rounded4(decimalSum(rs.map(r => r.discharge_kwh))),
        block === 0 ? "0:00" : block === 1 ? "24:00" : null,
        block === 0 ? rounded4(rows[0].soc_start_kwh) : block === 1 ?
rounded4(rows[143].soc_end_kwh) : null,
    ]);
}
const cUsed = charge.getRangeByIndexes(0, 0, cm.length, 5);
cUsed.values = cm;
styleSheet(charge, cUsed, [charge.getRange("B2:C7"), charge.getRange("E2:E3")]);
charge.getRange(`A1:E${cm.length}`).format.columnWidth = 18;
}

function addChargeSheet(wb, byDate) {
    const sheet = wb.worksheets.add("充放电量");
    const matrix = [["日期", "时间段", "充电量", "放电", "时刻", "储电量"]];
    for (const [date, rows] of byDate) {
        for (let block = 0; block < 6; block++) {
            const rs = rows.slice(block * 24, (block + 1) * 24);
            matrix.push([
                block === 0 ? excelDate(date) : null,
                `${block * 4}:00-${(block + 1) * 4}:00`,
                rounded4(decimalSum(rs.map(r => r.charge_kwh))),
                rounded4(decimalSum(rs.map(r => r.discharge_kwh))),
                block === 0 ? "0:00" : block === 1 ? "24:00" : null,
                block === 0 ? rounded4(rows[0].soc_start_kwh) : block === 1 ?
rounded4(rows[143].soc_end_kwh) : null,
            ]);
        }
    }
    const used = sheet.getRangeByIndexes(0, 0, matrix.length, 6);
    used.values = matrix;
    styleSheet(sheet, used, [sheet.getRangeByIndexes(1, 2, matrix.length - 1, 2),
sheet.getRangeByIndexes(1, 5, matrix.length - 1, 1)]);
    sheet.getRangeByIndexes(1, 0, matrix.length - 1, 1).format.numberFormat = "yyyy-mm-dd";
    sheet.freezePanes.freezeRows(1);
    sheet.getRange(`A1:A${matrix.length}`).format.columnWidth = 13;
    sheet.getRange(`B1:B${matrix.length}`).format.columnWidth = 16;
    sheet.getRange(`C1:F${matrix.length}`).format.columnWidth = 14;
}

function addEmergencySheet(wb, byDate, threshold) {
    const sheet = wb.worksheets.add("紧急购电量");
    const matrix = [["日期", "购电时间段", "购电量"]];
    for (const [date, rows] of byDate) {
        const groups = [];
        let current = null;
        for (const r of rows) {
            const pos = n(r.position), q = n(r.emergency_kwh);
            // The configured half-unit threshold implements ROUND_HALF_UP at four decimals.

```

```

    const shown = Math.abs(q) >= threshold;
    if (!shown) { if (current) { groups.push(current); current = null; } continue; }
    if (!current) current = { start: pos, end: pos, amounts: [r.emergency_kwh] };
    else if (pos === current.end + 1) { current.end = pos;
current.amounts.push(r.emergency_kwh); }
    else { groups.push(current); current = { start: pos, end: pos, amounts:
[r.emergency_kwh] }; }
    }
    if (current) groups.push(current);
    groups.forEach((g, i) => matrix.push([i === 0 ? excelDate(date) : null,
physicalLabel(g.start, g.end), rounded4(decimalSum(g.amounts))]);
    }
    const used = sheet.getRangeByIndexes(0, 0, matrix.length, 3);
    used.values = matrix;
    styleSheet(sheet, used, matrix.length > 1 ? [sheet.getRangeByIndexes(1, 2, matrix.length - 1,
1)] : []);
    if (matrix.length > 1) sheet.getRangeByIndexes(1, 0, matrix.length - 1,
1).format.numberFormat = "yyyy-mm-dd";
    sheet.freezePanes.freezeRows(1);
    sheet.getRange(`A1:A${matrix.length}`).format.columnWidth = 13;
    sheet.getRange(`B1:B${matrix.length}`).format.columnWidth = 18;
    sheet.getRange(`C1:C${matrix.length}`).format.columnWidth = 14;
    }

function addEvidenceSheets(wb, byDate, mode) {
    const wanted = new Set(["2025-03-20", "2025-06-21", "2025-09-23", "2025-12-21"]);
    const matrix = [["日期", "物理时间段", "0 点计划量", "最终购电量"]];
    for (const [day, rows] of byDate) {
        if (mode !== "q1" && !wanted.has(day)) continue;
        for (const hour of [10, 12, 14, 16, 18, 20]) {
            const row = rows.find(r => n(r.position) === hour * 6 + 1);
            matrix.push([excelDate(day), row.physical_interval, rounded4(row.plan_00_grid_kwh),
rounded4(row.final_adjusted_grid_kwh)]);
        }
    }
    const sheet = wb.worksheets.add("指定时段");
    const used = sheet.getRangeByIndexes(0,0,matrix.length,4); used.values=matrix;
    styleSheet(sheet, used, [sheet.getRangeByIndexes(1,2,matrix.length-1,2)]);
    sheet.getRange(`A2:A${matrix.length}`).format.numberFormat="yyyy-mm-dd";
    sheet.getRange(`A1:D${matrix.length}`).format.columnWidth=19;
    const mapping=wb.worksheets.add("时段映射");
    const data=[["位置", "源时间戳", "物理区间", "原模板标签"], ...[...byDate.values()]
[0].map(r=>[n(r.position),r.source_timestamp,r.physical_interval,r.official_label])];
    const
range=mapping.getRangeByIndexes(0,0,data.length,4);range.values=data;styleSheet(mapping
,range);
    mapping.getRange("A1:D145").format.columnWidth=21;
    }

```

```

async function exportAndVerify(wb, filename, sheetNames) {
  wb.recalculate();
  const file = path.join(workbookRoot, filename);
  const blob = await SpreadsheetFile.exportXlsx(wb);
  await blob.save(file);
  const reopened = await SpreadsheetFile.importXlsx(await FileBlob.load(file));
  const errors = await reopened.inspect({
    kind: "match",
    searchTerm: "#REF!|#DIV/0!|#VALUE!|#NAME\\?|#N/A|#NUM!|#NULL!|#SPILL!|#CALC!",
    options: { useRegex: true, maxResults: 100 },
    maxChars: 6000,
    summary: `${filename} formula error scan`,
  });
  const errorLines = (errors.ndjson || "").split("\n").filter(line => /"address"s*:.test(line));
  if (errorLines.length) throw new Error(`${filename}: formula error detected: $
{errorLines.join("\n")}`);
  // Some artifact-tool versions spill inspect output beside large workbooks.
  // It is a transient diagnostic, not part of the official submission.
  await fs.rm(`${file}.inspect.ndjson`, { force: true });
  const renderDir = path.join(qaRoot, filename.replace(/\.xlsx$/, ""));
  await fs.mkdir(renderDir, { recursive: true });
  for (const sheetName of sheetNames) {
    const sheet = reopened.worksheets.getItem(sheetName);
    const used = sheet.getUsedRange(true);
    const rowCount = used?.rowCount ?? 1;
    const colCount = used?.columnCount ?? 1;
    const ranges = [];
    if (colCount > 20) {
      ranges.push(`A1:P${Math.min(rowCount, 12)}`);
      ranges.push(`EI1:EQ${Math.min(rowCount, 12)}`);
      if (rowCount > 20) ranges.push(`A${Math.max(1, rowCount - 5)}:P${rowCount}`);
    } else {
      ranges.push(`A1:${String.fromCharCode(64 + Math.min(colCount, 26))}${Math.min(rowCount, 40)}`);
      if (rowCount > 45) ranges.push(`A${rowCount - 5}:${String.fromCharCode(64 + Math.min(colCount, 26))}${rowCount}`);
    }
    for (let i = 0; i < ranges.length; i++) {
      const png = await reopened.render({ sheetName, range: ranges[i], scale: 1.5, format:
"png" });
      const safe = sheetName.replace(/[\^p{L}\p{N}_-]/gu, "_");
      await fs.writeFile(path.join(renderDir, `${safe}_${i + 1}.png`), new Uint8Array(await
png.arrayBuffer()));
    }
  }
  return file;
}

```

```

async function main() {

```

```

await fs.mkdir(workbookRoot, { recursive: true });
await fs.mkdir(qaRoot, { recursive: true });
const cfg = JSON.parse(await fs.readFile(path.join(projectRoot, "config", "final.yaml"),
"utf8"));
const threshold = Number(cfg.audit.emergency_display_omit_threshold_kwh);

const specs = [
  ["q1_dispatch.csv", "result1.xlsx", "q1"],
  ["q2_dispatch.csv", "result2.xlsx", "q2"],
  ["q3_dispatch.csv", "result3.xlsx", "q3"],
  ["q4_2_dispatch.csv", "result4-2.xlsx", "q4_2"],
  ["q4_3_dispatch.csv", "result4-3.xlsx", "q4_3"],
];
const created = [];
for (const [csvName, xlsxName, mode] of specs) {
  const rows = await readCsv(csvName);
  const expectedDays = mode === "q1" ? 1 : 334;
  const byDate = validateDispatch(rows, mode, expectedDays);
  const wb = Workbook.create();
  let sheetNames;
  if (mode === "q1") {
    addQ1Sheets(wb, rows);
    sheetNames = ["计划购电量", "充放电量"];
  }
  else {
    const initialCostField = (mode === "q2" || mode === "q4_2") ?
"final_relative_cost_yuan" : "plan_cost_yuan";
    addPlanSheet(wb, "计划购电量", byDate, "plan_00_grid_kwh", initialCostField);
    if (mode === "q3" || mode === "q4_3") addPlanSheet(wb, "调整购电量", byDate,
"final_adjusted_grid_kwh", "final_relative_cost_yuan");
    addChargeSheet(wb, byDate);
    addEmergencySheet(wb, byDate, threshold);
    sheetNames = (mode === "q3" || mode === "q4_3")
      ? ["计划购电量", "调整购电量", "充放电量", "紧急购电量"]
      : ["计划购电量", "充放电量", "紧急购电量"];
  }
  addEvidenceSheets(wb, byDate, mode);
  sheetNames.push("指定时段", "时段映射");
  created.push(await exportAndVerify(wb, xlsxName, sheetNames));
}
await fs.writeFile(
  path.join(finalRoot, "qa", "workbook_build.json"),
  JSON.stringify({ created: created.map(file => path.relative(finalRoot, file)), threshold,
verified: true }, null, 2),
);
}

await main();

```

scripts/competition_checks.py

```
SHA256 6842e8294af118e3f923ae4c038c16356057f98d8234297301df1881d296bcba
"""Check verifiable submission properties without inventing human signoff."""
from pathlib import Path
import json,re,hashlib
from research_content import source_files
ROOT=Path(__file__).resolve().parents[1]
FINAL=ROOT/'outputs/c_final_v1'

def inspect(doc,pdf):
    checks=[]
    def add(name,ok,actual):checks.append({'check':name,'status':'PASS' if ok else
'FAIL','actual':actual})
    first="".join(pdf.pages[0].extract_text().split())
    add('abstract_dedicated_first_page','关键词' in first and '题目[1]中的微网' not in first,
{'first_page_has_keywords': '关键词' in first})
    body=next(p for p in doc.paragraphs if p.text=='1 问题与信息边界')

    add('body_explicit_new_page',bool(body.paragraph_format.page_break_before),str(body.para
graph_format.page_break_before))
    appendix=[i for i,p in enumerate(pdf.pages) if re.search(r'(?m)^\s*附录\s*$',p.extract_text())]
    add('main_body_not_over_30_pages',bool(appendix) and appendix[0]-1<=30,
{'appendix_start_pdf_page':appendix[0]+1 if appendix else None,'body_pages':appendix[0]-1
if appendix else None})

    add('all_margins_at_least_2_5cm',all(min(s.top_margin.cm,s.bottom_margin.cm,s.left_margi
n.cm,s.right_margin.cm)>=2.499 for s in doc.sections),len(doc.sections))
    expected={str(p.relative_to(ROOT)):p for p in
source_files()};seen={};active=None;started=False
    for p in doc.paragraphs:
        if p.text=='附录 C 完整源程序与运行配置':started=True
        if not started:continue
        if p.style.name=='Heading 3' and p.text in expected:
            active=p.text;seen[active]=[]
        elif active and not p.text.startswith('SHA256 '):seen[active].append(p.text)
    bad=[]
    for name,p in expected.items():
        want=[x if x else '' for x in p.read_text().splitlines()]
        if seen.get(name)!=want:bad.append(name)
    add('complete_source_appendix_exact_line_match',not bad and len(seen)==len(expected),
{'expected_files':len(expected),'actual_files':len(seen),'mismatch':bad})
    add('blank_author_metadata',not doc.core_properties.author and not
doc.core_properties.last_modified_by,{'author':doc.core_properties.author})
    research=json.loads((FINAL/'research/annual_independent_checks.json').read_text())
    add('four_full_year_independent_replays',len(research)==4 and all(x['status']=='PASS' and
x['rows']==48096 for x in research),research)
    before=[]
    for p in doc.paragraphs:
```

```

        if p.text=='附录':break
        before.append(p.text)
    text='\n'.join(before)
    add('seven_references_and_body_citations',all(f'[{n}]' in text.split('参考文献')[0] for n in
range(1,8)) and all(f'[{n}]' in text.split('参考文献')[-1] for n in range(1,8)),7)
    return checks

```

scripts/final_quality.py

```

SHA256 f014445eb34ddd51f5d113b2d48370fb14b9fcff4ce22dc35aadfe232be5b39
""""Validate generated paper tables against dispatch positions, plus PDF and assets.""""
from pathlib import Path
import json,re,hashlib
from datetime import datetime
import numpy as np,pandas as pd
from docx import Document
from pypdf import PdfReader
from docx.oxml.ns import qn
ROOT=Path(__file__).resolve().parents[1];FINAL=ROOT/'outputs/
c_final_v1';F=FINAL/'frozen'

def main():
    checks=[]
    def check(k,ok,value):checks.append(dict(check=k,status='PASS' if bool(ok) else
'FAIL',actual=value))

    audit=json.loads((FINAL/'audit/independent_validation.json').read_text());check('independent
_model_and_workbooks',audit['status']=='PASS',audit['status'])
    doc=Document(FINAL/'paper/C 题论文.docx');pdf=PdfReader(FINAL/'paper/C 题论
文.pdf');ai=PdfReader(FINAL/'paper/AI 工具使用详情.pdf')
    text='\n'.join(p.text for p in doc.paragraphs)+'\n'+'\n'.join(c.text for t in doc.tables for r in
t.rows for c in r.cells);pdftext='\n'.join(p.extract_text() for p in pdf.pages)
    # Literal template syntax inside the printed source appendix is source code,
    # not an unexpanded value in the paper. Check the rendered paper separately.
    body_paragraphs=[]
    for p in doc.paragraphs:
        if p.text=='附录 C 完整源程序与运行配置':break
        body_paragraphs.append(p.text)
    source_page=next((i for i,p in enumerate(pdf.pages) if '完整源程序与' in p.extract_text()
[:120]),len(pdf.pages))
    display_text='\n'.join(body_paragraphs)+'\n'+'\n'.join(c.text for t in doc.tables for r in t.rows
for c in r.cells)+'\n'+'.join(p.extract_text() for p in pdf.pages[:source_page])
    check('no_unresolved_placeholders',not re.search(r'\{\{^[^}\}]+\}\}
\}',display_text),len(re.findall(r'\{\{^[^}\}]+\}\}',display_text)))
    check('A4_pages',all(abs(float(p.mediabox.width)-595.276)<2 and
abs(float(p.mediabox.height)-841.89)<2 for p in pdf.pages),len(pdf.pages))
    check('PDF_under_20MB',(FINAL/'paper/C 题论文.pdf').stat().st_size<20*1024**2,
(FINAL/'paper/C 题论文.pdf').stat().st_size)

```

```

# Walk actual OOXML order; attach each appendix table to its section date.
active=None;seen={};errors=[];table_i=0;para_i=0
frames={mode:pd.read_csv(F/f{mode}_dispatch.csv') for mode in
['q1','q2','q3','q4_2','q4_3']}
def num(v):return float(v.replace(',').strip())
for el in doc.element.body:
    if el.tag==qn('w:p'):
        t="".join(el.itertext()) # itertext may repeat run text; regex still identifies section.
        m=re.search(r'(Q1|Q2|Q3|Q4_2|Q4_3) (2025-\d\d-\d\d) 指定结果',t)
        if m:active=(m.group(1).lower(),m.group(2));seen.setdefault(active,set())
    elif el.tag==qn('w:tbl'):
        table=doc.tables[table_i];table_i+=1
        if active is None:continue
        data=[[c.text for c in r.cells] for r in
table.rows];head=data[0];mode,day=active;z=frames[mode];z=z[z.date==day]
        if head==['物理区间','0 点计划量','最终承诺量']:
            seen[active].add('ten_minute')
            if len(data)!=7:errors.append((active,'ten_minute_count'))
            for row,h in zip(data[1:],[10,12,14,16,18,20]):
                r=z[z.position==h*6+1].iloc[0]
                if row[0]!=r.physical_interval or abs(num(row[1])-r.plan_00_grid_kwh)>.000051
or abs(num(row[2])-r.final_adjusted_grid_kwh)>.000051:errors.append((active,row))
            elif head==['物理区间','充电量','放电量']:
                seen[active].add('four_hour')
                for b,row in enumerate(data[1:]):
                    zz=z[z.position>b*24&(z.position<=(b+1)*24)]
                    if row[0]!=f'{b*4}:00-{(b+1)*4}:00' or abs(num(row[1])-
zz.charge_kwh.sum())>.000051 or abs(num(row[2])-
zz.discharge_kwh.sum())>.000051:errors.append((active,row))
            elif head==['0 点 SOC','24 点 SOC']:
                seen[active].add('soc')
                if abs(num(data[1][0])-z.soc_start_kwh.iloc[0])>.000051 or abs(num(data[1][1])-
z.soc_end_kwh.iloc[-1])>.000051:errors.append((active,'SOC'))
            elif head==['全天计划量','全天最终量','全天总购电费']:
                seen[active].add('daily')
                if abs(num(data[1][0])-z.plan_00_grid_kwh.sum())>.000051 or abs(num(data[1]
[1])-z.final_adjusted_grid_kwh.sum())>.000051 or abs(num(data[1][2])-
z.final_relative_cost_yuan.sum())>.0051:errors.append((active,'daily'))
            elif head==['紧急购电物理区间','紧急购电量']:
                seen[active].add('emergency');expected=[];start=None;total=0.;last=None
                for r in z.sort_values('position').itertuples():
                    if r.emergency_kwh>=.00005:
                        if start is None:start=(int(r.position)-1)*10
                        last=int(r.position)*10;total+=r.emergency_kwh
                    elif start is not None:
                        expected.append((f'{start//60}:{start%60:02d}-{last//60}:
{last%60:02d}',total));start=None;total=0.
                        if start is not None:expected.append((f'{start//60}:{start%60:02d}-{last//60}:
{last%60:02d}',total))

```

```

        if not expected:expected=[('无正缺口',0.))
        if len(data)-1!=len(expected):errors.append((active,'emergency_count'))
        for row,(span,value) in zip(data[1:],expected):
            if row[0]!=span or abs(num(row[1])-
value)>.000051:errors.append((active,'emergency',row))
            required={(mode,day) for mode in ['q2','q3','q4_2','q4_3'] for day in ['2025-03-20','2025-
06-21','2025-09-23','2025-12-21']};required.add(('q1','2025-01-01'))
            missing=[k for k in required if k not in seen or not ({'ten_minute','four_hour','soc','daily'}|
({'emergency'}) if k[0]!='q1' else set()))<=seen[k]]
            check('all_17_specified_date_groups',not missing,
{'groups':len(seen),'missing':missing});check('paper_numbers_independently_reconciled',not
errors,errors)
            missing_pdf_values=[]
            compact_pdf=re.sub(r'\s+',",",pdftext)
            for table in doc.tables:
                for row in table.rows:
                    for cell in row.cells:
                        value=cell.text.strip()
                        if re.fullmatch(r'-?[d,]+\.\d+',value) and value not in
compact_pdf:missing_pdf_values.append(value)
                        check('all_table_numeric_strings_survive_PDF',not
missing_pdf_values,missing_pdf_values)
                        # Confirm key corrected Q1 numbers survived the PDF conversion.
                        from decimal import Decimal,ROUND_HALF_UP
                        q1value=float(frames['q1'].loc[frames['q1'].position==73,'plan_00_grid_kwh'].iloc[0])
                        for value in
[format(Decimal(str(q1value)).quantize(Decimal('.0001'),rounding=ROUND_HALF_UP),'.4f
)]:
                            check('PDF_corrected_physical_value_'+value,value in pdftext,value)

check('five_workbooks',len(list((FINAL/'workbooks').glob('result*.xlsx')))==5,len(list((FINA
L/'workbooks').glob('result*.xlsx'))))
    check('AI_details_nonempty',len(ai.pages)>0 and len(".join(p.extract_text() for p in
ai.pages))>200,len(ai.pages))

check('charts_present',len(list((FINAL/'charts').glob('*.png')))>=4,len(list((FINAL/'charts').gl
ob('*.png'))))
    (FINAL/'qa/paper_text.txt').write_text(pdftext)
    from competition_checks import inspect
    checks.extend(inspect(doc,pdf))
    result=dict(status='PASS' if all(c['status']=='PASS' for c in checks) else
'FAIL',generated_at=datetime.now().astimezone().isoformat(),scope='current_run_numeric_d
ocument_and_export_checks',checks=checks,metrics=dict(pdf_pages=len(pdf.pages),docx_ta
bles=len(doc.tables),specified_date_groups=len(seen)),limitations=['Historical scenarios do
not guarantee unseen tail-risk coverage.','Refund and delivery-time pricing are declared
interpretations, not additional official rules.','Solver diagnostics cover 32 exact LP instances,
not full-year alternative-solver policy rollouts.','Team submission approval and authorship
signoff are not asserted.'])
    result['limitations']=['Finite historical scenarios and one-year replays do not guarantee
future tail-risk coverage.','Refund and delivery-time pricing remain declared

```

interpretations.', 'Annual alternative-solver checks cover S2 and S3 with a fixed common reference contract; no test-period policy selection.', 'Team authorship and manual review are pending; this PASS is not a submission eligibility certificate.']

```
result['submission_readiness']='PENDING_TEAM_MANUAL_REVIEW_AND_RULES_CONFIRMATION'
```

```
artifacts=list((FINAL/'paper').glob('*.*docx'))+list((FINAL/'paper').glob('*.*pdf'))
+list((FINAL/'workbooks').glob('*.*xlsx'))
```

```
result['artifact_hashes']=[dict(path=str(p.relative_to(FINAL)),sha256=hashlib.sha256(p.read_bytes()).hexdigest()) for p in artifacts]
```

```
(FINAL/'final_validation_v3.json').write_text(json.dumps(result,ensure_ascii=False,indent=2))
print(json.dumps(result,ensure_ascii=False,indent=2));return result
```

```
if __name__=='__main__':raise SystemExit(0 if main()['status']=='PASS' else 1)
```

scripts/freeze_codec.py

SHA256 e02914b278ec089c0d24c9612830fc4ca3504ce315fbb169b8e1dee00399fa45

""""Lossless column deduplication for the competition's support-file size limit.

All original CSV strings are preserved. The restored bytes must match SHA256; this is packaging only and never rounds or changes model observations/results.

""""

```
from pathlib import Path
import csv,hashlib,json,lzma,io,shutil,tarfile,tempfile
```

```
def sha(data):return hashlib.sha256(data).hexdigest()
```

```
def pack(source,target):
```

```
source=Path(source);target=Path(target);blobs=target/'columns';blobs.mkdir(parents=True,exist_ok=True)
```

```
manifest={'format':'lossless-csv-columns-v1','files':[]}
```

```
duplicates={'q3_decision_versions.csv','q4_3_decision_versions.csv','q3_S3_decisions.csv','q4_3_S3_decisions.csv','q3_S3_dispatch.csv','q4_3_S3_dispatch.csv'}
```

```
for p in sorted(source.iterdir()):
```

```
    if not p.is_file() or p.name in duplicates:continue
```

```
    if p.suffix!='.csv':shutil.copy2(p,target/p.name);continue
```

```
    raw=p.read_bytes();bom=raw.startswith(b'\xef\xbb\xbf');newline='\r\n' if b'\r\n' in raw
```

```
else '\n'
```

```
    reader=csv.reader(io.StringIO(raw.decode('utf-8-
```

```
sig'),newline="));header=next(reader);cols=[[ ] for _ in header];count=0
```

```
    for row in reader:
```

```
        if len(row)!=len(header):raise ValueError(('Nonrectangular CSV',p.name))
```

```
        count+=1
```

```
        for col,val in zip(cols,row):
```

```
            if '\n' in val or '\r' in val:raise ValueError(('Multiline value',p.name))
```

```

        col.append(val)
    ids=[]
    for col in cols:
        data='\n'.join(col).encode('utf-8');key=sha(data);out=blobs/(key+'.xz')
        if not out.exists():out.write_bytes(lzma.compress(data,preset=6))
        ids.append(key)

record=dict(name=p.name,sha256=sha(raw),bytes=len(raw),bom=bom,newline=newline,header=header,rows=count,columns=ids)
    # Verify the CSV dialect before recording a supposedly lossless manifest.
    if sha(reconstruct(record,cols))!=record['sha256']:raise ValueError(('CSV byte roundtrip failed',p.name))
    manifest['files'].append(record);print('Packed losslessly',p.name,flush=True)
    (target/'csv_manifest.json').write_text(json.dumps(manifest,ensure_ascii=False,indent=2))
    return manifest

def reconstruct(record,cols):

out=io.StringIO(newline="");writer=csv.writer(out,lineterminator=record['newline']);writer.writerow(record['header']);writer.writerows(zip(*cols))
    return out.getvalue().encode('utf-8-sig' if record['bom'] else 'utf-8')

def unpack(source,target):
    source=Path(source);target=Path(target);target.mkdir(parents=True,exist_ok=True)
    manifest=json.loads((source/'csv_manifest.json').read_text())
    solid={}
    if (source/'columns.tar.xz').exists():
        with tarfile.open(source/'columns.tar.xz','r:xz') as archive:
            for member in archive:
                if not member.isfile() or len(member.name)!=64:raise ValueError('Invalid column member')
                data=archive.extractfile(member).read()
                if sha(data)!=member.name:raise ValueError('Corrupt solid column')
                solid[member.name]=data
    for record in manifest['files']:
        cols=[]
        for key in record['columns']:
            data=solid[key] if solid else
lzma.decompress((source/'columns'/(key+'.xz')).read_bytes())
            if sha(data)!=key:raise ValueError('Corrupt column: '+key)
            vals=data.decode('utf-8').split('\n') if record['rows'] else []
            if len(vals)!=record['rows']:raise ValueError('Wrong row count')
            cols.append(vals)
        raw=reconstruct(record,cols)
        if sha(raw)!=record['sha256']:raise ValueError('Corrupt CSV: '+record['name'])
        (target/record['name']).write_bytes(raw)
    for p in source.iterdir():
        if p.is_file() and p.name not in ['csv_manifest.json','columns.tar.xz'] and p.parent!=target:shutil.copy2(p,target/p.name)
    return len(manifest['files'])

```

```

def solidify(source,target):
    source=Path(source)
    manifest=json.loads((source/'csv_manifest.json').read_text())
    labels={}
    for r in manifest['files']:
        for label,key in zip(r['header'],r['columns']):labels.setdefault(key,label)
    keys=sorted(labels,key=lambda key:(labels[key],key))
    with
lzma.open(target,'wb',format=lzma.FORMAT_XZ,filters=[{'id':lzma.FILTER_LZMA2,'prese
t':9,'dict_size':128*1024*1024}]) as dst:
        with tarfile.open(fileobj=dst,mode='w|') as archive:
            for key in keys:
                p=source/'columns'/(key+'.xz')

raw=lzma.decompress(p.read_bytes());info=tarfile.TarInfo(p.stem);info.size=len(raw);info.mt
ime=0;archive.addfile(info,io.BytesIO(raw))

if __name__=='__main__':
    import argparse

ap=argparse.ArgumentParser();ap.add_argument('operation',choices=['pack','unpack']);ap.add
_argument('source');ap.add_argument('target');a=ap.parse_args()
(pack if a.operation=='pack' else unpack)(a.source,a.target)

```

scripts/make_charts.py

```

SHA256 c2f3803d727c9d75ad253ae2b9bf5a7d954a529110bac539847f08e97c7f6879
from pathlib import Path
import sys
ROOT=Path(__file__).resolve().parents[1];sys.path.insert(0,str(ROOT/'work/python_deps'))
import matplotlib
matplotlib.use('Agg')
import matplotlib.pyplot as plt
import numpy as np,pandas as pd
F=ROOT/'outputs/c_final_v1/frozen';C=F.parent/'charts';C.mkdir(exist_ok=True)
plt.rcParams.update({'font.family':'DejaVu
Sans','font.size':10,'axes.spines.top':False,'axes.spines.right':False,'figure.dpi':160})
s=pd.read_csv(F/'experiment_summary.csv');official=s[(s.model_arm=='joint_empirical')&((s
['mode'].isin(['q2','q4_2']))|(s.scenario=='S3'))]
fig,ax=plt.subplots(figsize=(8,3.6));x=np.arange(len(official));bottom=np.zeros(len(x))
for col,label,color in [('plan_cost_yuan','Scheduled','#35658a'),('adjustment_cost_yuan','Net
revisions','#dfa04f'),('emergency_cost_yuan','Emergency','#b85450')]:
    vals=official[col].to_numpy()/1e6;ax.bar(x,vals,bottom=bottom,label=label,color=color);bott
om+=vals
ax.set_xticks(x,official['mode']);ax.set_ylabel('Realized cost / million
yuan');ax.legend(ncol=3,frameon=False);fig.tight_layout();fig.savefig(C/'cost_components.pn
g');plt.close(fig)
fig,axs=plt.subplots(1,2,figsize=(8,3.5))

```

```

for ax,mode in zip(axes,['q3','q4_3']):

d=s[(s['mode']==mode)&(s.model_arm=='joint_empirical')].sort_values('scenario');ax.plot(d.scenario,d.final_relative_cost_yuan/1e6,'o-',color='#35658a');ax.set_title(mode.upper());ax.set_ylabel('Main cost / million yuan');ax.grid(axis='y',alpha=.2)
fig.tight_layout();fig.savefig(C/'schedule_comparison.png');plt.close(fig)
fig,axes=plt.subplots(2,1,figsize=(8,4.8),sharex=True);d=pd.read_csv(F/'q3_dispatch.csv');d=d[d.date=='2025-06-21'];h=(d.position.to_numpy()-1)/6
axes[0].plot(h,d.plan_00_grid_kwh,label='Common day-ahead');axes[0].plot(h,d.final_adjusted_grid_kwh,label='Final commitment');axes[0].fill_between(h,0,d.emergency_kwh,color='#b85450',alpha=.5,label='Emergency');axes[0].set_ylabel('Energy / kWh');axes[0].legend(ncol=3,frameon=False,fontsize=8)
axes[1].plot(h,d.soc_start_kwh,color='#35658a');axes[1].axhline(1200,ls='--',color='gray');axes[1].axhline(10800,ls='--',color='gray');axes[1].set_ylabel('SOC / kWh');axes[1].set_xlabel('Physical hour on 2025-06-21');fig.tight_layout();fig.savefig(C/'representative_dispatch.png');plt.close(fig)
fig,ax=plt.subplots(figsize=(8,3.6))
for mode,color in [('q3','#35658a'),('q4_3','#b85450')]:
a=pd.read_csv(F/f'{mode}_S0_daily.csv');b=pd.read_csv(F/f'{mode}_S3_daily.csv');delta=b.final_relative_cost_yuan-a.final_relative_cost_yuan;ax.plot(pd.to_datetime(a.date),delta.cumsum()/1000,label=mode,color=color)
ax.axhline(0,color='gray',lw=.8);ax.set_ylabel('Cumulative S3 - S0 / thousand yuan');ax.legend(frameon=False);fig.autofmt_xdate();fig.tight_layout();fig.savefig(C/'cumulative_difference.png');plt.close(fig)
print('Four data-derived figures written',flush=True)
monthly=pd.read_csv(F.parent/'research/monthly_mechanisms.csv')
fig,axes=plt.subplots(2,1,figsize=(8,5.0),sharex=True)
for mode,color in zip(['q2','q4_2','q3','q4_3'], ['#35658a','#b85450','#548764','#9b6a9a']):
z=monthly[monthly['mode']==mode];x=np.arange(len(z))
axes[0].plot(x,z.cost/1e6,'o-',color=color,label=mode)
axes[1].plot(x,z.emergency_cost/z.cost*100,'o-',color=color)
axes[0].set_ylabel('Monthly cost / million yuan');axes[0].legend(ncol=4,frameon=False)
axes[1].set_ylabel('Emergency cost share / %');axes[1].set_xticks(x,z.month.str[5:]);axes[1].set_xlabel('Month in 2025')
for ax in axes:ax.grid(axis='y',alpha=.2)
fig.tight_layout();fig.savefig(C/'monthly_risk.png');plt.close(fig)

```

scripts/native_equations.py

SHA256 178b05030cd6abeb1ca97603ddfad843c100387694b29c9a0a789e69cac7b2dd
 """Small native OMML builder for the paper's reviewed equations.

Supports grouped subscripts/superscripts and fractions, without raster formulas or Unicode presentation subscripts that some DOCX converters overlap.
 """

```

from docx.oxml import OxmlElement
from docx.oxml.ns import qn
import unicodedata

```

```

EQUATIONS = {
1: r'F_τ = σ {观测结束时刻 ≤ τ; 预报发布时间 ≤ τ; 既有采购承诺}',
2: r'A_t + V_t + D_t + Q_t = L_t + C_t + U_t; Q_t, U_t ≥ 0',
3: r'E_{t+1} = E_t + 0.9 C_t - \frac{D_t}{0.9}',
4: r'1200 ≤ E_t ≤ 10800; 0 ≤ C_t, D_t ≤ \frac{5000}{6}',
5: r'\tilde{X}_t^s = \max\{0, \hat{X}_{t|\tau} + (X_{h^s} - \hat{X}_{h^s|\tau_h})\}',
6: r'G_t + D_t - C_t + Q_t^s ≥ L_t^s - V_t^s; Q_t^s ≥ 0, \forall s, t \in H_\tau',
7: r'G_t^s = G_t, C_t^s = C_t, D_t^s = D_t, E_t^s = E_t, \forall s',
8: r'\min \frac{1}{N_\tau} \sum_s \sum_t [p_t^s G_t + 5 p_t^s Q_t^s] + \lambda |E_{\text{末}} - 6000| + \epsilon \sum_t (C_t + D_t)',
9: r'f_t(x) = p_t x + 0.5 p_t |x|',
10: r'J_{\text{净}} = \sum_t [p_t B_t + f_t(A_t^{\text{末}} - B_t) + 5 p_t Q_t]',
11: r'J_{\text{逐次}} = \sum_t [p_t B_t + \sum_k f_t(A_t^k - A_t^{k-1}) + 5 p_t Q_t]',
12: r'J_{\text{逐次}} - J_{\text{净}} = 0.5 \sum_t p_t [\sum_k |\Delta A_t^k| - |\sum_k \Delta A_t^k|] ≥ 0',
}

```

```

def node(tag, children=()):
    n=OxmlElement('m:'+tag)
    for c in children:n.append(c)
    return n

```

```

def run(text):
    r=node('r');pr=node('rPr');sty=node('sty');sty.set(qn('m:val'),'p');pr.append(sty);r.append(pr)
    wr=OxmlElement('w:rPr');font=OxmlElement('w:rFonts')
    for k in ['ascii','hAnsi','eastAsia']:font.set(qn('w:'+k),'Arial Unicode MS')

    wr.append(font);size=OxmlElement('w:sz');size.set(qn('w:val'),'20');wr.append(size);r.append(wr)
    t=node('t');t.set(qn('xml:space'),'preserve');t.text=text;r.append(t);return r

```

```

def parse(text):
    i=0
    def atom():
        nonlocal i
        if text[i]=='{':
            i+=1;out=sequence(True);return out
        if text.startswith(r'\frac',i):
            i+=5;a=atom();b=atom();return [node('f',[node('num',a),node('den',b)])]
        c=text[i];i+=1
        while i<len(text) and unicodedata.combining(text[i]):c+=text[i];i+=1
        return [run(c)]
    def sequence(group=False):
        nonlocal i
        out=[]
        while i<len(text):
            if group and text[i]=='}':i+=1;return out

```

```

base=atom();sub=sup=None
while i<len(text) and text[i] in '_^':
    marker=text[i];i+=1;value=atom()
    if marker=='_':sub=value
    else:sup=value
    if sub is not None and sup is not None:out.append(node('sSubSup',
[node('e',base),node('sub',sub),node('sup',sup)]))
    elif sub is not None:out.append(node('sSub',[node('e',base),node('sub',sub)]))
    elif sup is not None:out.append(node('sSup',[node('e',base),node('sup',sup)]))
    else:out.extend(base)
return out
return sequence()

```

```

def equation(number):
    # Literal grouping braces outside scripts are displayed as parentheses.
    source=EQUATIONS[number]
    if number in
(1,5):source=source.replace('σ','σ').replace('max','max');source=source[:-1]+'
    return node('oMath',parse(source)+[run(' ('+str(number)+')')])

```

scripts/package_final.py

```

SHA256 a3318552522c7eda1848c923e59b87b3f4b5f33a25d97c6d7d33675026385ce6
from pathlib import Path
from datetime import datetime
import hashlib,json,lzma,shutil,zipfile,tempfile,os
import re
from freeze_codec import pack,solidify
ROOT=Path(__file__).resolve().parents[1];FINAL=ROOT/'outputs/c_final_v1'
def digest(p):return hashlib.sha256(p.read_bytes()).hexdigest()
def main():
    v=json.loads((FINAL/'final_validation_v3.json').read_text())
    if v['status']!='PASS':raise RuntimeError('Refusing to package failed current run')
    source=[]
    for folder in ['src','scripts','tests','paper','config']:
        for p in (ROOT/folder).rglob('*'):
            if p.is_file() and '__pycache__' not in
p.parts:source.append(dict(path=str(p.relative_to(ROOT)),sha256=digest(p)))

(FINAL/'evidence/active_source_hashes.json').write_text(json.dumps(source,ensure_ascii=False,indent=2))
provenance=json.loads((FINAL/'evidence/source_provenance.json').read_text())
original=[r for r in provenance if r.get('kind')=='original_input']
for r in original:
    source_root=os.environ.get('CUMCM_SOURCE_ROOT',json.loads((ROOT/'config/
final.yaml').read_text())['source_root'])
    if digest(Path(source_root)/r['path'])!=r['sha256']:raise RuntimeError('Original input
changed: '+r['path'])
    (FINAL/'evidence/source_provenance.json').write_text(json.dumps(original+
[dict(kind='current_source',**r) for r in source],ensure_ascii=False,indent=2))

```

```

cache=ROOT/'work/packed_frozen';manifest=cache/'csv_manifest.json';solid=ROOT/'work/
columns_solid.tar.xz';tag=ROOT/'work/columns_solid_manifest.sha256'

duplicates={'q3_decision_versions.csv','q4_3_decision_versions.csv','q3_S3_decisions.csv','q
4_3_S3_decisions.csv','q3_S3_dispatch.csv','q4_3_S3_dispatch.csv'}
expected={p.name:digest(p) for p in (FINAL/'frozen').glob('*.*csv') if p.name not in
duplicates}
valid=manifest.exists() and {r['name']:r['sha256'] for r in json.loads(manifest.read_text())
['files']}==expected
if not valid:pack(FINAL/'frozen',cache)
if not solid.exists() or not tag.exists() or tag.read_text()!=digest(manifest):
    solidify(cache,solid);tag.write_text(digest(manifest))
with tempfile.TemporaryDirectory(prefix='v3-package-') as td:
    stage=Path(td)
    for folder in ['src','scripts','tests','paper','config']:
        shutil.copytree(ROOT/folder,stage/folder,ignore=shutil.ignore_patterns('__pycache__'))
        for folder in ['audit','evidence','workbooks','charts','logs','research']:
            shutil.copytree(FINAL/folder,stage/folder)
        # Full research trajectories are in the common lossless column archive.
        for p in (stage/'research').glob('*_ipm.csv.xz'):p.unlink()
        for name in ['run_pipeline.sh','requirements.txt']:shutil.copy2(ROOT/name,stage/name)
        shutil.copy2(FINAL/'paper/AI 工具使用详情.pdf',stage/'AI 工具使用详
情.pdf');shutil.copy2(FINAL/'final_validation_v3.json',stage/'final_validation_v3.json')
        (stage/'frozen').mkdir()
        shutil.copy2(manifest,stage/'frozen/csv_manifest.json');shutil.copy2(solid,stage/'frozen/
columns.tar.xz')
        for p in (FINAL/'frozen').iterdir():
            if not p.is_file() or p.name in duplicates:continue
            if p.suffix!='.csv':shutil.copy2(p,stage/'frozen'/p.name)
        (stage/'README.md').write_text("# V3 闭环修订支撑材料")

```

原始输入为官方 C 题及附件，未修改观测。先在 config/final.yaml 配置 source_root（其下应含 C 题/附件），或设置 CUMCM_SOURCE_ROOT。运行 run_pipeline.sh 即可重建；运行环境需 Python 及 requirements.txt、Node 与 @oai/artifact-tool，DOCX 转 PDF 使用 Codex 文档运行时。对冻结结果只读检查可先执行 scripts/unpack_frozen.py。

frozen 保存全精度结果，CSV 按列去重后固实 XZ 压缩，未舍入任何数字。csv_manifest 保存每份 CSV 的原始 SHA256；unpack_frozen 将逐字节恢复并校验，另恢复重复 S3 轨迹及正式子账本，输出到 outputs/c_final_v1/frozen，随后可运行 src/audit/audit_final.py --workbooks。主结果 S3 与 S0/S1/S2 共享参考日前承诺；所有年度状态来自一月可核验启动账本。训练配置见 resolved_config，不能用未解析默认值替代。

论文费用按取消退原价保留 50%违约金、交付物理区间价解释；同路径逐次与不退款费用另列。不声称其他结算解释下最优。历史场景为预测分布，非伪造的实际数据。未宣称团队已经完成提交签核。

```

")
    for p in stage.rglob('*'):
        if p.is_file() and p.suffix in ['.log','.md','.json','.txt'] and not
p.is_relative_to(stage/'frozen'):
            content=p.read_text(encoding='utf-8')
            content=re.sub('/'+Users/['^\\s\\'\\'\\x27]+','<LOCAL_HOME>',content)
            p.write_text(content,encoding='utf-8')
            entries=[dict(path=str(p.relative_to(stage)),bytes=p.stat().st_size,sha256=digest(p)) for p in
stage.rglob('*') if p.is_file()]

(stage/'SHA256SUMS.json').write_text(json.dumps(dict(generated_at=datetime.now().astime
zone().isoformat(),files=entries),ensure_ascii=False,indent=2))
    with zipfile.ZipFile(FINAL/'支撑材料.zip','w',zipfile.ZIP_DEFLATED) as z:
        for p in stage.rglob('*'):
            if p.is_file():z.write(p,str(p.relative_to(stage)))
        size=(FINAL/'支撑材料.zip').stat().st_size
        if size>20_000_000:raise RuntimeError(f'Support archive exceeds 20MB: {size}')
        files=[dict(path=str(p.relative_to(FINAL)),bytes=p.stat().st_size,sha256=digest(p)) for p in
FINAL.rglob('*') if p.is_file() and p.name!='SHA256SUMS.json' and 'qa' not in
p.relative_to(FINAL).parts]

(FINAL/'SHA256SUMS.json').write_text(json.dumps(dict(generated_at=datetime.now().asti
mezone().isoformat(),files=files,archive_bytes=size),ensure_ascii=False,indent=2));print('Pac
kaged',size,'bytes',flush=True)
if __name__=='__main__':main()

```

scripts/paper_content.py

```

SHA256 a64bdd16d9e2f314521aa672578924eb4ec4381f2302c29d81eaff82547e7ec0
from pathlib import Path
from decimal import Decimal,ROUND_HALF_UP
import json,re
import pandas as pd,numpy as np
ROOT=Path(__file__).resolve().parents[1];F=ROOT/'outputs/c_final_v1/frozen'

def fmt(x,places=2):
    rounded=Decimal(str(float(x))).quantize(Decimal(1).scaleb(-
places),rounding=ROUND_HALF_UP)
    return format(abs(rounded) if rounded==0 else rounded,f',{places}f')

def table(headers,rows):
    return '\n'.join(['| '+'| '.join(headers)+'|','| '+'| '.join(['---']*len(headers))+| '+'|
|.join(map(str,r))+| ' for r in rows])

def emergency(z):
    groups=[];active=[]

```

```

def add():
    a=(active[0].position-1)*10;b=active[-1].position*10;groups.append([f'{a//60}:{a%60:02d}-{b//60}:{b%60:02d}',fmt(sum(x.emergency_kwh for x in active),4)])
    for x in z.itertuples():
        if x.emergency_kwh>=.00005:active.append(x)
        elif active:active.add();active=[]
    if active:active.add()
    return groups

def specified():
    lines=[];checks=[]
    for mode in ['q1','q2','q3','q4_2','q4_3']:
        d=pd.read_csv(F/f'{mode}_dispatch.csv');days=['2025-01-01'] if mode=='q1' else ['2025-03-20','2025-06-21','2025-09-23','2025-12-21']
        for day in days:
            z=d[d.date==day].sort_values('position');lines.append(f'##### {mode.upper()} {day}
指定结果')
            rows=[]
            for h in [10,12,14,16,18,20]:
                physical=f'{h:02d}:00-{h:02d}:10';selected=z[z.physical_interval==physical]
                if len(selected)!=1:raise ValueError('Nonunique physical interval')

x=selected.iloc[0];rows.append([physical,fmt(x.plan_00_grid_kwh,4),fmt(x.final_adjusted_grid_kwh,4)])

checks.append(dict(mode=mode,date=day,kind='ten_minute',key=physical,plan_grid=float(x.plan_00_grid_kwh),final_grid=float(x.final_adjusted_grid_kwh)))
    lines.append(table(['物理区间','0点计划量','最终承诺量'],rows))
    lines.append(table(['全天计划量','全天最终量','全天总购电费'],
[[fmt(z.plan_00_grid_kwh.sum(),4),fmt(z.final_adjusted_grid_kwh.sum(),4),fmt(z.final_relative_cost_yuan.sum())]]))
    rows=[]
    for b in range(6):
        zz=z[(z.position>b*24)&(z.position<=(b+1)*24)];span=f'{b*4}:00-{(b+1)*4}:00';rows.append([span,fmt(zz.charge_kwh.sum(),4),fmt(zz.discharge_kwh.sum(),4)])
checks.append(dict(mode=mode,date=day,kind='four_hour',key=span,charge=float(zz.charge_kwh.sum()),discharge=float(zz.discharge_kwh.sum())))
        lines.append(table(['物理区间','充电量','放电量'],rows));lines.append(table(['0点SOC','24点SOC'],[[fmt(z.soc_start_kwh.iloc[0],4),fmt(z.soc_end_kwh.iloc[-1],4)]]))
        if mode!='q1':lines.append(table(['紧急购电物理区间','紧急购电量'],emergency(z) or ['无正缺口','0.0000'])))

(F/'paper_specified_values.json').write_text(json.dumps(checks,ensure_ascii=False,indent=2))
    return '\n\n'.join(lines)

def generate():
    cfg=json.loads((F/'resolved_config.json').read_text());exp=pd.read_csv(F/'experiment_summa

```

```

ry.csv');official=pd.read_csv(F/'official_summary.csv');training=pd.read_csv(F.parent/'eviden
ce/
january_parameter_validation.csv');perfect=pd.read_csv(F/'perfect_information_summary.csv
');jan=pd.read_csv(F/'startup_summary.csv');ci=pd.read_csv(F/'comparison_stats.csv');settle=
pd.read_csv(F/'settlement_sensitivity.csv');solver=pd.read_csv(F/'solver_instance_sensitivity.
csv');audit=json.loads((F.parent/'audit/
independent_validation.json').read_text());q1=pd.read_csv(F/'q1_dispatch.csv')
def row(mode,scenario='S0',arm='joint_empirical'):
    z=exp[(exp['mode']==mode)&(exp.scenario==scenario)&(exp.model_arm==arm)]
    if len(z)!=1:raise ValueError((mode,scenario,arm,len(z)))
    return z.iloc[0]

values={'Q1_COST':fmt(q1.final_relative_cost_yuan.sum()),'Q2_COST':fmt(row('q2').final_r
relative_cost_yuan),'Q42_COST':fmt(row('q4_2').final_relative_cost_yuan),'Q3_DIFF':fmt(ro
w('q3','S3').final_relative_cost_yuan-
row('q3').final_relative_cost_yuan),'Q43_DIFF':fmt(row('q4_3','S3').final_relative_cost_yuan-
row('q4_3').final_relative_cost_yuan),'SELECTED_MODEL':{'raw_history':'原始历史联合
场景','causal_residual':'逐日前推因果残差场景'}[cfg['forecast']
['scenario_model']], 'SELECTED_MULT':str(cfg['rolling']
['selected_soft_penalty_multiplier'])}
baseline=float(np.sum(q1.settlement_price*np.maximum(q1.actual_load_kwh-
q1.actual_pv_kwh,0)))
values['Q1_TABLE']=table(['方案','全天购电量','全天费用'],[['储能
LP',fmt(q1.plan_00_grid_kwh.sum(),4),values['Q1_COST']],['无储
能',fmt(np.maximum(q1.actual_load_kwh-q1.actual_pv_kwh,0).sum(),4),fmt(baseline)]]))

tr=training.groupby(['scenario_model','penalty_multiplier']).final_relative_cost_yuan.sum();v
alues['TRAIN_TABLE']=table(['场景候选','惩罚倍数','四模式一月验证合计费用'],
[[sm,str(mult),fmt(cost)] for (sm,mult),cost in tr.items()])
rows=[]
for mode in ['q2','q4_2']:
    for arm,label in [('joint_empirical','联合场景风险采购'),('mean_forecast','均值预测'),
('no_storage','无储能')]:
r=row(mode,arm=arm);rows.append([mode,label,fmt(r.final_relative_cost_yuan),fmt(r.emerg
ency_kwh,4)])
p=perfect[perfect['mode']==mode].iloc[0];rows.append([mode,'完全信息事后下
界',fmt(p.purchase_cost_yuan),'0.0000'])
values['RISK_TABLE']=table(['模式','方案','正式期总费用','紧急购电量'],rows)
values['SETTLEMENT_TABLE']=table(['模式','最终净额','同链逐次','不退原价同路径'],
[[r['mode'],fmt(r.main_cost),fmt(r.same_chain_sequential_cost),fmt(r.no_refund_same_path_
cost)] for _,r in settle.iterrows()])
values['STARTUP_TABLE']=table(['模式','一月启动费用','一月末 SOC','二月初 SOC'],
[[r['mode'],fmt(r.cost_yuan),fmt(r.terminal_soc_kwh,4),fmt(pd.read_csv(F/f'{r["mode"]}
_dispatch.csv').soc_start_kwh.iloc[0],4)] for _,r in jan.iterrows()])

```

```

values['OFFICIAL_TABLE']=table(['模式','计划费','调整净费','紧急费','总费用'],
[[r['mode'],fmt(r.plan_cost_yuan),fmt(r.adjustment_cost_yuan),fmt(r.emergency_cost_yuan),f
mt(r.final_relative_cost_yuan)] for _,r in official.iterrows()])
values['SCHEDULE_TABLE']=table(['模式','时点策略','主费用','逐次费用','紧急购电
量'],
[[mode,sc,fmt(row(mode,sc).final_relative_cost_yuan),fmt(row(mode,sc).sequential_cost_yua
n),fmt(row(mode,sc).emergency_kwh,4)] for mode in ['q3','q4_3'] for sc in
['S0','S1','S2','S3']])
statements=[]
for mode in ['q3','q4_3']:
    costs={sc:float(row(mode,sc).final_relative_cost_yuan) for sc in
['S0','S1','S2','S3']};winner=min(costs,key=costs.get);diff=costs['S3']-
costs['S0'];statements.append(f'{mode.upper()}在本样本共同日前合同下，主费用最低为
{winner}，费用{fmt(costs[winner])}元；S3 相对 S0'+('增加' if diff>=0 else '减少')
+f{fmt(abs(diff))}元。该差额来自本算法在更新信息下选择的后续采购和储能，不能推
广成信息本身必然有害或必然有益。')

```

```

values['SCHEDULE_NARRATIVE']='\n\n'.join(statements);values['CI_TABLE']=table(['模
式','比较','配对差额','95%区间下端','95%区间上端'],
[[r['mode'],r.comparison,fmt(r.difference_yuan),fmt(r.ci_low),fmt(r.ci_high)] for _,r in
ci.iterrows()])

```

values['SOLVER_NARRATIVE']=f'共完成{len(solver)}次固定实例复求。相对 HiGHS 自动法的目标最大绝对差为{solver.objective_difference_vs_highs.abs().max():.6g}元，窗口采购量最大 L1 差为{solver.grid_l1_difference.max():.6g} kWh；最大 SOC 递推残差为{solver.soc_residual.max():.3e} kWh。目标接近而动作不一致时应视为并列解敏感性，全年策略排序仍需额外完整回放才能确认。'

failures=[c for c in audit['checks'] if c['status']!=
='PASS'];values['AUDIT_NARRATIVE']=f'本轮独立检查共{len(audit["checks"])}项，失败{len(failures)}项。检查覆盖逐时段原数据匹配、物理区间、设备边界、真实缺口及费用、正式承诺连续性、共同日前合同和一月衔接。测试还覆盖未来观测隔离、分段状态不重置、不同充放电效率和年末边界。该结果是所列检查范围内的证据，不是模型最优性或外推有效性的证明。'

if failures:raise RuntimeError('Refusing to publish paper over failed independent checks')

```

values['SPECIFIED_TABLES']=specified();values['REPRODUCTION_TABLE']=table(['证
据','位置'],[['正式明细','frozen/q*_dispatch.csv（解包后）'],['真实承诺链','frozen/
decision_versions.csv（解包后）'],['训练与解析配置','evidence/
january_parameter_validation.csv 及 frozen/resolved_config.json'],['独立复核','audit/
independent_validation.json'],['源码与输入校验','evidence/source_provenance.json'],['可执
行入口','run_pipeline.sh'],['数学约束测试','tests/test_closed_loop.py']])
text=(ROOT/'paper/main.md').read_text()
for key,val in values.items():text=text.replace('{{'+key+'}}',val)
if re.search(r'\{\{[^}]+\}\}',text):raise ValueError('Unresolved result macro')
ai=""# AI 工具使用详情

```

使用范围与责任边界

本轮在 OpenAI Codex 任务中使用基于 GPT-6 的智能体，辅助独立复核、程序修订、实验组织、数值审计及文档生成。此前会话的具体模型版本无法由现存材料完整验证，未猜填。用户于本轮明确授权修复，并要求使用真实数据、避免未来信息泄漏、遵守数学模型约束。

输入与操作

输入包括原题 PDF、附件 1 至 5、V3 源码与交付文件、两份历史审查报告。本轮保留修订前快照，停止旧模型后台重算后建立新的正式运行。核心改动包括因果联合场景、共享决策约束、真实承诺逐次计费、共同日前计划比较、可核验一月启动、物理时间抽取及独立交付校验。

数据与预测

原始观测不作伪造或未来回填。残差场景由已观测历史构造，属于明确标记的预测分布；移动块重采样用于统计区间，不当作新观测数据。没有历史时不使用当天实际曲线预测当天。一月训练选择完成后评价二月至十二月。

可核查记录

运行日志位于 logs；配置为 frozen/resolved_config.json；训练表为 evidence/january_parameter_validation.csv；一月和正式期轨迹、各策略承诺均位于 frozen；独立复算位于 audit；源码与原始输入哈希在 evidence/source_provenance.json；数学约束和反事实测试见 tests/test_closed_loop.py。

尚需人工审阅的内容

退费和交付时段价格是题意解释，不能冒充额外给定的市场规则。经验场景与有限窗口不保证覆盖所有极端情形。团队需人工审阅假设、策略选择、公式、程序、引文与最终提交要求。本文件只登记可验证的本轮操作，不声称此前全部记录已恢复，也不声称已完成团队签核。

'''

return text,ai

scripts/rebuild_control_details.py

```
SHA256 1c634a0c2b71c370ffbd2c7186dda4abea47039e8c612d7b6a0f7da8562afd90
""""Recover full control trajectories for a run initially saved with daily controls."""
from pathlib import Path
import sys,json
```

```

import pandas as pd
ROOT=Path(__file__).resolve().parents[1];sys.path.insert(0,str(ROOT))
from src.model.config import load_config
from src.model.data import load_data
from src.model.experiments import write_csv
from src.model.rolling import simulate,RunSpec
cfg=load_config('config/
final.yaml');F=Path(cfg['_output_root'])/'frozen';cfg.update(json.loads((F/'resolved_config.json').read_text()));data=load_data(cfg)
for mode in ['q2','q4_2']:
    for arm,point,nostore in [('mean_forecast',True,False),('no_storage',False,True)]:
        initial=float(pd.read_csv(F/f'{mode}
_january_dispatch.csv').soc_end_kwh.iloc[-1]);r=simulate(data,cfg,RunSpec(mode,'S0',arm,collect_detail=False),initial_soc=initial,point=point,no_storage=nostore)
        old=pd.read_csv(F/f'{mode}_{arm}
_daily.csv');expected=float(old.final_relative_cost_yuan.sum());actual=r['summary']
['final_relative_cost_yuan']
        if abs(expected-actual)>1e-6:raise AssertionError((mode,arm,expected,actual))
        write_csv(F/f'{mode}_{arm}_dispatch.csv',r['dispatch']);write_csv(F/f'{mode}_{arm}
_solver.csv',r['solve_stats']);print('Recovered verified control',mode,arm,actual,flush=True)

```

scripts/research_content.py

```

SHA256 ab5ee95b15d1f1ee6d4b00eeced9c56b701d930476c6368dd885a9feb81c85d4
""""Inject measured diagnostics and a byte-verifiable complete source appendix.""""
from pathlib import Path
import json, hashlib
import numpy as np
import pandas as pd
from docx import Document
from docx.shared import Pt
from docx.oxml.ns import qn
from paper_content import table,fmt
ROOT=Path(__file__).resolve().parents[1]
FINAL=ROOT/'outputs/c_final_v1'
R=FINAL/'research'

def source_files():
    files=[]
    for folder in ['src','scripts','tests']:
        files.extend(p for p in (ROOT/folder).rglob('*') if p.is_file() and p.suffix in ['.py','.mjs']
and '__pycache__' not in p.parts)
    files.extend([ROOT/'run_pipeline.sh',ROOT/'requirements.txt',ROOT/'config/final.yaml'])
    return sorted(files,key=lambda p:str(p.relative_to(ROOT)))

def append_sources(path):
    doc=Document(path)
    h=doc.add_paragraph('附录 C 完整源程序与运行配置',style='Heading
2');h.paragraph_format.page_break_before=True

```

doc.add_paragraph('以下逐文件收录当前工程自编的全部 Python、JavaScript、Shell 程序和配置。第三方依赖按 requirements.txt 安装；工作簿导出还需 Node 与 @oai/artifact-tool，文档转换需配套 LibreOffice 运行环境。程序以支撑材料内的原始文件运行，版面换行不改变源文件。独立数值复验仅依赖 Python，不要求文档转换环境。')

```
records=[]
for p in source_files():
    name=str(p.relative_to(ROOT));raw=p.read_bytes();content=raw.decode('utf-8')
    doc.add_paragraph(name,style='Heading 3')
    note=doc.add_paragraph('SHA256 '+hashlib.sha256(raw).hexdigest())
    note.paragraph_format.space_after=Pt(2)
    for run in note.runs:run.font.size=Pt(7)
    for line in content.splitlines():
```

```
para=doc.add_paragraph();para.paragraph_format.space_after=Pt(0);para.paragraph_format.space_before=Pt(0)
```

```
    para.paragraph_format.line_spacing=1.0;para.paragraph_format.keep_together=False
    run=para.add_run(line if line else ' ');run.font.name='Consolas';run.font.size=Pt(7.5)
    run._element.get_or_add_rPr().rFonts.set(qn('w: eastAsia'),'Arial Unicode MS')
```

```
records.append({'path':name,'sha256':hashlib.sha256(raw).hexdigest(),'lines':len(content.splitlines())})
```

```
    doc.save(path)
```

```
(FINAL/'evidence/source_appendix_manifest.json').write_text(json.dumps(records,ensure_ascii=False,indent=2))
```

```
def enrich(text,ai):
```

```
annual=pd.read_csv(R'/annual_solver_comparison.csv');tail=pd.read_csv(R'/tail_risk.csv');err=pd.read_csv(R'/prediction_error.csv');blocks=pd.read_csv(R'/block_sensitivity.csv');months=pd.read_csv(R'/monthly_mechanisms.csv')
```

```
start=text.index('针对负载');end=text.index("\n\n 关键词",start)
```

text=text[:start]+""针对微网负载、光伏与电价不确定性，本文以十分钟物理区间为统一尺度，将储能调度、非对称缺口损失与购电承诺建立为可实施的联合优化模型。数据仅来自题目附件，所有预测输入满足观测完成和预报发布时间约束。

问题一采用含效率、功率与储电量边界的线性规划，得到日购电费 35,126.95 元。问题二和问题四的日前模型把五倍价紧急采购直接纳入经验场景目标；历史负载、光伏和电价按同日联合取样，候选场景与终端惩罚只在一月选定。二月至十二月费用分别为 14,794,949.72 元和 15,623,134.83 元，相对均值预测控制分别减少 1,662,260.45 元和 1,668,740.06 元。

问题三及其波动电价扩展采用 24 小时滚动窗口。固定共同日前承诺后，0、6、12、18 点更新的 S3 方案相对仅 0 点更新的 S0 方案分别减少 123,506.61 元和 155,342.73 元；同

时报告 S2、不同求解器和计费解释的结果，不把单一路径排序推广为信息价值定理。对真实承诺链逐笔结算，逐次调整相对最终净额分别多 96,160.53 元和 98,826.62 元。

为避免无历史首日的信息泄漏，一月采用明示的保守启动合同，并从 4464 个实际执行时段衔接二月 SOC。以独立物理复算、共同基线、全年替代求解器回放、月度机制分析及块长度敏感性检验形成证据链。结果支持在当前数据与计费解释下显式处理尾部缺口损失和日内信息更新；单个有限窗口的最优解不等于全年策略最优，经验尾部统计也不能保证未知极端场景下的供电表现。"""text[end:]

text=text.replace('微网由小区负载、光伏电源、储能和外部电网构成。','题目[1]中的微网由小区负载、光伏电源、储能和外部电网构成。',1)

intro="""

1.1 方法依据与本题贡献

线性规划提供物理可行域和费用目标的求解基础[2]；样本平均近似以有限场景表示不确定量[3]；滚动控制在新观测到达后更新有限窗口决策[4]。这些是已有方法，本文不把算法名称本身列为创新。本题的针对性工作是：将五倍缺口代价、联合历史误差和储能跨时段耦合放入同一规划；以共同日前合同隔离日内更新频次；记录每次承诺变化，使净额与逐次结算可在同一执行路径上核验。

本文分别回答三个问题：给定信息下决策是否可实施，计费与物理结果是否可复算，以及相对可行基线是否改善经济结果。前两者不能替代第三者；对有限窗口的求解最优性也不作全年策略最优性解释。

1.2 技术路线

原始附件先按日期和物理区间对齐并统一量纲；每次决策截取已完成历史和已发布预报，生成联合场景；在 SOC 和承诺约束下求解，仅执行至下一允许更新时点；区间结束后记录实际缺口、费用和下一状态。训练期选择、正式回放、对照实验及文稿生成分别留存输出，避免把预测样本当成实测数据。本文使用历史数据回放，因此因果信息检查证明的是逐次决策的信息边界，不宣称真实部署过该系统。

"""

text=text.replace("\n## 2 假设与符号',intro+"\n## 2 假设与符号')

proof="""

上述互斥处理具有支配关系。若 $z \geq 0$ ，则 $C' = C - D/(\eta c \eta d)$ 、 $D' = 0$ ，减少的母线需求为 $D(1/(\eta c \eta d) - 1) \geq 0$ ；若 $z < 0$ ，则 $C' = 0$ 、 $D' = D - \eta c \eta d C$ ，减少的母线需求为 $C(1 - \eta c \eta d) \geq 0$ 。两种情形均有 $C' \leq C$ 、 $D' \leq D$ ，故额定功率上界保留，SOC 轨迹不变，采购需求不增。由于本题允许不用富余且不设最小充放功率，此变换给出不劣的可行互斥解。若另加禁止弃电、负电价收益或最小运行时间等条件，需要重新论证，不能直接沿用该结论。

"""

text=text.replace("\n 问题一以 $Q=0$ ',proof+"\n 问题一以 $Q=0$ ')

text=text.replace('每一历史误差也必须相对当时仅用更早日期生成的预测计算，不能用当前的拟合曲线回填过去。','每一历史误差也必须相对当时仅用更早日期生成的预测计算，不能用当前的拟合曲线回填过去。这遵循逐次向前验证的时间切分原则[5]。')

text=text.replace('本文使用完整 LP 进行正式调度。','本文使用完整 LP 进行正式调度。更具体地，单期目标对采购 q 的导数为 $p-5p \cdot \Pr(N>q)$ ，令其为零得到 0.8 分位点；离散经验分布取对应分位区间即可。若价格和缺口相关，期望应对联合分布计算，不能先把价格与需求独立平均。')

text=text.replace('\n## 6 连续启动与因果检验','

5.3 信息价值与实际收益的区别

如果扩大的信息集允许完全忽略新增预报，且优化覆盖相同状态合同下的全部可行策略，则较丰富信息的最优期望费用不会更高：旧策略仍属于新策略集合。这是可行策略集合包含关系的结果，而不是某次实现路径上的费用保证。本文采用有限窗口、有限经验场景和数值求解，因此既不保证每次更新降低实际费用，也不以 S2 略低于 S3 否定信息的一般价值。是否引入更新应依据受控费用、缺口和实现负担共同判断。

6 连续启动与因果检验")

text=text.replace('逐日配对差使用 7 天移动块','逐日配对差参考块自助法处理局部时间相关的思路[6]，使用 7 天移动块')

text=text.replace('分别用 HiGHS 自动、对偶单纯形和内点法复求。','分别用 HiGHS 自动、对偶单纯形和内点法复求。HiGHS 求解器及其对偶单纯形实现的来源见文献[7]。')

text=text.replace('全年策略排序仍需额外完整回放才能确认。','其全年影响进一步通过第 7.3 节的连续回放检验。')

rows=[]

for (mode,method),g in annual.groupby(['mode','method'],sort=False):

a=g.set_index('scenario').cost;rows.append([mode,method,fmt(a['S2']),fmt(a['S3']),fmt(a['S3']-a['S2'])])

diagnostics='\n\n### 7.1 缺口与尾部成本\n\n'

diagnostics+='紧急购电补足实际缺口，以下指标衡量额外采购暴露，不等同于停电概率。每日费用最高的 17 天定义为本样本上尾约 5%，其均值仅作描述性尾部指标；本文没有把目标函数改为 CVaR。'\n\n'

diagnostics+=table(['模式','紧急时段占比','最高日费/元','上尾 17 日均费/元','紧急费占比'],

[[r['mode'],fmt(r.emergency_interval_fraction*100)+'%',fmt(r.max_daily_cost),fmt(r.tail5_mean_cost),fmt(r.emergency_cost_share*100)+'%'] for _,r in tail.iterrows()])

diagnostics+='\n\n### 7.2 预测误差与月度机制\n\n预测误差针对实际执行时段使用的场景均值，Q3 包含多个发布时刻，不能将其与 Q2 的差异单独归因为预测模型。

MAE=mean(|预测-实测|)，RMSE=sqrt(mean((预测-实测)²))；夜间光伏和接近零的量不适合用 MAPE。原始观测未被改动。'\n\n'

```

diagnostics+=table(['模式','变量','MAE','RMSE','偏差'],[[r['mode']],{'load':'负载/
kWh','pv':'光伏/kWh','price':'电价/元每 kWh'}
[r.variable],fmt(r.mae,4),fmt(r.rmse,4),fmt(r.bias,4)] for _,r in err.iterrows() if r.variable!
='price' or r['mode'].startswith('q4'))
diagnostics+='\n\n 月度统计见图 5：成本波动同时取决于负荷、电价和充放电轨迹，
不能仅由一个误差指标解释。\\n\\n[[FIGURE:monthly_risk.png|正式期月度费用与紧急采
购占比]]\\n\n'
for mode in ['q2','q4_2','q3','q4_3']:

z=months[months['mode']==mode];a=z.loc[z.cost.idxmax()];b=z.loc[(z.emergency_cost/
z.cost).idxmax()]
diagnostics+=f'{mode}月总费最高为{a.month}，{fmt(a.cost)}元；紧急费用占比最
高为{b.month}，{fmt(b.emergency_cost/b.cost*100)}%。\\n\n'
diagnostics+= '### 7.3 全年替代求解器回放\\n\n 在补充回放前固定协议，以相同一月
末 SOC、同一组 S0 参考日前承诺和一月已选定参数，分别对两种电价的 S2、S3 运行
HiGHS 内点法。每组连续回放 334 天，不在每日重置状态；保留每段动作并从原始附
件重算费用。四组结果全部报告，不据本实验更换正式方案。\\n\n'+table(['模式','求解方
法','S2 费用/元','S3 费用/元','S3 减 S2/元'],rows)
for mode in ['q3','q4_3']:

g=annual[annual['mode']==mode].pivot(index='method',columns='scenario',values='cost');delt
a=g.S3-g.S2
diagnostics+=f'\n\n{mode} 中，默认方法的 S3 减 S2 为{fmt(delta["highs"])}元，内点
法为{fmt(delta["highs-ipm"])}元。'+('两种方法的排序方向一致，但差距大小随数值方法
变化。' if np.sign(delta['highs'])==np.sign(delta['highs-ipm']) else '两种方法出现排序反
转，不能保留 S2 严格优于 S3 的稳健性结论。')
diagnostics+= '\n\n### 7.4 时间相关与块长度敏感性\\n\n 在同一 334 天费用差上，预先
列出 3、7、14、28 天四种移动块长度，每种重采样 2000 次，使用固定随机种子
20260913。表中列 S3 减 S2，正数表示 S3 较贵；完整 S3 减 S0 区间也随支撑材料交
付。年份存在季节变化，块重采样不保证严格平稳条件，区间用于判断样本内结论对
重采样尺度的敏感性，不解释为下一年收益保证。\\n\n'
z=blocks[blocks.comparison=='S3-S2'];diagnostics+=table(['模式','块长/天','实际差
额/元','95%下端/元','95%上端/元'],
[[r['mode'],int(r.block_days),fmt(r.difference),fmt(r.ci_low),fmt(r.ci_high)] for _,r in
z.iterrows()])
diagnostics+= '\n\n 四种块长度下，S3 相对 S0 的费用差区间均低于零，S3 相对 S2 的
区间均跨零。日内更新具有本样本经济价值，但尚不足以认定增加 18 点更新严格改善
经济性。将更新计算或交易操作成本加入后，还需重新计算其盈亏阈值。\\n\n'
text=text.replace('\n## 8 证据链与结论',diagnostics+'\n## 8 证据链与结论')
text=text.replace('文档构建从持久化 paper/main.md 读取内容，全部结果宏由当前运行
值替换；生成后再核对指定区间和四小时聚合。','文档构建从持久化正文与补充诊断生
成器读取内容，全部表格由对应冻结值生成；生成后再核对指定区间和四小时聚合。')
a=text.index('## 参考材料');b=text.index('## 附录',a)

```

```
text=text[:a]+""## 参考文献
```

[1] 全国大学生数学建模竞赛组委会. 2026 年高教社杯全国大学生数学建模竞赛 C 题: 微网与外部电网电力调控策略及附件[Z]. 2026.

[2] BOYD S, VANDENBERGHE L. Convex Optimization[M]. Cambridge: Cambridge University Press, 2004. <https://web.stanford.edu/~boyd/cvxbook/>.

[3] SHAPIRO A, DENTCHEVA D, RUSZCZYNSKI A. Lectures on Stochastic Programming: Modeling and Theory[M]. 3rd ed. Philadelphia: SIAM, 2021. DOI:10.1137/1.9781611976595.

[4] RAWLINGS J B, MAYNE D Q, DIEHL M M. Model Predictive Control: Theory, Computation, and Design[M]. 2nd ed. Santa Barbara: Nob Hill Publishing, 2017. <https://sites.engineering.ucsb.edu/~jbrow/mpc/>.

[5] HYNDMAN R J, ATHANASOPOULOS G. Forecasting: Principles and Practice[M/OL]. 3rd ed. Melbourne: OTexts, 2021[2026-09-12]. <https://otexts.com/fpp3/tscv.html>.

[6] KUNSCH H R. The jackknife and the bootstrap for general stationary observations[J]. The Annals of Statistics, 1989, 17(3): 1217-1241. DOI:10.1214/aos/1176347265.

[7] HUANGFU Q, HALL J A J. Parallelizing the dual revised simplex method[J]. Mathematical Programming Computation, 2018, 10(1): 119-142. DOI:10.1007/s12532-017-0130-5.

```
""+text[b:]
```

```
files=source_files()
```

```
text+='\n\n 支撑材料的必要文件按功能列示如下：每个源码文件在附录 C 逐一给出，完整文件哈希清单见包内 SHA256SUMS.json.\n\n'+table(['目录或文件','内容'],[['src/model','全部模型与数据加载程序'],['src/audit','独立逐时段及工作簿审核程序'],['scripts','运行、补充实验、图表、文稿、打包程序'],['tests','数学约束和未来观测扰动测试'],['config/final.yaml 与 requirements.txt','配置及依赖'],['frozen 与 workbooks','原正式结果及五份指定工作簿'],['research','补充全年回放、尾部、月度、误差与统计诊断'],['evidence 与 audit','来源、源码附录清单及复验记录'],['charts 与 logs','图像及运行日志'],['AI 工具使用详情.pdf','真实使用范围及仍需人工核验事项']])
```

```
ai+=""
```

```
## 本轮继续修订的真实提示与采用情况
```

用户要求“我同意 我们要解决问题 不用假的数据 不用未来函数 建模不做成通俗化建模 注意各种数学建模的约束 开始做”，后续要求“能继续做吗 继续做呗 给过你链路 记得我要高分啊”。本轮据此恢复已交付快照，补做两种电价下 S2 和 S3 的全年

内点法回放，独立复算其物理与费用指标；补充月度、尾部、预测误差及块长度诊断，修订论文和完整源码附录。

提供的两份自查 DOCX 及速成 PDF 仅作为参考材料。未采用其中示例数值、固定评分权重和获奖保证；适用的修改方向包括有数据支持的检验、文献对应和完整代码。学术文献由作者或出版社公开页面核对，文稿不声称这些既有方法由本队原创。

人工核验状态

现有可核实人工行为为用户提出约束、审阅对话结论并授权修订和仓库同步。核心代码修订、实验组织和本轮正文补充由 AI 实际参与完成，未降格描述为仅改错别字。尚无证据表明参赛队员已经逐项独立复核模型推导、程序和结果，因此这些项登记为待核验；不能由 AI 自检代替人工主导，也不能以本说明冒充团队完成签核。此前会话完整提示及采用修改记录存在缺口，参赛队需据真实记录补齐。

'''

return text,ai

scripts/run_model.py

SHA256 d18883b0bce2bcd10a73e1126443f420fca7b1d31247a3cb9ccaaab0b741b30e

#!/usr/bin/env python3

from src.model.cli import main

if __name__ == "__main__":

raise SystemExit(main())

scripts/strengthen_evidence.py

SHA256 7b64dcd4a9fc3bd8b342883f2739275add73444bd1f00846f968811397885d33

"""Prespecified diagnostic replays; never select the submitted policy on test data."""

from pathlib import Path

import sys, json, hashlib, time

import numpy as np

import pandas as pd

ROOT=Path(__file__).resolve().parents[1]

sys.path.insert(0,str(ROOT))

from src.model.config import load_config

from src.model.data import load_data

from src.model.rolling import RunSpec, simulate

F=ROOT/'outputs/c_final_v1/frozen'

R=F.parent/'research'

R.mkdir(exist_ok=True)

def main():

cfg=load_config('config/final.yaml')

selected=json.loads((F/'resolved_config.json').read_text())

```

for k in ('study','storage','forecast','rolling','cost','experiments'):cfg[k]=selected[k]
data=load_data(cfg)
protocol={'role':'post_selection_diagnostic; no policy replacement',
'methods':['highs (existing frozen run)','highs-ipm'],
'modes':['q3','q4_3'],'schedules':['S2','S3'],
'period':['2025-02-01','2025-12-31'],'initial_soc':6000,
'common_plan':'existing S0 reference plan, fixed across all methods and schedules',
'selection':'January-selected configuration is frozen; all four IPM runs reported',
'recorded_at':time.strftime('%Y-%m-%dT%H:%M:%S%z'),
'source_hashes':{str(p.relative_to(ROOT)):hashlib.sha256(p.read_bytes()).hexdigest() for
p in (ROOT/'src/model').glob('*.py')}}
protocol_path=R/'protocol.json'
if not
protocol_path.exists():protocol_path.write_text(json.dumps(protocol,ensure_ascii=False,inde
nt=2))
summaries=[];checks=[]
for mode in ['q3','q4_3']:
b=pd.read_csv(F/f{mode}_S0_dispatch.csv')
plans={day:z.plan_00_grid_kwh.to_numpy() for day,z in b.groupby('date',sort=False)}
for sc in ['S2','S3']:
existing=pd.read_csv(F/f{mode}_{sc}_daily.csv')

summaries.append({'mode':mode,'scenario':sc,'method':'highs','cost':float(existing.final_relati
ve_cost_yuan.sum()),'emergency_kwh':float(existing.emergency_kwh.sum())})
path=R/f{mode}_{sc}_ipm.csv.xz'
if not path.exists():
print('START',mode,sc,'full-year IPM',flush=True)
result=simulate(data,cfg,RunSpec(mode,sc,method='highs-
ipm',collect_detail=False),initial_soc=6000,common_plans=plans)
d=pd.DataFrame(result['dispatch'])

cols=['date','position','final_adjusted_grid_kwh','charge_kwh','discharge_kwh','soc_start_kwh',
'soc_end_kwh','emergency_kwh','final_relative_cost_yuan']
d[cols].to_csv(path,index=False,compression={'method':'xz','preset':9})
pd.DataFrame(result['daily']).to_csv(R/f{mode}_{sc}_ipm_daily.csv',index=False)
pd.DataFrame(result['solve_stats']).to_csv(R/f{mode}_{sc}
_ipm_solvers.csv',index=False)
d=pd.read_csv(path)
# Independently reconstruct physical constraints and cost from raw arrays.
ids=np.array([data.date_index[___import__('datetime').date.fromisoformat(x)] for x in
d.date])
pos=d.position.to_numpy()-1
L=data.load_kw[ids,pos]/6;V=data.pv_kw[ids,pos]/6
price=data.realtime_price[ids,pos] if mode.startswith('q4') else data.fixed_price[pos]
B=np.concatenate(list(plans.values()));A=d.final_adjusted_grid_kwh.to_numpy()

c=d.charge_kwh.to_numpy();dd=d.discharge_kwh.to_numpy();s=d.soc_start_kwh.to_numpy(
);e=d.soc_end_kwh.to_numpy()
q=np.maximum(L+c-V-dd-A,0)
cost=price*(B+(A-B)+.5*np.abs(A-B)+5*q)

```

```

        metrics={'rows':len(d),'cost_max_error':float(abs(cost-
d.final_relative_cost_yuan).max()),'emergency_max_error':float(abs(q-
d.emergency_kwh).max()),
        'soc_residual':float(abs(e-s-.9*c+dd/.9).max()),'continuity_residual':float(abs(s[1:]-
e[:-1]).max()),
        'soc_min':float(min(s.min(),e.min())), 'soc_max':float(max(s.max(),e.max())),

'power_max':float(max(c.max(),dd.max())), 'simultaneous_max':float(np.minimum(c,dd).max(
)),
        'start_soc':float(s[0]), 'end_soc':float(e[-1])}
        ok=(len(d)==48096 and all(metrics[k]<1e-7 for k in
['cost_max_error','emergency_max_error','soc_residual','continuity_residual','simultaneous_m
ax']))
        and metrics['soc_min']>=1200-1e-7 and metrics['soc_max']<=10800+1e-7 and
metrics['power_max']<=5000/6+1e-7
        and abs(s[0]-6000)<1e-7 and abs(e[-1]-6000)<1e-7
        assert ok,metrics
        checks.append({'mode':mode,'scenario':sc,'status':'PASS',**metrics})
        summaries.append({'mode':mode,'scenario':sc,'method':'highs-
ipm','cost':float(cost.sum()),'emergency_kwh':float(q.sum())})
        print('DONE',mode,sc,'cost',cost.sum(),flush=True)
        pd.DataFrame(summaries).to_csv(R'/annual_solver_comparison.csv',index=False)
        (R'/annual_independent_checks.json').write_text(json.dumps(checks,indent=2))
        descriptive()

```

```
def descriptive():
```

```

    monthly=[];risks=[];calibration=[];bootstrap=[]
    for mode in ['q2','q4_2','q3','q4_3']:
        d=pd.read_csv(F/f'{mode}_dispatch.csv')
        for month,z in d.groupby(d.date.str[:7]):

```

```

        monthly.append({'mode':mode,'month':month,'cost':z.final_relative_cost_yuan.sum(),'emerge
ncy_cost':z.emergency_cost_yuan.sum(),'emergency_kwh':z.emergency_kwh.sum(),
        'unused_kwh':z.unused_surplus_kwh.sum(),'throughput_kwh':
(z.charge_kwh+z.discharge_kwh).sum(),'low_soc_fraction':float((z.soc_start_kwh<1200.001)
.mean())})

```

```

daily=d.groupby('date').agg(cost=('final_relative_cost_yuan','sum'),q=('emergency_kwh','sum'
),ecost=('emergency_cost_yuan','sum'))
costs=daily.cost.to_numpy();n_tail=max(1,int(np.ceil(.05*len(costs))))

```

```

risks.append({'mode':mode,'emergency_interval_fraction':float((d.emergency_kwh>1e-7).me
an()),'emergency_days':int((daily.q>1e-7).sum()),'max_daily_cost':float(costs.max()),'p95_dai
ly_cost':float(np.quantile(costs,.95)),
        'tail5_mean_cost':float(np.sort(costs)[-
n_tail:].mean()),'emergency_cost_share':float(d.emergency_cost_yuan.sum()/
d.final_relative_cost_yuan.sum())})
        for var,a,b in [('load','forecast_load_kwh','actual_load_kwh'),
('pv','forecast_pv_kwh','actual_pv_kwh'),('price','forecast_price','settlement_price')]:

```

```

err=d[a]-
d[b];calibration.append({'mode':mode,'variable':var,'mae':float(abs(err).mean()),'rmse':float(n
p.sqrt((err**2).mean())),'bias':float(err.mean()),'scope':'executed-interval scenario mean;
mixed available decision times'})
rng=np.random.default_rng(20260913)
for mode in ['q3','q4_3']:
    s0=pd.read_csv(F/f'{mode}_S0_daily.csv').final_relative_cost_yuan.to_numpy()
    s2=pd.read_csv(F/f'{mode}_S2_daily.csv').final_relative_cost_yuan.to_numpy()
    s3=pd.read_csv(F/f'{mode}_S3_daily.csv').final_relative_cost_yuan.to_numpy()
    for name,delta in [('S3-S0',s3-s0),('S3-S2',s3-s2)]:
        for block in [3,7,14,28]:
            starts=rng.integers(0,len(delta)-block+1,size=(2000,(len(delta)+block-1)//block))
            ix=(starts[:,None]+np.arange(block)).reshape(2000,-1)[:,:len(delta)]
            sampled=delta[ix].sum(axis=1)

bootstrap.append({'mode':mode,'comparison':name,'block_days':block,'replicates':2000,'differ
ence':float(delta.sum()),'ci_low':float(np.quantile(sampled,.025)), 'ci_high':float(np.quantile(sa
mpled,.975))})
    for name,rows in [('monthly_mechanisms',monthly),('tail_risk',risks),
('prediction_error',calibration),('block_sensitivity',bootstrap)]:pd.DataFrame(rows).to_csv(R/
f'{name}.csv',index=False)
import lzma,shutil
for p in R.glob('*_ipm.csv.xz'):
    with lzma.open(p,'rb') as src,(F/('research_'+p.name.removesuffix('.csv.xz')
+'_dispatch.csv')).open('wb') as dst:shutil.copyfileobj(src,dst)
print('All diagnostic outputs completed',flush=True)

if __name__=='__main__':main()

```

scripts/unpack_frozen.py

```

SHA256 5f709e3aa4097da14c354d5ef1d8f82226c52d41d2f7633ab7ff53b198d3ce29
from pathlib import Path
import lzma,shutil, csv,json
ROOT=Path(__file__).resolve().parents[1]
cfg=json.loads((ROOT/'config/final.yaml').read_text())
FINAL=ROOT/cfg['output_root'];F=FINAL/'frozen';F.mkdir(parents=True,exist_ok=True)
SOURCE=ROOT/'frozen'
if not SOURCE.exists():SOURCE=F
if (SOURCE/'csv_manifest.json').exists():
    from freeze_codec import unpack
    unpack(SOURCE,F)
for p in SOURCE.iterdir():
    if p.name.endswith('.csv.xz'):
        with lzma.open(p,'rb') as src,(F/p.with_suffix('').name).open('wb') as
dst:shutil.copyfileobj(src,dst)
    elif p.is_file() and p.parent!=F and p.name not in
['csv_manifest.json','columns.tar.xz']:shutil.copy2(p,F/p.name)
for folder in ['workbooks','evidence','charts','research']:
    if (ROOT/folder).exists():shutil.copytree(ROOT/folder,FINAL/folder,dirs_exist_ok=True)

```

```

for mode in ['q3','q4_3']:
    shutil.copy2(F/f{mode}_dispatch.csv',F/f{mode}_S3_dispatch.csv')
    with (F/'decision_versions.csv').open(encoding='utf-8-sig',newline='') as src,(F/f{mode}_S3_decisions.csv').open('w',encoding='utf-8-sig',newline='') as dst:

r=csv.DictReader(src);w=csv.DictWriter(dst,fieldnames=r.fieldnames);w.writeheader();w.wri
terows(row for row in r if row['mode']==mode)
    shutil.copy2(F/f{mode}_S3_decisions.csv',F/f{mode}_decision_versions.csv')
print('Frozen CSVs restored')
for p in F.glob('research_*_dispatch.csv'):
    name=p.name.removeprefix('research_').removesuffix('_dispatch.csv')+'.csv.xz'
    with p.open('rb') as src,lzma.open(FINAL/'research'/name,'wb') as
dst:shutil.copyfileobj(src,dst)

```

scripts/write_improvement_report.py

```

SHA256 18201ad61edd6c3d0d5064c58762b46e130c266d76e2618d694eab6a389ce618
from pathlib import Path
import json,hashlib
import pandas as pd
ROOT=Path(__file__).resolve().parents[1];F=ROOT/'outputs/c_final_v1';R=F/'research'

def main():
    v=json.loads((F/'final_validation_v3.json').read_text())
    a=pd.read_csv(R/'annual_solver_comparison.csv');b=pd.read_csv(R/'block_sensitivity.csv')
    text="## V3 研究论证与提交格式修订报告

```

本次保留原正式模型及五份工作簿，补充独立诊断并修订论文。新增实验是在已查看正式结果后明确登记的诊断，不作为新的盲测或调参集；不因替代求解器的费用更低而替换正式结果。

实际新增工作

1. 两种电价下 S2、S3 各完成一次连续 334 天内点法回放，合计 192384 个十分钟执行时段。共同日前合同、初始 SOC 及模型参数固定；新增每段动作从原始附件独立复核供需、费用、SOC、功率和端点。
2. 计算四种模式的月度分解、预测误差、紧急时段比例、最高日费和上尾 17 日均费；以 3/7/14/28 天块长度重复检验 S3 相对 S0 和 S2 的费用差。
3. 论文补充方法来源、储能互斥变换的证明、信息价值与实现费用的区别；加入真实出版物引文及结果解释。
4. 摘要与正文分页，图表连续编号，源码逐文件完整收入附录，并检查其每一行与实际程序一致。代码原文件仍以支撑材料内版本运行。

全年替代求解器费用

模式	方法	S2/元	S3/元	S3 减 S2/元
----	----	------	------	-----------

```
|---|---|---:|---:|---:|
'''
```

```
    for (mode,method),g in a.groupby(['mode','method'],sort=False):
        x=g.set_index('scenario').cost;text+=f'|{mode}|{method}|{x.S2:,.6f}|{x.S3:,.6f}|{x.S3-
x.S2:,.6f}|\n'
    text+=""
```

两种求解方法下 S2 均略低于 S3，但费用差的大小发生变化；原正式轨迹的 S3 减 S2 在全部四个块长度下区间均跨零。故保留日内更新相对 S0 的样本证据，不把 18 点更新或 S2/S3 的微小排序写成稳定的一般结论。

复核与状态

```
'''
```

```
    text+=f'文档及交付检查状态：{v["status"]}；共{len(v["checks"])}项。论文 PDF 共
{v["metrics"]["pdf_pages"]}页，含完整源码附录；正文页数与摘要分页另有专门检查。
数值复核记录见 audit/independent_validation.json，补充实验复核见 research/
annual_independent_checks.json。\\n'
    text+=""
```

新增参考书为 75 页扫描 PDF，本轮实际查看目录与优化、检验、评价和附录相关页；采用其要求给出实证检验和完整来源的方向，不采用示例数据、保证获奖和缺乏依据的固定格式阈值。原有参考书未复制到交付包。

仍需真实人工完成的事项

本轮不承诺分数或奖项。参赛队员须独立核对题意、推导、代码、数值与 AI 采用记录，按真实贡献确认人工主导及符合本年度参赛规则；当前没有证据证明该人工核验已完成。AI 使用详情如实说明 AI 参与核心修订。单年历史场景、保守一月启动和交易时刻电价解释仍是模型限制。格式检查通过不能覆盖参赛资格、队员主导、匿名及其他纪律要求。

本轮所有路径调整仅为可移植配置，未修改原始观测；正式结果与上次交付的差异核对见交付清单。支撑包中的完整源码与压缩轨迹可恢复复算。

```
'''
```

```
    (F/'evidence/本轮研究与交付修订报告.md').write_text(text)
    print('Improvement report written')
```

```
if __name__=='__main__':main()
```

scripts/write_repair_report.py

```
SHA256 c3be81844147eab5040e50396e82cc7ac151b960e444e4f578352e9aef0e72a6
""""Current-run repair report, with computed effects and explicit limits.""""
from pathlib import Path
from datetime import datetime
import json,pandas as pd
```

```

ROOT=Path(__file__).resolve().parents[1];O=ROOT/'outputs/c_final_v1';F=O/'frozen'

def main():

audit=json.loads((O/'audit/independent_validation.json').read_text());quality=json.loads((O/'fi
nal_validation_v3.json').read_text())

exp=pd.read_csv(F/'experiment_summary.csv');official=pd.read_csv(F/'official_summary.csv'
);settle=pd.read_csv(F/'settlement_sensitivity.csv');jan=pd.read_csv(F/'startup_summary.csv');
q1=pd.read_csv(F/'q1_dispatch.csv')
    def get(mode,sc='S0',arm='joint_empirical'):
        return exp[(exp['mode']==mode)&(exp.scenario==sc)&(exp.model_arm==arm)].iloc[0]
    def f(x,n=2):return f'{x:.{n}f}'

old={'q2':17646794.819761,'q3':16662138.664212,'q4_2':18660362.332063,'q4_3':17539062.
517961}
    rows='\n'.join(f' {r["mode"]} | {f(old[r["mode"]])} | {f(r.final_relative_cost_yuan)} |
{f(r.final_relative_cost_yuan-old[r["mode"]])} |' for _,r in official.iterrows())
    risk='\n'.join(f' {mode} | {f(get(mode,arm="mean_forecast").final_relative_cost_yuan)} |
{f(get(mode).final_relative_cost_yuan)} |
{f(get(mode,arm="mean_forecast").final_relative_cost_yuan-
get(mode).final_relative_cost_yuan)} | {f(get(mode).emergency_kwh,4)} |' for mode in
['q2','q4_2'])
    ledger='\n'.join(f' {r["mode"]} | {f(r.main_cost)} | {f(r.same_chain_sequential_cost)} |
{f(r.same_chain_sequential_cost-r.main_cost)} | {f(r.no_refund_same_path_cost)} |' for _,r in
settle.iterrows() if r['mode'] in ['q3','q4_3'])
    schedule='\n'.join(f' {mode.upper()} | {f(get(mode).final_relative_cost_yuan)} |
{f(get(mode,"S3").final_relative_cost_yuan)} | {f(get(mode,"S3").final_relative_cost_yuan-
get(mode).final_relative_cost_yuan)} | {f(get(mode,"S3").final_relative_cost_yuan-
get(mode,"S2").final_relative_cost_yuan)} |' for mode in ['q3','q4_3'])
    times='\n'.join(f' {h}:00-{h}:10 | {int(h*6+1)} |
{f(q1.loc[q1.position==h*6+1,"plan_00_grid_kwh"].iloc[0],6)} |' for h in
[10,12,14,16,18,20])
    text=f"""# V3 闭环修复与独立复验报告

```

生成时间：{datetime.now().astimezone().isoformat()}。对象：桌面“最新版本V3”。用户本轮已明确授权实际修复，要求保留真实数据、禁止未来信息泄漏并遵守数学约束。此前只读审查报告描述修复前快照；本报告使用新的正式运行。

当前结论

本轮已实际重写风险规划、滚动承诺和启动逻辑，重跑一月选型及二月至十二月正式实验，重新导出论文和五份结果工作簿。原始附件未修改。独立数值及工作簿检查{len(audit['checks'])}项，状态{audit['status']}；文档与导出检查{len(quality['checks'])}项，状态{quality['status']}。这些检查证明下述实现、约束和证据一致性，不证明模型在所有解释和所有未来数据下最优。

没有用旧报告数值充当本轮结果，没有把历史预测场景伪装成实际观测，也没有仅修改 PASS 字段。旧流水线被停止，修订前完整备份保留在本次 Codex 任务的 work/before_repair；项目自身旧输出保留在 work/pre_repair_output。

1. 已修复事项与数值影响

五倍缺口成本已进入日前规划

`src/model/risk.py:cases/solve`使用完整观测的历史联合轨迹构造经验预测分布。所有采购、充放电和状态在场景之间共享；仅紧急采购随场景变化，目标直接含五倍价缺口损失。没有提前给优化器未来实际负载、光伏或电价。残差的基准预测也在对应历史日只使用更早日期生成。

`experiments.py`只用 1 月 8—31 日比较两种场景构造与 0.5/1/1.5 终端惩罚倍数，锁定因果残差场景、倍数 1.0。已解析配置在`frozen/resolved\_config.json`；候选默认值不能代替实际解析配置。正式期 334 天不参与参数选择。

相同日期与状态合同下，均值预测控制与新风险策略比较如下，差额是本数据实现结果：

模式	均值预测费/元	风险策略费/元	费用降低/元	风险策略紧急量/kWh
---	---	---	---	---
{risk}				

无储能控制费用分别为 $\{f(\text{get}('q2', \text{arm}='no_storage').\text{final_relative_cost_yuan})\}$ 、 $\{f(\text{get}('q4_2', \text{arm}='no_storage').\text{final_relative_cost_yuan})\}$ 元。紧急购电是题面允许的缺口补足机制，不能把该量称为最终停电量。有限历史分布仍可能漏掉极端风险。

逐次结算已改为同一真实承诺链

`rolling.py:simulate`记录每次正式更新的旧承诺、新承诺、交付位置和发布时间。后续预览不记为正式订单；已交付段不允许重写。现有同链逐次结果如下：

模式	净额费用/元	同链逐次费用/元	逐次减净额/元	不退原价同路径/元
---	---	---	---	---
{ledger}				

独立审计从版本记录重新累积每一笔调整，并单独从原始曲线重算同一执行轨迹的紧急费。旧版以另一条确定性路径计算调整费、再拼正式路径紧急费的做法已删除。相同非负交付价下，`逐次-净额 = 0.5Σp(Σ|ΔA| - |ΣΔA|) ≥ 0`；100→80→100、价格 1 元的例子，逐次较净额多 20 元。退费和交付价是明示解释；其他定价解释须重新定义模型。

十分钟错位和 Q2—Q4 缺表已修复

以右端点数据代表前十分钟区间为统一离散化口径。论文按`physical\_interval`抽取，官方模板按原列序写入；新增“时段映射”页保留原标签与物理键的区别。Q1 费用仍为 $\{f(q1.final_relative_cost_yuan.sum(),6)\}$ 元；不以目标相同掩盖时段错误。

题面物理区间	正确位置	当前计划量/kWh（六位展示）
{times}

例如 12:00 应取位置 73、原模板 B74，旧论文误取位置 72。六个时段已在 DOCX、PDF、工作簿中核对。Q2/Q3/Q4-2/Q4-3 四类×四指定日期均包括六个十分钟值、六组四小时充放电、0/24 点 SOC、全天量/费用和连续紧急区间。含 Q1 共 17 组；文档存在的占位符数量为 0。修复前本次审查也已查明“剩 80 个占位”的历史说法不适用于当时快照，因此不把这一旧数字算作本轮成果。

S0/S3 比较已固定共同日前承诺

先用因果的 S0 参考过程形成各日 B。四种频次使用同一 B、相同一月启动和年末状态；滚动差异来自后续允许信息更新。全部计划动作在场景中共享完整窗口，避免旧版午夜分支共享范围不一致。

模式	S0 费用/元	S3 费用/元	S3 减 S0/元	S3 减 S2/元
{schedule}

本次 S2 略低于 S3，S3 仍作为题设四时点机制的正式结果，而非被称为全局最优。32 个相同 LP 实例、3 种求解法共 96 次诊断中，目标最大差约 7.28×10^{-12} 元，但采购轨迹 L1 最大差 292.46 kWh、SOC 轨迹最大差 2577.19 kWh。存在并列解敏感性；没有据此宣称替代求解器的全年策略排序已经稳定。逐日配对 7 天块重采样区间已列论文，重采样不被当作新观测。

一月 SOC 衔接与无历史启动已重建

删除“首日无历史时用当天完整实际曲线”的未来信息兜底。采用明示、可行的一月启动合同：电池待机 $C=D=0$ ；首日计划为 0，实际缺口按紧急合同支付；以后只用已经完成的历史日作采购分位数决策。逐段记录 4464 个时段，因此一月末 6000 是状态递推的结果，二月初值从账本读取，不是空填 6000。Q2/Q3 一月启动费各 $\{f(jan[jan['mode']=='q2'].cost_yuan.iloc[0])\}$ 元，Q4-2/Q4-3 各 $\{f(jan[jan['mode']=='q4_2'].cost_yuan.iloc[0])\}$ 元，均披露在论文。

这是一项保守的启动策略，不声称一月最优。离线选型回放与真实启动合同区分记录。普通分段结束不强行回到 6000；仅明确评价末端（训练子实验或 12 月 31 日）施加终端条件。首日、跨日、分段恢复、不同充放电效率和年末边界已测试。

2. 与修复前已交付版本的差异

模式	修复前已交付费/元	当前正式费/元	当前减旧版/元
---	---	---	---

{rows}

旧版数值来自本轮修复前的独立复算，不是历史报告直接抄录的结论。修复前后台另有 Q3 从 1200 起步的未同步结果 16,664,426.073252 元；它与当时 6000 起步的交付稿不一致。本次所有分支已统一启动合同。表中总差额同时包含风险模型、预测、合同与边界改变，不能全部归因为一个修复点；模型改进效果优先看第 1 节受控比较。

3. 数学约束与信息隔离证据

- 真实负载、光伏、价格直接从附件 1、2、4 读取重算，逐时段匹配；源功率转电量除以 6。
- 独立重算供需、紧急量、未用富余、费用和 $E_{next}=E+0.9C-D/0.9$ ；检查 SOC 在 [1200,10800]、每段充放电 $\leq 5000/6$ 、不同时充放电、前后状态连续。
- Q1 另构建一套 LP 约束矩阵验证最优目标，未调用原模型费用函数。
- 未来观测扰动保持决策已知信息不变；改变未发生的负载、光伏、电价后，场景和当前决策不变。更晚发布的预报不能影响 0 点结果。测试范围见 `tests/test\_closed\_loop.py`，共 7 个测试。
- 对全部 8 条 S0—S3 承诺链独立回放，核对旧值连续、没有改写过去、最终值等于正式执行及同路径费用不等式。
- 五份工作簿检查原始列序、逐时段量、日期、日总量/费用、四小时汇总、端点 SOC 及全部紧急区间，不只查文件存在。
- DOCX/PDF 指定表逐值与全精度轨迹核对；PDF 中的表格数字文字与 DOCX 一致，公式、图及各页进行了渲染检查。

4. 可复算证据链与运行

证据环节	当前文件/入口
---	---
原题与原附件	原题 C 题.pdf、附件 1—5；evidence/source_provenance.json 保存原始 SHA256
信息与模型	src/model/risk.py、forecast.py、rolling.py、experiments.py
参数来源	evidence/january_parameter_validation.csv、frozen/resolved_config.json
执行与合同	frozen 中正式/控制/各频次逐时段轨迹、decision_versions、各 S0—S3 账本

| 独立复算 | src/audit/audit_final.py、audit/independent_validation.json |
| 表格及文稿 | 五份 result 工作簿、paper/main.md、scripts/paper_content.py、
native_equations.py |
| 最终检查 | final_validation_v3.json、SHA256SUMS.json；哈希清单排除自身 |

工程完整运行：在支撑材料目录执行`./run\_pipeline.sh`。可通过
CODEX_PYTHON_BIN、CODEX_NODE_BIN 指定解释器；原附件根路径配置在
config/final.yaml 或 CUMCM_SOURCE_ROOT。Python 依赖见 requirements.txt；表
格/文档构建还使用已安装的 Codex 工作区运行时。

压缩交付使用列去重与 XZ 无损归档，全精度 CSV 文本不舍入。运行`scripts/
unpack_frozen.py`后恢复到配置指定的 outputs/c_final_v1/frozen，并逐文件校验原始
SHA256；再运行`src/audit/audit\_final.py --workbooks`即可独立复验。压缩仅为满足支
撑材料体积约束，不丢弃不利结果或实验分支。

5. 仍需讨论的模型取舍

已确认的错位、混链、未来信息冷启动、对照不一致及文档缺表已在本轮修复。剩余
的是需要明确承认的模型边界：

1. 退费和价格时点存在题意解释空间；论文已列同路径敏感性，不能冒充官方唯一解
释。
2. 一月待机启动虽严格可行，经济上保守；若进一步优化一月，应使用可用信息重新
运行全年，不能仅把启动末 SOC 改成更有利数值。
3. 20 条历史经验场景、全窗口共同动作及有限期终端惩罚可能偏保守，不能保证未见
极端风险或全局最优。
4. S2 与 S3 的差额很小，若要声称更新频次排序稳健，修复顺序应是先增加有明确二级
经济含义的选解准则，再做完整年度替代求解器回放；目前只作名义样本比较。

下一轮如继续提升，优先讨论上述结算解释与一月启动目标，其次才是提高预测/风险
分布质量和全年排序稳定性。禁止通过挑正式期参数、选择性省略成本或改写实际数
据获得更好结果。

'''

```
(O/'evidence/闭环修复与独立复验报告.md').write_text(text)
print('Repair report written')
```

```
if __name__ == '__main__':main()
```

src/audit/_init_.py

```
SHA256 0e5daf34a6792487bce8e142d9f81487b6c2edb4b3ceba6019ce985674182432
""""Independent audit gate for the C-problem delivery."""
```

src/audit/audit_final.py

```
SHA256 6318fce7aa9336723fb22e8fc194df9b26e970a10c02f0b9e71e3c42b5cf4367
"""Independent validation from original workbooks and public dispatch/commitment data.
Does not import the model, scenario generator or model settlement functions.
"""

from pathlib import Path
from datetime import datetime
import json, csv, argparse, hashlib, os
import numpy as np
import pandas as pd
import openpyxl
from scipy.optimize import linprog
ROOT=Path(__file__).resolve().parents[2]

def run(workbooks=False):

    cfg=json.loads((ROOT/'config/final.yaml').read_text());F=ROOT/cfg['output_root']/'frozen';O
    =F.parent/'audit';O.mkdir(parents=True,exist_ok=True);checks=[];metrics={}
    def check(name,ok,actual):checks.append(dict(check=name,status='PASS' if bool(ok) else
    'FAIL',actual=actual))
    P=Path(os.environ.get('CUMCM_SOURCE_ROOT',cfg['source_root']))/'C 题/附件'
    def matrix(path,sheet):

        w=openpyxl.load_workbook(path,read_only=True,data_only=True);rows=list(w[sheet].value
        s);w.close();return np.array(rows[1:],object)[:,1:145].astype(float)
        w=openpyxl.load_workbook(P/'附件
        1.xlsx',read_only=True,data_only=True);a=np.array(list(w.active.values
        [1:],object);w.close();price=a[:,1].astype(float);qL=a[:,2].astype(float)/
        6;qV=a[:,3].astype(float)/6
        L=matrix(P/'附件 2.xlsx','小区负载')/6;V=matrix(P/'附件 2.xlsx','光伏发电实际功率')/
        6;price_rt=matrix(P/'附件 4.xlsx','Sheet1');models={};tol=1e-7
        for path in sorted(F.glob('*dispatch.csv')):
            if 'january' in path.name or path.name.startswith('research_'):continue
            d=pd.read_csv(path);name=path.stem;mode='q1' if name.startswith('q1') else ('q4_3' if
            name.startswith('q4_3') else 'q4_2' if name.startswith('q4_2') else name[:2]);models[name]=d
            if mode=='q1':l,v,p=qL,qV,price
            else:
                i=(pd.to_datetime(d.date)-pd.Timestamp('2025-01-
                01')).dt.days.to_numpy();j=d.position.to_numpy()-1;l=L[i,j];v=V[i,j];p=price_rt[i,j] if
                mode.startswith('q4') else price[j]

            c=d.charge_kwh.to_numpy();dis=d.discharge_kwh.to_numpy();g=d.final_adjusted_grid_kwh.
            to_numpy();b=d.plan_00_grid_kwh.to_numpy();s0=d.soc_start_kwh.to_numpy();s1=d.soc_e
            nd_kwh.to_numpy();q=np.maximum(l+c-v-dis-g,0);u=np.maximum(g+v+dis-l-c,0);delta=g-b
            cost=p*b+p*(delta+.5*np.abs(delta))+5*p*q
            maxerr=lambda x:float(np.max(np.abs(x)))
```

```

    check(name+'.count',len(d)==(144 if mode=='q1' else 48096) and not
d.duplicated(['date','position']).any() and all(len(z)==144 and
set(z.position)==set(range(1,145)) for _,z in d.groupby('date')),len(d))
    check(name+'.raw_load_pv_price',max(maxerr(l-d.actual_load_kwh),maxerr(v-
d.actual_pv_kwh),maxerr(p-d.settlement_price))<tol,dict(load=maxerr(l-
d.actual_load_kwh),pv=maxerr(v-d.actual_pv_kwh),price=maxerr(p-d.settlement_price)))
    labels=[f'{(int(x)-1)*10//60:02d}':{(int(x)-1)*10%60:02d}-{int(x)*10//60:02d}:
{int(x)*10%60:02d}' for x in d.position]
    check(name+'.physical_interval',labels==d.physical_interval.tolist(),sum(x!=y for x,y in
zip(labels,d.physical_interval)))
    sr=maxerr(s1-s0-.9*c+dis/.9);continuity=maxerr(s0[1:]-s1[:-1]);check(name+'.soc',sr<tol
and continuity<tol,dict(recursion=sr,continuity=continuity))
    check(name+'.device_constraints',min(s0.min(),s1.min())>=1200-tol and
max(s0.max(),s1.max())<=10800+tol and c.min()>=-tol and dis.min()>=-tol and
max(c.max(),dis.max())<=5000/6+tol and not
np.any((c>tol)&(dis>tol)),dict(min_soc=float(min(s0.min(),s1.min())),max_soc=float(max(s0.
max(),s1.max())),max_charge=float(c.max()),max_discharge=float(dis.max())))
    check(name+'.realized_balance',max(maxerr(q-d.emergency_kwh),maxerr(u-
d.unused_surplus_kwh))<tol,maxerr(q-d.emergency_kwh))
    check(name+'.main_cost',maxerr(cost-d.final_relative_cost_yuan)<tol,maxerr(cost-
d.final_relative_cost_yuan))
    check(name+'.same_path_inequality',float((d.sequential_cost_yuan-
d.final_relative_cost_yuan).min())>=-tol,float((d.sequential_cost_yuan-
d.final_relative_cost_yuan).min()))
    check(name+'.end_soc',abs(s1[-1]-6000)<tol,float(s1[-1]))
    if mode!='q1':
        jan=pd.read_csv(F/f'{mode}
_january_dispatch.csv');check(name+'.January_link',abs(s0[0]-
jan.soc_end_kwh.iloc[-1])<tol,dict(january_end=float(jan.soc_end_kwh.iloc[-1]),february_sta
rt=float(s0[0])))

metrics[name]=dict(total_cost=float(cost.sum()),plan_cost=float((p*b).sum()),adjustment_cos
t=float((p*(delta+.5*abs(delta))).sum()),emergency_cost=float((5*p*q).sum()),emergency_k
wh=float(q.sum()),emergency_intervals=int((q>=.00005).sum()),max_emergency_kwh=float(
q.max()))
    summary=pd.read_csv(F/'experiment_summary.csv')
    for r in summary.itertuples():
        key=(r.mode+'_dispatch') if r.mode in ['q2','q4_2'] and r.model_arm=='joint_empirical'
else (r.mode+'_'+r.model_arm+'_dispatch' if r.mode in ['q2','q4_2'] else
r.mode+'_'+r.scenario+'_dispatch')
        m=metrics[key];err=max(abs(m['total_cost']-
r.final_relative_cost_yuan),abs(m['emergency_kwh']-r.emergency_kwh),abs(m['plan_cost']-
r.plan_cost_yuan),abs(m['adjustment_cost']-
r.adjustment_cost_yuan),abs(m['emergency_cost']-r.emergency_cost_yuan))
        check(key+'.summary_reconciliation',err<1e-6,err)
    # Reconstruct every saved schedule's formal chain, including repeated revisions.
    ledgers=[]
    for mode in ['q3','q4_3']:
        base=models[f'{mode}_S0_dispatch'].plan_00_grid_kwh.to_numpy()
        for sc in ['S0','S1','S2','S3']:

```

```

d=models[f'{mode}_{sc}_dispatch'];check(f'{mode}_{sc}.common_plan',np.max(np.abs(base-
d.plan_00_grid_kwh))<tol,float(np.max(np.abs(base-d.plan_00_grid_kwh))))
dd=d.set_index(['date','position']);v=pd.read_csv(F/f'{mode}_{sc}_
_decisions.csv',low_memory=False);chain={};fees={};bad=0;past=0
prices={(r.date,int(r.position)):float(r.settlement_price) for r in d.itertuples()}
for r in v.itertuples():
    if r.is_preview_next_day:continue
    k=(r.target_date,int(r.position));new=float(r.new_commitment_kwh)
    delivery=pd.Timestamp(r.target_date)
+pd.Timedelta(minutes=(int(r.position)-1)*10)
    past+=int(delivery<pd.Timestamp(r.decision_time))
    if k in chain:
        bad+=int(abs(chain[k]-r.old_commitment_kwh)>tol);delta=new-
chain[k];fees[k]=fees.get(k,0)+prices[k]*(delta+.5*abs(delta))
        chain[k]=new
    last=np.array([chain[k] for k in dd.index]);computed=dd.plan_cost_yuan.to_numpy()
+dd.emergency_cost_yuan.to_numpy()+np.array([fees.get(k,0) for k in
dd.index]);err=float(np.max(abs(computed-dd.sequential_cost_yuan.to_numpy()))
    check(f'{mode}_{sc}.formal_chain',bad==0 and past==0 and np.max(abs(last-
dd.final_adjusted_grid_kwh.to_numpy()))<tol and
err<tol,dict(old_commitment_errors=bad,past_rewrites=past,final_error=float(np.max(abs(las
t-dd.final_adjusted_grid_kwh.to_numpy()))),sequential_error=err))

ledgers.append(dict(mode=mode,scenario=sc,formal_chain_cost_yuan=float(computed.sum()
),main_cost_yuan=float(dd.final_relative_cost_yuan.sum()),difference_yuan=float(computed.
sum()-dd.final_relative_cost_yuan.sum()))
    for mode in ['q2','q4_2','q3','q4_3']:
        d=pd.read_csv(F/f'{mode}_january_dispatch.csv');i=(pd.to_datetime(d.date)-
pd.Timestamp('2025-01-01')).dt.days.to_numpy();j=d.position.to_numpy()-1;p=price_rt[i,j] if
mode.startswith('q4') else price[j];q=np.maximum(L[i,j]-V[i,j]-
d.plan_grid_kwh,0);cost=p*d.plan_grid_kwh+5*p*q
        history_ok=all(all(h<r.date for h in str(r.history_dates).split('|') if h and h!='nan') for r in
d.itertuples())
        check(mode+'.January_startup',len(d)==4464 and history_ok and
np.all(d.charge_kwh==0) and np.all(d.discharge_kwh==0) and
np.max(abs(d.soc_start_kwh-6000))<tol and np.max(abs(d.soc_end_kwh-6000))<tol and
np.max(abs(cost-
d.cost_yuan))<tol,dict(rows=len(d),history_before_decision=history_ok,cost_error=float(np.
max(abs(cost-d.cost_yuan))),cost=float(cost.sum()))

stats=pd.read_csv(F/'solver_decisions.csv');bad=int((pd.to_datetime(stats.latest_observation_t
ime)>pd.to_datetime(stats.decision_time)).sum());check('observation_availability',bad==0,bad)
    # Independent Q1 LP with raw input and physical units, separate construction.
    n=144;M=np.zeros((2*n,5*n));rhs=np.r_[qL-qV,np.zeros(n)];rhs[n]=6000
    for t in range(n):
        M[t,t]=1;M[t,n+t]=-1;M[t,2*n+t]=1;M[t,3*n+t]=-1;M[n+t,n+t]=-.9;M[n+t,2*n+t]=1/.9;M[n+t,
4*n+t]=1

```

```

if t:M[n+t,4*n+t-1]=-1
bounds=[(0,None)]*n+[(0,5000/6)]*(2*n)+[(0,x) for x in qV]+
[(1200,10800)]*n;bounds[-1]=(6000,6000)

opt=linprog(np.r_[price,np.zeros(4*n)],A_eq=M,b_eq=rhs,bounds=bounds,method='highs');c
heck('Q1.independent_optimum',opt.success and abs(opt.fun-metrics['q1_dispatch']
['total_cost'])<1e-6,float(opt.fun))
if workbooks:
    for mode,name in [('q1','result1'),('q2','result2'),('q3','result3'),('q4_2','result4-2'),
('q4_3','result4-3')]:
        d=models[mode+'_dispatch'];w=openpyxl.load_workbook(F.parent/f'workbooks/
{name}.xlsx',read_only=True,data_only=True)
        required={'计划购电量','充放电量','指定时段','时段映射'}
        if mode!='q1':required.add('紧急购电量')
        if mode in ['q3','q4_3']:required.add('调整购电量')
        check(name+'.required_sheets',required<=set(w.sheetnames),sorted(w.sheetnames))
        for sheet,col in [('计划购电量','plan_00_grid_kwh'),('调整购电
量','final_adjusted_grid_kwh')]:
            if sheet not in w.sheetnames:continue
            rows=list(w[sheet].values)
            vals=np.array([r[1] for r in rows[1:145]],float) if mode=='q1' else np.array([r[1:145]
for r in rows[1:335]],float).reshape(-1)
            e=float(np.max(abs(vals-d[col].to_numpy())));check(f'{name}.
{sheet}',e<=.000051,e)
            if mode!='q1':
                cost_col='plan_cost_yuan' if sheet=='计划购电量' and mode in ['q3','q4_3'] else
'final_relative_cost_yuan'
                expected=d.groupby('date',sort=False)[[col,cost_col]].sum().to_numpy()
                actual=np.array([r[145:147] for r in rows[1:335]],float)
                err=float(np.max(abs(expected-actual)))
                dates_ok=[str(r[0])[10] for r in rows[1:335]]==list(d.date.unique())
                check(f'{name}.{sheet}.daily_totals_and_dates',err<=.000051 and
dates_ok,dict(max_error=err,dates_match=dates_ok))
                # All storage blocks and SOC endpoints; all emergency interval keys.
                dates=d.date.unique();srows=list(w['充放电量'].values)[1:];errs=[]
                for di,day in enumerate(dates):
                    z=d[d.date==day]
                    for b in range(6):
                        rr=srows[di*6+b];off=1 if mode=='q1' else 2;zz=z.iloc[b*24:(b+1)*24]
                        errs.extend([abs(float(rr[off])-zz.charge_kwh.sum()),abs(float(rr[off+1])-
zz.discharge_kwh.sum())])
                        if b<2:errs.append(abs(float(rr[off+3])-(z.soc_start_kwh.iloc[0] if b==0 else
z.soc_end_kwh.iloc[-1])))
                    check(f'{name}.all_storage_blocks',max(errs)<=.000051,float(max(errs)))
                if mode!='q1':
                    expected=[]
                    for day,z in d.groupby('date'):
                        active=[]
                        def emit():

```

```

        a=(active[0].position-1)*10;b=active[-1].position*10
        expected.append((day,f {a//60}:{a%60:02d}-{b//60}:{
{b%60:02d}',sum(x.emergency_kwh for x in active)))
        for r in z.itertuples():
            if r.emergency_kwh>=.00005:active.append(r)
            elif active:emit();active=[]
            if active:emit()
        actual=[];day=None
        for r in list(w['紧急购电量'].values)[1:]:
            if r[0] is not None:day=str(r[0])[10]
            actual.append((day,r[1],float(r[2])))
        keys_ok=[x[:2] for x in actual]==[x[:2] for x in expected]
        err=max([abs(x[2]-y[2]) for x,y in zip(actual,expected)] or [0])
        check(name+'.emergency_intervals',keys_ok and
err<=.000051,dict(rows=len(actual),expected=len(expected),value_error=err))
        specified=list(w['指定时段'].values)[1:];specified_error=[]
        for rr in specified:
            day=str(rr[0])[10];zz=d[(d.date==day)&(d.physical_interval==rr[1])]
            if len(zz)!=1:specified_error.append('missing key')
            else:
                x=zz.iloc[0]
                if abs(float(rr[2])-x.plan_00_grid_kwh)>.000051 or abs(float(rr[3])-
x.final_adjusted_grid_kwh)>.000051:specified_error.append((day,rr[1]))
                check(name+'.specified_physical_values',not specified_error and len(specified)==(6 if
mode=='q1' else 24),specified_error)
        w.close()
        out=dict(status='PASS' if all(r['status']=='PASS' for r in checks) else
'FAIL',generated_at=datetime.now().astimezone().isoformat(),checks=checks,metrics=metrics
,formal_settlement=ledgers)

(O/'independent_validation.json').write_text(json.dumps(out,ensure_ascii=False,indent=2));pd
.DataFrame(ledgers).to_csv(O/'settlement_recalculation.csv',index=False)
    print(json.dumps(dict(status=out['status'],checks=len(checks),failures=[r for r in checks if
r['status']=='FAIL']),ensure_ascii=False),flush=True)
    return out

if __name__=='__main__':
    ap=argparse.ArgumentParser();ap.add_argument('--
workbooks',action='store_true');a=ap.parse_args();r=run(a.workbooks);raise SystemExit(0 if
r['status']=='PASS' else 1)

```

src/model/_init_.py

SHA256 5cb253a6bf42d59171e8d2a280901387c14f4942303ddd19a8b4b46f3c634cfe
 """"C-problem causal microgrid model and reproducible rolling experiments.""""

```

from .config import load_config
from .data import DataBundle, load_data
from .rolling import RunSpec, simulate

```

```
__all__ = ["DataBundle", "RunSpec", "load_config", "load_data", "simulate"]
```

src/model/cli.py

```
SHA256 44f9863d234037e56260c544ac3637b85f5274d451fb9afb65bb968820b6bebf
```

```
import argparse
from .config import load_config
from .data import load_data
from .experiments import run_all

def main():
    ap=argparse.ArgumentParser();ap.add_argument('--config',default='config/
final.yaml');ap.add_argument('--
scope',choices=['all','train','official'],default='all');a=ap.parse_args();cfg=load_config(a.config)
;run_all(load_data(cfg),cfg,a.scope);return 0
```

src/model/config.py

```
SHA256 8f9800ac73bb699fa1d65cd5eea8ea4ac1d8949812fed19f930288b8ad171776
```

```
from __future__ import annotations
```

```
import json
import os
from pathlib import Path
from typing import Any
```

```
PROJECT_ROOT = Path(__file__).resolve().parents[2]
```

```
def load_config(path: str | Path) -> dict[str, Any]:
    """Load final.yaml.
```

```

    The file intentionally uses JSON syntax, which is a strict subset of YAML. This
    keeps the pipeline dependency-free while remaining readable by YAML tooling.
    """
```

```
    p = Path(path)
    if not p.is_absolute():
        p = PROJECT_ROOT / p
    cfg = json.loads(p.read_text(encoding="utf-8"))
    cfg["_config_path"] = str(p.resolve())
    source_value=os.environ.get("CUMCM_SOURCE_ROOT",cfg["source_root"])
    source=Path(source_value)
    if not source.is_absolute(): source=PROJECT_ROOT/source
    cfg["_source_root"]=str(source.resolve())
    out = Path(cfg["output_root"])
    if not out.is_absolute():
        out = PROJECT_ROOT / out
```

```

cfg["_output_root"] = str(out.resolve())
return cfg

```

src/model/data.py

```

SHA256 192c71b4505cf89305ddda219a898304c607445f32f0a559925ca4496ed24e6c

```

```

from __future__ import annotations

```

```

from dataclasses import dataclass
from datetime import date, datetime, time, timedelta
from pathlib import Path
from typing import Any

```

```

import numpy as np
from openpyxl import load_workbook

```

```

def _date(value: Any) -> date:
    if isinstance(value, datetime):
        return value.date()
    if isinstance(value, date):
        return value
    return datetime.strptime(str(value).strip().replace(".", "-").replace("/", "-"), "%Y-%m-%d").date()

```

```

def _hm(value: Any) -> str:
    if isinstance(value, time):
        return f"{value.hour:02d}:{value.minute:02d}"
    return str(value).strip()

```

```

def _numeric(values: list[Any], label: str) -> np.ndarray:
    bad = [i for i, v in enumerate(values) if v is None or v == "" or isinstance(v, bool)]
    if bad:
        raise ValueError(f"{label}: blank/non-numeric cells at offsets {bad[:10]}")
    try:
        arr = np.asarray(values, dtype=float)
    except (TypeError, ValueError) as exc:
        raise ValueError(f"{label}: non-numeric cell present") from exc
    if not np.all(np.isfinite(arr)):
        raise ValueError(f"{label}: NaN/Inf present")
    return arr

```

```

@dataclass(frozen=True)
class DataBundle:
    dates: tuple[date, ...]

```

```

source_timestamps: tuple[str, ...]
official_labels: tuple[str, ...]
fixed_price: np.ndarray
q1_load_kw: np.ndarray
q1_pv_kw: np.ndarray
load_kw: np.ndarray
pv_kw: np.ndarray
realtime_price: np.ndarray
pv_forecast_kw: dict[tuple[date, int], np.ndarray]

@property
def date_index(self) -> dict[date, int]:
    return {d: i for i, d in enumerate(self.dates)}

def time_rows(self) -> list[dict[str, Any]]:
    rows: list[dict[str, Any]] = []
    for di, day in enumerate(self.dates):
        for pos in range(144):
            start = pos * 10
            end = (pos + 1) * 10
            rows.append({
                "date": day.isoformat(),
                "position": pos + 1,
                "global_position": di * 144 + pos,
                "source_timestamp": self.source_timestamps[pos],
                "physical_interval": f"{start//60:02d}:{start%60:02d}-{end//60:02d}:{end%60:02d}",
                "official_label": self.official_labels[pos],
            })
    return rows

def _read_matrix(path: Path, sheet: str) -> tuple[tuple[str, ...], tuple[date, ...], np.ndarray]:
    ws = load_workbook(path, data_only=True, read_only=True)[sheet]
    rows = ws.iter_rows(values_only=True)
    header = next(rows)
    if len(header) < 145:
        raise ValueError(f"{path.name}/{sheet}: expected 145 columns")
    headers = tuple(_hm(x) for x in header[1:145])
    dates: list[date] = []
    matrix: list[np.ndarray] = []
    for ri, row in enumerate(rows, start=2):
        if row[0] in (None, ""):
            raise ValueError(f"{path.name}/{sheet}!A{ri}: blank date")
        dates.append(_date(row[0]))
        matrix.append(_numeric(list(row[1:145]), f"{path.name}/{sheet} row {ri}"))
    out = np.vstack(matrix)
    if out.shape != (365, 144) or len(set(dates)) != 365:

```

```

        raise ValueError(f'{path.name}/{sheet}: expected 365 unique days x 144 intervals, got {out.shape}')
    return headers, tuple(dates), out

```

```

def load_data(cfg: dict[str, Any]) -> DataBundle:
    root = Path(cfg['_source_root']) / "C 题" / "附件"
    ws1 = load_workbook(root / "附件 1.xlsx", data_only=True, read_only=True).active
    rows1 = list(ws1.iter_rows(values_only=True))
    if len(rows1) != 145:
        raise ValueError("附件 1: expected header + 144 intervals")
    source_timestamps = tuple(_hm(r[0]) for r in rows1[1:])
    fixed_price = _numeric([r[1] for r in rows1[1:]], "附件 1/电价")
    q1_load = _numeric([r[2] for r in rows1[1:]], "附件 1/负载")
    q1_pv = _numeric([r[3] for r in rows1[1:]], "附件 1/光伏")

    h1, dates, load = _read_matrix(root / "附件 2.xlsx", "小区负载")
    h2, dates2, pv = _read_matrix(root / "附件 2.xlsx", "光伏发电实际功率")
    h3, dates3, rt = _read_matrix(root / "附件 4.xlsx", "Sheet1")
    if not (dates == dates2 == dates3 and h1 == h2 == h3):
        raise ValueError("附件 2/4 date or timestamp axes differ")
    expected = tuple(date(2025, 1, 1) + timedelta(days=i) for i in range(365))
    if dates != expected:
        raise ValueError("附件 2/4 dates are not exactly 2025-01-01..2025-12-31")

    template = load_workbook(root / "附件 5" / "result2.xlsx", data_only=True,
read_only=True)["计划购电量"]
    official = tuple(str(x.value) for x in next(template.iter_rows())[1:145])
    if len(official) != 144 or len(set(official)) != 144:
        raise ValueError("official result template does not contain 144 unique interval labels")

    ws3 = load_workbook(root / "附件 3.xlsx", data_only=True, read_only=True).active
    forecasts: dict[tuple[date, int], np.ndarray] = {}
    current: date | None = None
    for ri, row in enumerate(ws3.iter_rows(min_row=2, values_only=True), start=2):
        if row[0] not in (None, ""):
            current = _date(row[0])
        if current is None:
            raise ValueError(f'附件 3 row {ri}: missing carried date')
        issue_text = str(row[1]).strip()
        try:
            issue = int(issue_text.split(":", 1)[0])
        except Exception as exc:
            raise ValueError(f'附件 3 row {ri}: invalid issue time {row[1]!r}') from exc
        vals = _numeric(list(row[2:26]), f'附件 3 row {ri}')

```

```

    key = (current, issue)
    if key in forecasts:
        raise ValueError(f'附件 3 duplicate forecast {key}')
    forecasts[key] = vals
    expected_keys = {(d, h) for d in dates for h in (0, 6, 12, 18)}
    if set(forecasts) != expected_keys:
        raise ValueError(f'附件 3 forecast coverage mismatch: missing={len(expected_keys-
set(forecasts))}, extra={len(set(forecasts)-expected_keys)}')
    return DataBundle(dates, source_timestamps, official, fixed_price, q1_load, q1_pv, load,
pv, rt, forecasts)

```

src/model/experiments.py

SHA256 16629a5fbfac9794b8f5d6a742d636fb988efd59aa8f9229dde00e47e2bb5a05

```

from __future__ import annotations
import csv,json,hashlib,platform,sys,copy
from datetime import date,datetime,timedelta
from pathlib import Path
import numpy as np
from .rolling import simulate,RunSpec,summarize,interval
from .lp import solve_deterministic

def write_csv(path,rows):
    path=Path(path);path.parent.mkdir(parents=True,exist_ok=True)
    if not rows:raise ValueError(f'No evidence rows: {path}')
    with path.open('w',newline='',encoding='utf-8-sig') as f:
        w=csv.DictWriter(f,fieldnames=list(rows[0]));w.writeheader();w.writerows(rows)

def q1(data,cfg):

    sol=solve_deterministic(data.q1_load_kw/6,data.q1_pv_kw/6,data.fixed_price,6000,np.full(1
44,np.nan),'force_target',6000,0,cfg)
    rows=[];starts=np.r_[6000,sol.soc_end[:-1]]
    for j in range(144):

        g=float(sol.grid[j]);c=float(sol.charge[j]);d=float(sol.discharge[j]);L=float(data.q1_load_kw[j]
/6);V=float(data.q1_pv_kw[j]/6);p=float(data.fixed_price[j]);u=max(g+V+d-L-c,0)
        rows.append(dict(date='2025-01-
01',position=j+1,source_timestamp=data.source_timestamps[j],physical_interval=interval(j),o
fficial_label=data.official_labels[j],plan_00_grid_kwh=g,final_adjusted_grid_kwh=g,adjustm
ent_kwh=0,charge_kwh=c,discharge_kwh=d,soc_start_kwh=float(starts[j]),soc_end_kwh=flo
at(sol.soc_end[j]),curtail_pv_kwh=min(u,V),planned_unused_nonpv_kwh=max(u-
V,0),forecast_load_kwh=L,forecast_pv_kwh=V,actual_load_kwh=L,actual_pv_kwh=V,forec
ast_price=p,settlement_price=p,emergency_kwh=0,unused_surplus_kwh=u,plan_cost_yuan=
p*g,adjustment_cost_yuan=0,emergency_cost_yuan=0,final_relative_cost_yuan=p*g,sequentia
l_cost_yuan=p*g,decision_hour=0,history_cutoff_date='',model_arm='deterministic_q1',sce
nario='Q1'))
    return rows

```

```

def startup(data,cfg,mode):
    """Declared common January commissioning policy, independent of fitted model.
    Battery idle. With no completed day, scheduled purchase is zero and actual
    deficits are covered by the emergency contract. Thereafter use the empirical
    price-weighted 80% net-load quantile of completed days. No manufactured prior.
    """
    rows=[];soc=float(cfg['study']['initial_soc_kwh'])
    for di,day in enumerate(data.dates):
        if day>=date.fromisoformat(cfg['study']['official_start']):break
        hist=list(range(max(0,di-int(cfg['forecast']['scenario_count'])),di));B=np.zeros(144)
        if hist:
            nets=np.maximum((data.load_kw[hist]-data.pv_kw[hist])/6,0)
            P=data.realtime_price[hist] if mode.startswith('q4') else np.tile(data.fixed_price,
            (len(hist),1))
            for j in range(144):

order=np.argsort(nets[:,j]);cum=np.cumsum(P[order,j]);ix=np.searchsorted(cum,.8*cum[-1]);
B[j]=nets[order[ix],j]
            for j in range(144):

L=float(data.load_kw[di,j]/6);V=float(data.pv_kw[di,j]/6);p=float(data.realtime_price[di,j] if
mode.startswith('q4') else data.fixed_price[j]);Q=max(L-V-B[j],0);U=max(B[j]+V-L,0)

rows.append(dict(mode=mode,date=str(day),position=j+1,physical_interval=interval(j),plan_
grid_kwh=float(B[j]),charge_kwh=0.,discharge_kwh=0.,soc_start_kwh=soc,soc_end_kwh=soc,
c,actual_load_kwh=L,actual_pv_kwh=V,emergency_kwh=Q,unused_surplus_kwh=U,settleme
nt_price=p,cost_yuan=p*B[j]+5*p*Q,history_dates=''.join(str(data.dates[i]) for i in
hist),policy='zero_plan_no_history' if not hist else 'empirical_newsvendor_idle_battery'))
    return rows

def perfect(data,cfg):
    out=[];i=data.date_index[date.fromisoformat(cfg['study']
['official_start'])];n=(len(data.dates)-i)*144
    for mode in ['q2','q4_2']:
        p=data.realtime_price[i:].reshape(-1) if mode.startswith('q4') else
np.tile(data.fixed_price,len(data.dates)-i)
        sol=solve_deterministic(data.load_kw[i:].reshape(-1)/6,data.pv_kw[i:].reshape(-1)/
6,p,6000,np.full(n,np.nan),'force_target',6000,0,cfg)

out.append(dict(mode=mode,purchase_cost_yuan=float(p@sol.grid),initial_soc_kwh=6000,te
rminal_soc_kwh=6000,days=len(data.dates)-
i,information='ex_post_lower_bound_not_executable'))
    return out

def run_all(data,cfg,stage='all'):

```

```

root=Path(cfg['_output_root']);frozen=root/'frozen';ev=root/'evidence';frozen.mkdir(parents=True,exist_ok=True);ev.mkdir(parents=True,exist_ok=True)
# The input configuration never changes in place. Chosen January parameters
# live in a resolved copy and are reused by every later branch.
cfg=copy.deepcopy(cfg)
if stage in ('all','train'):
    training=[]
    for scenario_model in ['raw_history','causal_residual']:
        for mult in cfg['rolling']['soft_penalty_multipliers']:
            trial=copy.deepcopy(cfg);trial['rolling']=
['selected_soft_penalty_multiplier']=mult;trial['forecast']['scenario_model']=scenario_model
            for mode in ['q2','q3','q4_2','q4_3']:

r=simulate(data,trial,RunSpec(mode,'S0',collect_detail=False),initial_soc=6000,start_date=date(2025,1,8),end_date=date(2025,1,31),evaluation_terminal=True)

training.append(dict(scenario_model=scenario_model,penalty_multiplier=mult,**r['summary']
));print('TRAIN',scenario_model,mode,mult,r['summary']
['final_relative_cost_yuan'],flush=True)
    scenario_model,selected=min([(sm,m) for sm in ['raw_history','causal_residual'] for m in
cfg['rolling']['soft_penalty_multipliers']],key=lambda pair:sum(r['final_relative_cost_yuan']
for r in training if r['scenario_model']==pair[0] and r['penalty_multiplier']==pair[1]))
    cfg['forecast']['scenario_model']=scenario_model
    cfg['rolling']=
['selected_soft_penalty_multiplier']=selected;write_csv(ev/'january_parameter_validation.csv',training)
    (frozen/'resolved_config.json').write_text(json.dumps({k:v for k,v in cfg.items() if not
k.startswith('_')},ensure_ascii=False,indent=2))
    if stage=='train':return
    else:
        cfg.update(json.loads((frozen/'resolved_config.json').read_text()))
    if stage in ('all','official'):

write_csv(frozen/'time_index.csv',data.time_rows());write_csv(frozen/'q1_dispatch.csv',q1(data,
a,cfg));write_csv(frozen/'perfect_information_summary.csv',perfect(data,cfg))
    summary=[];versions=[];stats=[];startup_summary=[]
    for mode in ['q2','q4_2','q3','q4_3']:
        jan=startup(data,cfg,mode);write_csv(frozen/f'{mode}
_january_dispatch.csv',jan);initial=jan[-1]
['soc_end_kwh'];startup_summary.append(dict(mode=mode,initial_soc_kwh=jan[0]
['soc_start_kwh'],terminal_soc_kwh=initial,cost_yuan=sum(x['cost_yuan'] for x in
jan),policy='common_causal_commissioning'))
        if mode in ('q2','q4_2'):
            r=simulate(data,cfg,RunSpec(mode,'S0'),initial_soc=initial);write_csv(frozen/
f'{mode}
_dispatch.csv',r['dispatch']);summary.append(r['summary']);stats.extend(r['solve_stats'])
            for arm,point,nostore in [('mean_forecast',True,False),('no_storage',False,True)]:

control=simulate(data,cfg,RunSpec(mode,'S0',arm,collect_detail=False),initial_soc=initial,poi

```

```

nt=point,no_storage=nostore);write_csv(frozen/f {mode} _ {arm}
_daily.csv',control['daily']);write_csv(frozen/f {mode} _ {arm}
_dispatch.csv',control['dispatch']);write_csv(frozen/f {mode} _ {arm}
_solver.csv',control['solve_stats']);summary.append(control['summary'])
print('OFFICIAL',mode,r['summary']['final_relative_cost_yuan'],flush=True)
else:
    ref=simulate(data,cfg,RunSpec(mode,'S0'),initial_soc=initial);write_csv(frozen/
f {mode} _ S0_dispatch.csv',ref['dispatch']);write_csv(frozen/f {mode}
_S0_daily.csv',ref['daily']);write_csv(frozen/f {mode}
_S0_decisions.csv',ref['decision_versions']);summary.append(ref['summary']);stats.extend(ref
['solve_stats'])
    for sc in ['S1','S2','S3']:

r=simulate(data,cfg,RunSpec(mode,sc),initial_soc=initial,common_plans=ref['plans']);write_c
sv(frozen/f {mode} _ {sc} _ dispatch.csv',r['dispatch']);write_csv(frozen/f {mode} _ {sc}
_daily.csv',r['daily']);write_csv(frozen/f {mode} _ {sc}
_decisions.csv',r['decision_versions']);summary.append(r['summary']);stats.extend(r['solve_st
ats'])
        if sc=='S3':write_csv(frozen/f {mode}
_dispatch.csv',r['dispatch']);write_csv(frozen/f {mode}
_decision_versions.csv',r['decision_versions']);versions.extend(r['decision_versions'])
        print('OFFICIAL',mode,sc,r['summary']['final_relative_cost_yuan'],flush=True)

write_csv(frozen/'decision_versions.csv',versions);write_csv(frozen/'experiment_summary.cs
v',summary);write_csv(frozen/'official_summary.csv',[r for r in summary if
r['model_arm']=='joint_empirical' and (r['mode'] in ('q2','q4_2') or
r['scenario']=='S3')]);write_csv(frozen/'startup_summary.csv',startup_summary);write_csv(fro
zen/'solver_decisions.csv',stats)

(frozen/'model_run_summary.json').write_text(json.dumps(dict(generated_at=datetime.now().
astimezone().isoformat(),environment=dict(python=sys.version,platform=platform.platform()
),results=summary,training_selected_multiplier=cfg['rolling']
['selected_soft_penalty_multiplier'],official_scenario='S3',comparison_contract='identical_ref
erence_midnight_grid_plans; common_January_idle_battery_startup;
same_terminal_SOC'),ensure_ascii=False,indent=2))
return cfg

```

src/model/forecast.py

```

SHA256 c148e88935f278fea55cd8aceb86b86781bc37bb2c6e770ae28a01d80cf764bb
"""Strict causal forecast interface. See risk.py for joint scenario provenance."""
from datetime import date
import numpy as np

def history_days(data,target_day:date,cutoff_day:date,count=4):
    ids=[i for i,d in enumerate(data.dates) if d<cutoff_day and
d.weekday()==target_day.weekday()]
    if not ids:ids=[i for i,d in enumerate(data.dates) if d<cutoff_day]
    if not ids:raise ValueError('No causal history: startup policy is required')
    return ids[-count:]

```

```

def profile_horizon(data,start_global,length,cutoff_day,kind,count):
    matrix={'load':data.load_kw,'pv':data.pv_kw,'price':data.realtime_price}
    [kind];out=[];used=set()
    for k in range(length):

di,pos=divmod(start_global+k,144);hist=history_days(data,data.dates[di],cutoff_day,count);u
sed.update(hist);out.append(matrix[hist,pos].mean())
    return np.array(out),[data.dates[i] for i in sorted(used)]

def published_pv_horizon(data,day,issue_hour,length):
    return np.repeat(data.pv_forecast_kw[(day,issue_hour)],6)[:length].copy()

```

src/model/lp.py

```

SHA256 9d84289c658d300124ce3c862359880a407f7a3154db01a65c146581d8d4c6e7
from __future__ import annotations

```

```

from dataclasses import dataclass
import time
from typing import Any

```

```

import numpy as np
from scipy.optimize import linprog
from scipy.sparse import coo_matrix

```

```

@dataclass
class DispatchSolution:
    grid: np.ndarray
    charge: np.ndarray
    discharge: np.ndarray
    curtail: np.ndarray
    emergency: np.ndarray
    soc_end: np.ndarray
    objective: float
    elapsed_seconds: float
    iterations: int
    max_balance_residual: float
    max_soc_residual: float

```

```

def _one_way(charge:np.ndarray,discharge:np.ndarray,unused:np.ndarray,eta:float) ->
tuple[np.ndarray,np.ndarray,np.ndarray]:
    """Remove simultaneous cycling while preserving SOC change and energy balance."""
    delta=eta*charge-discharge/eta
    c=np.maximum(delta,0)/eta
    d=np.maximum(-delta,0)*eta
    u=unused+(d-discharge)-(c-charge)

```

```

u[np.abs(u)<1e-10]=0.0
if np.min(u)<-1e-7: raise AssertionError(f"one-way normalization produced negative
unused energy {np.min(u)}")
return c,d,np.maximum(u,0.0)

def _method_options(method: str, tolerance: float) -> dict[str, Any]:
    options: dict[str, Any] = {"presolve": True}
    if method in {"highs", "highs-ds"}:
        options.update({"primal_feasibility_tolerance": tolerance, "dual_feasibility_tolerance":
tolerance})
    if method in {"highs", "highs-ipm"}:
        options["ipm_optimality_tolerance"] = max(tolerance, 1e-10)
    return options

```

```

def solve_deterministic(
    load_kwh: np.ndarray,
    pv_kwh: np.ndarray,
    price: np.ndarray,
    initial_soc: float,
    baseline: np.ndarray,
    terminal_rule: str,
    terminal_target: float,
    terminal_penalty: float,
    cfg: dict[str, Any],
    method: str = "highs",
    tiebreak: float | None = None,
) -> DispatchSolution:
    load = np.asarray(load_kwh, float); pv = np.asarray(pv_kwh, float); price =
np.asarray(price, float)
    base = np.asarray(baseline, float)
    T = len(load)
    if not (len(pv) == len(price) == len(base) == T):
        raise ValueError("LP input lengths differ")
    G=slice(0,T); C=slice(T,2*T); D=slice(2*T,3*T); U=slice(3*T,4*T)
    E=slice(4*T,5*T); DN=slice(5*T,6*T); UP=slice(6*T,7*T)
    ZP=7*T; ZN=7*T+1; n=7*T+2
    committed=np.isfinite(base)
    c=np.zeros(n)
    c[G]=np.where(committed,0.0,price)
    c[DN]=np.where(committed,-cfg["cost"]["down_adjustment_multiplier"]*price,0.0)
    c[UP]=np.where(committed,cfg["cost"]["up_adjustment_multiplier"]*price,0.0)
    tb=float(cfg["cost"]["throughput_tiebreak_yuan_per_kwh"]) if tiebreak is None else
tiebreak)
    c[C]=tb; c[D]=tb
    if terminal_rule == "soft_target": c[ZP]=terminal_penalty; c[ZN]=terminal_penalty
    rr:list[int]=[]; cc:list[int]=[]; vv:list[float]=[]; rhs:list[float]=[]; row=0
    for t in range(T):

```

```

    rr += [row]*4; cc += [G.start+t,C.start+t,D.start+t,U.start+t]; vv += [1,-1,1,-1]
    rhs.append(float(load[t]-pv[t])); row+=1
eta=float(cfg["storage"]["charge_efficiency"])
for t in range(T):
    rr += [row]*3; cc += [C.start+t,D.start+t,E.start+t]; vv += [-eta,1/eta,1]
    if t: rr.append(row); cc.append(E.start+t-1); vv.append(-1); rhs.append(0.0)
    else: rhs.append(float(initial_soc))
    row+=1
for t in np.flatnonzero(committed):
    rr += [row]*3; cc += [G.start+t,DN.start+t,UP.start+t]; vv += [1,1,-1]
    rhs.append(float(base[t])); row+=1
if terminal_rule == "soft_target":
    rr += [row]*3; cc += [E.stop-1,ZP,ZN]; vv += [1,-1,1];
rhs.append(float(terminal_target)); row+=1
Aeq=coo_matrix((vv,(rr,cc)),shape=(row,n)).tocsr()
st=cfg["storage"]; step=float(st["power_kw"])*float(st["interval_minutes"])/60
bounds=(
    [(0,None)]*T+[(0,step)]*T+[(0,step)]*T+[(0,float(x)) for x in pv]
    +[(float(st["soc_min_kwh"]),float(st["soc_max_kwh"]))]*T
    +[(0,float(base[t])) if committed[t] else (0,0) for t in range(T)]
    +[(0,None) if committed[t] else (0,0) for t in range(T)]
    +([(0,None),(0,None)] if terminal_rule == "soft_target" else [(0,0),(0,0)])
)
if terminal_rule in {"daily_closed","state_cycle","force_target"}:
    target = initial_soc if terminal_rule == "state_cycle" else terminal_target
    bounds[E.stop-1]=(float(target),float(target))
started=time.perf_counter()
res=linprog(c,A_eq=Aeq,b_eq=np.asarray(rhs),bounds=bounds,method=method,
            options=_method_options(method,float(cfg["audit"]["feasibility_tolerance"])))
elapsed=time.perf_counter()-started
if not res.success: raise RuntimeError(f"LP infeasible ({method}): {res.message}")
x=res.x; charge,discharge,unused=_one_way(x[C],x[D],x[U],eta);
e0=np.r_[initial_soc,x[E][-1]]
bal=x[G]+pv+discharge-load-charge-unused
soc=x[E]-e0-eta*charge+discharge/eta
return
DispatchSolution(x[G],charge,discharge,unused,np.zeros(T),x[E],float(res.fun),elapsed,int(res
.nit),float(np.max(np.abs(bal))),float(np.max(np.abs(soc))))

def solve_stochastic(
    load_kwh: np.ndarray,
    pv_scenarios_kwh: np.ndarray,
    price: np.ndarray,
    initial_soc: float,
    baseline: np.ndarray,
    execute_intervals: int,
    terminal_rule: str,
    terminal_target: float,
    terminal_penalty: float,

```

```

    cfg: dict[str, Any],
    method: str = "highs",
    tiebreak: float | None = None,
) -> DispatchSolution:
    """Two-stage extensive-form LP; current executable block is nonanticipative."""
    load=np.asarray(load_kwh,float); pvs=np.asarray(pv_scenarios_kwh,float);
    price=np.asarray(price,float); base=np.asarray(baseline,float)
    S,T=pvs.shape
    if len(load)!=T or len(price)!=T or len(base)!=T: raise ValueError("stochastic LP input
lengths differ")
    # Per scenario: grid, charge, discharge, curtail/unused, emergency, SOC, down, up.
    per=8*T; soft=terminal_rule=="soft_target"; n=S*per+(2*S if soft else 0)
    committed=np.isfinite(base); c=np.zeros(n); prob=1.0/S
    st=cfg["storage"]; eta=float(st["charge_efficiency"]);
    step=float(st["power_kw"])*float(st["interval_minutes"])/60
    down_mult=float(cfg["cost"]["down_adjustment_multiplier"]); up_mult=float(cfg["cost"]
["up_adjustment_multiplier"]); em_mult=float(cfg["cost"]["emergency_multiplier"])
    tb=float(cfg["cost"]["throughput_tiebreak_yuan_per_kwh"]) if tiebreak is None else
    tiebreak)
    bounds:list[tuple[float,float|None]]=[]
    def slices(s:int):
        o=s*per
        return tuple(slice(o+i*T,o+(i+1)*T) for i in range(8))
    for s in range(S):
        G,C,D,U,H,E,DN,UP=slices(s)
        c[G]=prob*np.where(committed,0.0,price); c[C]=prob*tb; c[D]=prob*tb
        c[H]=prob*em_mult*price
        c[DN]=prob*np.where(committed,-down_mult*price,0.0);
    c[UP]=prob*np.where(committed,up_mult*price,0.0)
    bounds += [(0,None)]*T+[(0,step)]*T+[(0,step)]*T+[(0,None)]*T+[(0,None)]*T
    bounds += [(float(st["soc_min_kwh"]),float(st["soc_max_kwh"]))]*T
    bounds += [(0,float(base[t])) if committed[t] else (0,0) for t in range(T)]
    bounds += [(0,None) if committed[t] else (0,0) for t in range(T)]
    if soft:
        for s in range(S):
            c[S*per+2*s:S*per+2*s+2]=prob*terminal_penalty
            bounds += [(0,None),(0,None)]
    rr:list[int]=[]; cc:list[int]=[]; vv:list[float]=[]; rhs:list[float]=[]; row=0
    for s in range(S):
        G,C,D,U,H,E,DN,UP=slices(s)
        for t in range(T):
            rr += [row]*5; cc += [G.start+t,C.start+t,D.start+t,U.start+t,H.start+t]; vv += [1,-1,1,-
1,1]
            rhs.append(float(load[t]-pvs[s,t])); row+=1
        for t in range(T):
            rr += [row]*3; cc += [C.start+t,D.start+t,E.start+t]; vv += [-eta,1/eta,1]
            if t: rr.append(row); cc.append(E.start+t-1); vv.append(-1); rhs.append(0.0)
            else: rhs.append(float(initial_soc))
            row+=1
        for t in np.flatnonzero(committed):

```

```

rr += [row]*3; cc += [G.start+t,DN.start+t,UP.start+t]; vv += [1,1,-1]
rhs.append(float(base[t])); row+=1
if soft:
    rr += [row]*3; cc += [E.stop-1,S*per+2*s,S*per+2*s+1]; vv += [1,-1,1]
    rhs.append(float(terminal_target)); row+=1
elif terminal_rule in {"daily_closed","state_cycle","force_target":
    target=initial_soc if terminal_rule=="state_cycle" else terminal_target
    bounds[E.stop-1]=(float(target),float(target))
k=min(int(execute_intervals),T)
for s in range(1,S):
    a=slices(0); b=slices(s)
    for block in (0,1,2):
        for t in range(k):
            rr += [row,row]; cc += [a[block].start+t,b[block].start+t]; vv += [1,-1];
rhs.append(0.0); row+=1
Aeq=coo_matrix((vv,(rr,cc)),shape=(row,n)).tocsr()
started=time.perf_counter()
res=linprog(c,A_eq=Aeq,b_eq=np.asarray(rhs),bounds=bounds,method=method,
            options=_method_options(method,float(cfg["audit"])[["feasibility_tolerance"]]))
elapsed=time.perf_counter()-started
if not res.success: raise RuntimeError(f"stochastic LP infeasible ({method}):
{res.message}")
x=res.x; G,C,D,U,H,E,_,_=slices(0);
charge,discharge,unused=_one_way(x[C],x[D],x[U],eta); e0=np.r_[initial_soc,x[E][:1]]
bal=x[G]+pvs[0]+discharge+x[H]-load-charge-unused
soc=x[E]-e0-eta*charge+discharge/eta
return
DispatchSolution(x[G],charge,discharge,unused,x[H],x[E],float(res.fun),elapsed,int(res.nit),fl
oat(np.max(np.abs(bal))),float(np.max(np.abs(soc))))

```

src/model/risk.py

SHA256 93a7bf5c1a8572ee99e4098d67ba03fac164fdea33f9ea8198eab654fff0591a
 """Causal empirical scenarios and a nonanticipative stochastic storage LP.

Only emergency procurement is scenario recourse. All scheduled grid, charge, discharge and SOC variables are shared across scenarios over the full horizon.

```

from __future__ import annotations
from dataclasses import dataclass
from datetime import date,datetime,time,timedelta
import time as timer
import numpy as np
from scipy.optimize import linprog
from scipy.sparse import coo_matrix
from .data import DataBundle

```

```

@dataclass
class Cases:
    load: np.ndarray

```

```

pv: np.ndarray
price: np.ndarray
histories: list[str]
observed_until: list[str]
issue_time: str
kind: str

def causal_profile(data, matrix, target_i, cutoff_i):
    ids=[i for i in range(cutoff_i) if data.dates[i].weekday()==data.dates[target_i].weekday()]
    [-4:]
    if not ids:ids=list(range(max(0,cutoff_i-4),cutoff_i))
    if not ids:raise ValueError("No prequential baseline before first observation")
    return matrix[ids].mean(axis=0)

def point_horizon(data, matrix, day_i, issue, length):
    first=causal_profile(data, matrix, day_i, day_i)[issue*6:]
    if length>len(first):first=np.r_[first,causal_profile(data, matrix, day_i+1, day_i)]
    return first[:length]

def cases(data:DataBundle, cfg:dict, day:date, issue:int, length:int, mode:str, point:bool=False)-
>Cases:
    di=data.date_index[day];now=di*144+issue*6
    histories=[];ends=[];loads=[];pvs=[];prices=[]
    n=int(cfg['forecast']['scenario_count']);
    calibrated=cfg['forecast'].get('scenario_model')=='causal_residual'
    if calibrated:
        current_load=point_horizon(data, data.load_kw, di, issue, length)
        current_pv=point_horizon(data, data.pv_kw, di, issue, length) if mode in ('q2', 'q4_2') else
None
        current_price=point_horizon(data, data.realtime_price, di, issue, length) if
mode.startswith('q4') else None
    # Each donor is an actually observed 24h trajectory ending no later than now.
    for old_i in range(max(0, di-n), di):
        old=data.dates[old_i];g=old_i*144+issue*6
        if calibrated and old_i==0:continue
        if g+144>now:continue
        sl=slice(g, g+length)
        l=data.load_kw.reshape(-1)[sl]/6
        if calibrated:l=np.maximum(current_load+6*l-
point_horizon(data, data.load_kw, old_i, issue, length), 0)/6
        if mode in ('q3', 'q4_3'):
            old_actual=data.pv_kw.reshape(-1)[g:g+144].reshape(24,6).mean(axis=1)
            residual=old_actual-data.pv_forecast_kw[(old, issue)]
            # Forecast lead 1 represents the hour after publication (right endpoint
            # zero-order hold). No interpolation using future actual observations.
            pp=np.maximum(np.repeat(data.pv_forecast_kw[(day, issue)]+residual, 6)[:length], 0)/6

```

```

else:
    pp=data.pv_kw.reshape(-1)[sl]/6
    if calibrated:pp=np.maximum(current_pv+6*pp-
point_horizon(data,data.pv_kw,old_i,issue,length),0)/6
    p=data.realtime_price.reshape(-1)[sl] if mode.startswith('q4') else
np.asarray([data.fixed_price[(issue*6+k)%144] for k in range(length)])
    if calibrated and mode.startswith('q4'):p=np.maximum(current_price+p-
point_horizon(data,data.realtime_price,old_i,issue,length),0)
    loads.append(l);pvs.append(pp);prices.append(p);histories.append(old.isoformat())
    ends.append((datetime.combine(old,time())+timedelta(hours=issue+24)).isoformat(sep='
',timespec='minutes'))
    if not loads:raise ValueError('No observed history: execute the declared startup policy;
never substitute future actuals')
    L=np.array(loads);V=np.array(pvs);P=np.array(prices)
    if
point:L=L.mean(axis=0,keepdims=True);V=V.mean(axis=0,keepdims=True);P=P.mean(axis
=0,keepdims=True)
    return Cases(L,V,P,histories,ends,f'{day} {issue:02d}:00','historical_joint' if mode in
('q2','q4_2') else 'published_pv_plus_observed_residual')

```

```

@dataclass
class Solution:
    grid:np.ndarray
    charge:np.ndarray
    discharge:np.ndarray
    soc_end:np.ndarray
    objective:float
    seconds:float
    balance_residual:float
    soc_residual:float
    model_scenarios:int

```

```

def
solve(cs:Cases,soc0:float,baseline:np.ndarray,cfg:dict,terminal:bool=False,lock:np.ndarray|
None=None,method:str='highs',no_storage:bool=False)->Solution:
    L,V,P=cs.load,cs.pv,cs.price;S,T=L.shape;meanp=P.mean(axis=0)
    # Shared G,C,D,E,down,up, then scenario emergency variables and terminal slacks.
    n=(6+S)*T+2;G=0;C=T;D=2*T;E=3*T;DN=4*T;UP=5*T;Q=6*T;Z=n-2
    committed=np.isfinite(baseline);obj=np.zeros(n)
    obj[G:G+T]=np.where(committed,0,meanp)
    obj[DN:DN+T]=np.where(committed,-.5*meanp,0)
    obj[UP:UP+T]=np.where(committed,1.5*meanp,0)
    obj[Q:Q+S*T]=(5*P/S).reshape(-1)
    tb=float(cfg['cost'] ['throughput_tiebreak_yuan_per_kwh']);obj[C:C+T]=tb;obj[D:D+T]=tb
    penalty=float(np.median(meanp))*float(cfg['rolling'] ['selected_soft_penalty_multiplier'])
    obj[Z:Z+2]=penalty
    eta_c=float(cfg['storage'] ['charge_efficiency']);eta_d=float(cfg['storage']
['discharge_efficiency'])

```

```

cap=float(cfg['storage']['power_kw']/6;low=cfg['storage']
['soc_min_kwh'];high=cfg['storage']['soc_max_kwh']
bounds=[(0,None)]*T+[(0,0 if no_storage else cap)]*(2*T)+[(low,high)]*T
bounds += [(0,float(baseline[t])) if committed[t] else (0,0) for t in range(T)]
bounds += [(0,None) if committed[t] else (0,0) for t in range(T)]
bounds += [(0,None)]*(S*T+2)
if lock is not None:
    for t,v in enumerate(lock):
        if np.isfinite(v):bounds[t]=(float(v),float(v))
    if terminal:bounds[E+T-1]=(cfg['study']['terminal_soc_kwh'],cfg['study']
['terminal_soc_kwh'])
    er=[];ec=[];ev=[];b=[]
    def eq(pairs,rhs):
        r=len(b);b.append(rhs)
        for col,val in pairs:er.append(r);ec.append(col);ev.append(val)
    for t in range(T):
        pairs=[(E+t,1),(C+t,-eta_c),(D+t,1/eta_d)]
        if t:pairs.append((E+t-1,-1))
        eq(pairs,soc0 if t==0 else 0)
    for t in np.flatnonzero(committed):eq([(G+t,1),(DN+t,1),(UP+t,-1)],float(baseline[t]))
    eq([(E+T-1,1),(Z,-1),(Z+1,1)],float(cfg['study']['terminal_soc_kwh']))
    # G + D - C + Q_s >= L_s - PV_s. Surplus is allowed, never sold.
    rr=[];cc=[];vv=[]
    for s in range(S):
        for t in range(T):
            r=s*T+t;rr.extend([r]*4);cc.extend([G+t,D+t,C+t,Q+s*T+t]);vv.extend([-1,-1,1,-1])
        A=coo_matrix((vv,(rr,cc)),shape=(S*T,n)).tocsr();Aeq=coo_matrix((ev,
(er,ec)),shape=(len(b),n)).tocsr()
        start=timer.perf_counter()
        res=linprog(obj,A_ub=A,b_ub=(V-
L).reshape(-1),A_eq=Aeq,b_eq=np.array(b),bounds=bounds,method=method,options={'prima
l_feasibility_tolerance':1e-8,'dual_feasibility_tolerance':1e-8,'ipm_optimality_tolerance':1e-9}
)
        seconds=timer.perf_counter()-start
        if not res.success:raise RuntimeError(res.message)
        x=res.x;c=x[C:C+T].copy();d=x[D:D+T].copy();e=x[E:E+T].copy()
        # Exact dominance transformation: preserve SOC, reduce purchases needed by
        # removing simultaneous cycling; free disposal and no minimum flow assumed.
        delta=eta_c*c-d/eta_d;c=np.maximum(delta,0)/eta_c;d=np.maximum(-delta,0)*eta_d
        q=x[Q:Q+S*T].reshape(S,T)
        violation=float(np.maximum(L+c[None,:]-V-d[None,:]-x[:T][None,:]-q,0).max())
        sr=float(np.abs(e-np.r_[soc0,e[:-1]]-eta_c*c+d/eta_d).max())
        if violation>1e-7 or sr>1e-7:raise AssertionError('Post-solve physical residual')
        return Solution(x[:T].copy(),c,d,e,float(res.fun),seconds,violation,sr,S)

```

src/model/rolling.py

```

SHA256 eab5712bae13320e2c960f9c9c07e55fd6814911ba2899992d56bf06cec3668e
from __future__ import annotations
from dataclasses import dataclass
from datetime import date,timedelta

```

```
import numpy as np
from .risk import cases,solve
```

```
@dataclass(frozen=True)
class RunSpec:
    mode:str
    scenario:str='S3'
    arm:str='joint_empirical'
    method:str='highs'
    start_date:date|None=None
    end_date:date|None=None
    collect_detail:bool=True
```

```
def interval(pos):
    a=pos*10;b=a+10
    return f'{a//60:02d}:{a%60:02d}-{b//60:02d}:{b%60:02d}'
```

```
def make_row(data,day,j,B,A,c,d,s0,s1,fl,fp,pricehat,owner,mode,scenario,seq,history,arm):
    di=data.date_index[day];L=float(data.load_kw[di,j]/6);V=float(data.pv_kw[di,j]/6)
    p=float(data.realtime_price[di,j] if mode.startswith('q4') else data.fixed_price[j])
    deficit=L+c-V-d-A;Q=max(deficit,0);U=max(-deficit,0);delta=A-B
    pc=p*B;ac=p*(delta+.5*abs(delta));em=5*p*Q
    forecast_unused=max(A+fp+d-fl-c,0)
    return
dict(date=day.isoformat(),position=j+1,source_timestamp=data.source_timestamps[j],physical_interval=interval(j),official_label=data.official_labels[j],
plan_00_grid_kwh=float(B),final_adjusted_grid_kwh=float(A),adjustment_kwh=float(delta),
charge_kwh=float(c),discharge_kwh=float(d),soc_start_kwh=float(s0),soc_end_kwh=float(s1)
),
curtail_pv_kwh=min(forecast_unused,fp),planned_unused_nonpv_kwh=max(forecast_unused
-
fp,0),forecast_load_kwh=float(fl),forecast_pv_kwh=float(fp),actual_load_kwh=L,actual_pv_
kwh=V,
forecast_price=float(pricehat),settlement_price=p,emergency_kwh=Q,unused_surplus_kwh=
U,plan_cost_yuan=pc,adjustment_cost_yuan=ac,emergency_cost_yuan=em,
final_relative_cost_yuan=pc+ac+em,sequential_cost_yuan=pc+seq+em,decision_hour=owner
,history_cutoff_date=(day-
timedelta(days=1)).isoformat(),model_arm=arm,scenario=scenario)
```

```
def summarize(rows,mode,scenario,arm):
```

```

    sums={k:float(sum(r[k] for r in rows)) for k in
['plan_cost_yuan','adjustment_cost_yuan','emergency_cost_yuan','final_relative_cost_yuan','s
equential_cost_yuan','emergency_kwh','unused_surplus_kwh']}
    return dict(mode=mode,scenario=scenario,model_arm=arm,days=len({r['date'] for r in
rows})),initial_soc_kwh=rows[0]['soc_start_kwh'],terminal_soc_kwh=rows[-1]
['soc_end_kwh'],**sums)

```

```

def
simulate(data,cfg,spec:RunSpec,*,initial_soc:float,start_date=None,end_date=None,common
_plans=None,point=False,no_storage=False,evaluation_terminal=False):
    start=start_date or spec.start_date or date.fromisoformat(cfg['study']
['official_start']);end=end_date or spec.end_date or date.fromisoformat(cfg['study']
['official_end'])
    if initial_soc is None:raise ValueError('Explicit state from startup ledger required')
    soc=float(initial_soc);rows=[];versions=[];daily=[];stats=[];plans={}
    schedule=cfg['experiments']['schedules'][spec.scenario] if spec.mode in ('q3','q4_3') else [0]
    total=(data.date_index[end]+1)*144 if evaluation_terminal else len(data.dates)*144
    for day in data.dates:
        if not start<=day<=end:continue
        di=data.date_index[day];B=None;committed=None;seq=np.zeros(144);dayrows=[]
        for ix,issue in enumerate(schedule):
            pos=issue*6;remaining=144-pos;length=min(144,total-(di*144+pos)) if spec.mode in
('q3','q4_3') else 144
            cs=cases(data,cfg,day,issue,length,spec.mode,point)
            base=np.full(length,np.nan);lock=np.full(length,np.nan)
            if issue==0 and common_plans is not None:
                B=np.asarray(common_plans[day.isoformat()],float).copy();base[:144]=B;lock[:144]=B
                elif issue>0:base[:remaining]=B[pos:]
                terminal=di*144+pos+length==total

sol=solve(cs,soc,base,cfg,terminal=terminal,lock=lock,method=spec.method,no_storage=no_
storage)
    if issue==0:
        if B is None:B=sol.grid[:144].copy()
        committed=B.copy();plans[day.isoformat()]=B.copy()
    else:
        old=committed[pos:].copy();delta=sol.grid[:remaining]-old
        p=data.realtime_price[di,pos:] if spec.mode.startswith('q4') else
data.fixed_price[pos:]
        seq[pos:]+=p*(delta+.5*np.abs(delta))
        # Every shared current-day quantity is a genuine commitment; tomorrow
        # is a nonbinding preview. No deterministic alternate settlement path.
        for k in range(length):
            tdi,tj=divmod(di*144+pos+k,144);target=data.dates[tdi];preview=target!=day
            value=(B[k] if issue==0 else sol.grid[k]) if not preview else "
            previous=" if preview or issue==0 else float(committed[pos+k])
            if spec.collect_detail:

```

```

versions.append(dict(date=str(day),target_date=str(target),mode=spec.mode,scenario=spec.scenario,model_arm=spec.arm,decision_time=cs.issue_time,forecast_issue_time=cs.issue_time,
                    effective_start=cs.issue_time,effective_end=f'{day+timedelta(days=1)}
00:00',history_cutoff_date=str(day-
timedelta(days=1)),position=tj+1,physical_interval=interval(tj),official_label=data.official_labels[tj],
                    old_commitment_kwh=previous,new_commitment_kwh=float(value) if value!
=" else
",sensitivity_old_commitment_kwh=previous,sensitivity_new_commitment_kwh=float(value
) if value!=" else ",preview_grid_kwh=float(sol.grid[k]) if preview else ",
                    decision_basis='joint_nonanticipative_lp' if not preview else
'next_day_preview',soc_at_decision_kwh=soc,
                    load_history_dates=''.join(cs.histories) if k==0 else
",price_history_dates=''.join(cs.histories) if k==0 and spec.mode.startswith('q4') else
",pv_bias_history_dates=''.join(cs.histories) if k==0 else
",residual_scenario_dates=''.join(cs.histories) if k==0 else ",

latest_observation_time=max(cs.observed_until),is_preview_next_day=int(preview),is_previe
w_recourse=int(preview)))
    if issue>0:committed[pos:]=sol.grid[:remaining]
    nxt=schedule[ix+1] if ix+1<len(schedule) else 24;execute=(nxt-issue)*6
    sstarts=np.r_[soc,sol.soc_end[:-1]]
    for k in range(execute):
        j=pos+k

    dayrows.append(make_row(data,day,j,B[j],sol.grid[k],sol.charge[k],sol.discharge[k],sstarts[k]
,sol.soc_end[k],cs.load[:,k].mean(),cs.pv[:,k].mean(),cs.price[:,k].mean(),issue,spec.mode,spe
c.scenario,seq[j],cs.histories,spec.arm))
        soc=float(sol.soc_end[execute-1])

stats.append(dict(mode=spec.mode,scenario=spec.scenario,date=str(day),issue_hour=issue,sol
ver_method=spec.method,seconds=sol.seconds,objective=sol.objective,max_balance_residual
_kwh=sol.balance_residual,max_soc_residual_kwh=sol.soc_residual,scenario_count=sol.mod
el_scenarios,latest_observation_time=max(cs.observed_until),decision_time=cs.issue_time,hi
story_dates=''.join(cs.histories),common_action_intervals=length))

rows.extend(dayrows);daily.append(dict(date=str(day),**summarize(dayrows,spec.mode,spec
.scenario,spec.arm)))

summary=summarize(rows,spec.mode,spec.scenario,spec.arm);summary.update(solver_secon
ds=sum(s['seconds'] for s in
stats),solver_runs=len(stats),constraint_violation_count=sum(s['max_balance_residual_kwh']
>1e-7 or s['max_soc_residual_kwh']>1e-7 for s in stats))
    return
dict(dispatch=rows,decision_versions=versions,daily=daily,summary=summary,solve_stats=s
tats,plans=plans)

```

tests/test_closed_loop.py

SHA256 1f61a806e5df4a92ca5d11ef707618216086f85e54fd890dc008f11ee5583017

```

import unittest,copy,json
from dataclasses import replace
from datetime import date
from pathlib import Path
import numpy as np
from src.model.config import load_config
from src.model.data import load_data
from src.model.forecast import history_days
from src.model.risk import cases,solve
from src.model.rolling import simulate,RunSpec
from src.model.experiments import startup

class ClosedLoopTests(unittest.TestCase):
    @classmethod
    def setUpClass(cls):
        cls.cfg=load_config('config/final.yaml');
resolved=Path(cls.cfg['_output_root'])/'frozen/resolved_config.json'
        if resolved.exists():cls.cfg.update(json.loads(resolved.read_text()))
        cls.data=load_data(cls.cfg)
    def test_no_future_cold_start(self):
        with self.assertRaises(ValueError):history_days(self.data,date(2025,1,1),date(2025,1,1))
        data=self.data;mut=data.load_kw.copy();mut[0]+=2000
        a=startup(data,self.cfg,'q2')[:144];b=startup(replace(data,load_kw=mut),self.cfg,'q2')
[:144]
        self.assertEqual([r['plan_grid_kwh'] for r in a],[r['plan_grid_kwh'] for r in
b]);self.assertTrue(all(r['soc_end_kwh']==6000 for r in a))
    def test_future_observations_cannot_change_decision(self):
        for mode,issue in [('q2',0),('q4_2',0),('q3',12),('q4_3',18)]:
            day=date(2025,2,1);di=self.data.date_index[day];cut=di*144+issue*6;kw={}
            for field in ['load_kw','pv_kw','realtime_price']:
                x=getattr(self.data,field).copy();x.reshape(-1)[cut:]+=321;kw[field]=x

alt=replace(self.data,**kw);a=cases(self.data,self.cfg,day,issue,144,mode);b=cases(alt,self.cfg
,day,issue,144,mode)
        for field in
['load','pv','price']:np.testing.assert_array_equal(getattr(a,field),getattr(b,field))

baseline=np.full(144,np.nan);x=solve(a,6000,baseline,self.cfg);y=solve(b,6000,baseline,self.c
fg);np.testing.assert_array_equal(x.grid,y.grid)
    def test_later_forecasts_not_visible(self):
        day=date(2025,2,1); forecasts={k:v.copy() for k,v in self.data.pv_forecast_kw.items()}
        for key in forecasts:
            if key[0]>day or (key[0]==day and key[1]>0):forecasts[key]+=9876
        alt=replace(self.data,pv_forecast_kw=forecasts)

a=cases(self.data,self.cfg,day,0,144,'q3');b=cases(alt,self.cfg,day,0,144,'q3');np.testing.assert_
array_equal(a.pv,b.pv)

    def test_nonanticipativity_and_formal_ledger(self):
        data=self.data;cfg=self.cfg;st=date(2025,2,1);en=date(2025,2,2)

```

```

ref=simulate(data,cfg,RunSpec('q3','S0'),initial_soc=6000,start_date=st,end_date=en)

r=simulate(data,cfg,RunSpec('q3','S3'),initial_soc=6000,start_date=st,end_date=en,common_
plans=ref['plans'])
    np.testing.assert_allclose([x['plan_00_grid_kwh'] for x in ref['dispatch']],
[x['plan_00_grid_kwh'] for x in r['dispatch']],atol=1e-8)
    commitments={},fees={}
    for v in r['decision_versions']:
        if v['is_preview_next_day']:continue
        key=(v['target_date'],v['position']);new=v['new_commitment_kwh']
        if key in commitments:
            self.assertAlmostEqual(commitments[key],v['old_commitment_kwh']);delta=new-
commitments[key];p=data.fixed_price[key[1]-
1];fees[key]=fees.get(key,0)+p*(delta+.5*abs(delta))
            commitments[key]=new
        for x in r['dispatch']:

key=(x['date'],x['position']);self.assertAlmostEqual(commitments[key],x['final_adjusted_grid
_kwh']);self.assertAlmostEqual(x['sequential_cost_yuan'],x['plan_cost_yuan']
+fees.get(key,0)+x['emergency_cost_yuan']);self.assertGreaterEqual(x['sequential_cost_yuan']
-x['final_relative_cost_yuan'],-1e-7)
        self.assertAlmostEqual(x['soc_end_kwh'],x['soc_start_kwh']+.9*x['charge_kwh']-
x['discharge_kwh']/.9);self.assertLess(min(x['charge_kwh'],x['discharge_kwh']),1e-7)
        def test_partition_does_not_impose_terminal_reset(self):
            st=date(2025,1,30);en=date(2025,1,31);cfg=self.cfg;d=self.data
            full=simulate(d,cfg,RunSpec('q2','S0'),initial_soc=6000,start_date=st,end_date=en)
            first=simulate(d,cfg,RunSpec('q2','S0'),initial_soc=6000,start_date=st,end_date=st)
            second=simulate(d,cfg,RunSpec('q2','S0'),initial_soc=first['summary']
['terminal_soc_kwh'],start_date=en,end_date=en)
            np.testing.assert_allclose([r['final_adjusted_grid_kwh'] for r in full['dispatch']],
[r['final_adjusted_grid_kwh'] for r in first['dispatch']+second['dispatch']],atol=1e-8)
            def test_separate_charge_discharge_efficiencies(self):
                cfg=copy.deepcopy(self.cfg);cfg['storage']['discharge_efficiency']=.85

cs=cases(self.data,cfg,date(2025,2,1),0,144,'q2');x=solve(cs,6000,np.full(144,np.nan),cfg)
        np.testing.assert_allclose(x.soc_end-np.r_[6000,x.soc_end[:-1]],.9*x.charge-
x.discharge/.85,atol=1e-8)
        def test_charge_limit_and_year_end(self):

r=simulate(self.data,self.cfg,RunSpec('q4_3','S3'),initial_soc=1200,start_date=date(2025,12,3
1),end_date=date(2025,12,31))
        self.assertAlmostEqual(r['summary']['terminal_soc_kwh'],6000)
        for x in r['dispatch']:

self.assertLessEqual(x['charge_kwh'],5000/6+1e-7);self.assertLessEqual(x['discharge_kwh'],5
000/6+1e-7);self.assertGreaterEqual(x['soc_end_kwh'],1200-
1e-7);self.assertLessEqual(x['soc_end_kwh'],10800+1e-7)

if __name__=='__main__':unittest.main(verbosity=2)

```

附录3 公式推导与补充实验记录

3.1 充放电互斥变换的可行性

令 $z = \eta_c C - D / \eta_d$ ，其中 $C, D \geq 0$ ， $0 < \eta_c, \eta_d \leq 1$ 。取 $C' = \max(z, 0) / \eta_c$ ， $D' = \eta_d \max(-z, 0)$ ，则 $\eta_c C' - D' / \eta_d = z$ ，故变换前后储电量增量相同，且 $C'D' = 0$ 。

当 $z \geq 0$ 时， $C' = C - D / (\eta_c \eta_d)$ ， $D' = 0$ ，因此 $0 \leq C' \leq C$ ；母线净需求减少 $D(1/(\eta_c \eta_d) - 1) \geq 0$ 。当 $z < 0$ 时， $C' = 0$ ， $D' = D - \eta_c \eta_d C$ ，因此 $0 \leq D' \leq D$ ；母线净需求减少 $C(1 - \eta_c \eta_d) \geq 0$ 。原功率上界与储电量边界均保留。允许免费处置富余电量时，可通过增加未用富余或减少采购恢复供需平衡，费用不会增加。该结论依赖本文关于弃电、价格与运行约束的假设。

3.2 单期采购分位点与结算差额

设无储能单期净需求为 N ，采购 q ，固定价格 $p > 0$ ，目标 $g(q) = pq + 5pE[(N - q)^+]$ 。连续分布下 $g'(q) = p - 5p\Pr(N > q)$ ，驻点满足 $\Pr(N \leq q) = 0.8$ ；目标凸，故该分位点为最优解。离散分布下对应条件为 $F(q-) \leq 0.8 \leq F(q)$ 。该结论仅解释单期非对称损失，正式模型仍求解耦合储能的线性规划。

对每个交付时段 t ，令 $\Delta A_t^k = A_t^k - A_t^{k-1}$ ，则各次变化之和为最终承诺与初始承诺之差。主结算和逐次结算中的初始计划费、线性变化费与紧急采购费相消，只剩 $0.5 \sum_t p_t (\sum_k |\Delta A_t^k| - |\sum_k \Delta A_t^k|)$ 。当同一时段各次采用相同非负价格时，由三角不等式，该差额非负，即式 (12)。

3.3 参数调试与额外实验索引

参数候选、选择规则与费用见正文表 3 及支撑材料 `/evidence/january_parameter_validation.csv`；锁定配置见 `config/final.yaml` 与 `frozen/resolved_config.json`。指定日期求解器复求、全年替代求解器回放、月度费用、尾部风险、预测误差和块长度敏感性记录见 `research` 目录，相关程序见 `scripts/strengthen_evidence.py`。正文图 1—5 和表 9—13 给出主要结果，完整精度记录随支撑材料保存。